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Editorial Correspondence

FOLIA MEDICA,
MU, Library and Information Center, Tel. ++359 32 / 602 224
15 A Vassil Aprilov St., 4002 Plovdiv, Bulgaria e-mail: folia_medica@meduniversity-plovdiv.bg

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ORIGINAL ARTICLES

Clinical Investigations

FLUCTUATING ASYMMETRY IN DERMATOGLYPHIC TRAITS IN SCHIZOPHRENIC PATIENTS

Stefan T. Sivkov, Valentin H. Akabaliev[1], Katerina V. Akabalieva

Department of Anatomy, Histology and Embryology, [1]Department of Psychiatry and Medical psychology, Medical University, Plovdiv

ABSTRACT

Aim: The aim of the present study was to establish the validity of fluctuating asymmetry in dermatoglyphic traits as a sign of prenatal injury of schizophrenic patients.

Material and methods. The subjects for this study were 76 schizophrenic inpatients (43 men, 33 women, mean age 31.47 yrs) who satisfied DSM-IV criteria for a diagnosis of schizophrenia and 82 mentally healthy subjects of Bulgarian origin (mean age 39.24 yrs). Fingerprint variables obtained by the ink technique were examined. The fingerprints were read using the method of Cummins and Midlo.

Results: The schizophrenics tended to show higher degree of discordance in the fingerprint patterns and ridge counts on homologous fingers than the control subjects. There were evident sex-related differences between the schizophrenic patients and the control group of subjects.

Conclusion: Fluctuating asymmetry appears a promising method for study of schizophrenia, which could contribute to the establishment of connection between prenatal exogenous influences and structural brain alterations.

Key words: *fingerprints, fluctuating asymmetry, schizophrenia, neurodevelopment*

INTRODUCTION

Etiology of schizophrenia still remains a puzzle. However, many researchers believe suggest that the disease is a consequence of genetic and epigenetic factors disrupting the brain development.[1-4] As prenatal development of brain is impossible to examine directly, researchers refocused their efforts to search for adequate biological markers such as minor physical anomalies and dermatoglyphic traits.[5,6]

The biological value of dermatoglyphic traits stems from the consistency in the organization of dermal ridges on the fingers, toes, palms and soles. Once formed in the embryonal period, they never change in later life.[7] If adversities of various nature interfere in the prenatal ridge formation they disrupt the normal ridge differentiation. Apparently, disorders in the ridge structure and pattern can only be due to genetic or prenatally operating exogenous factors.

Dermatoglyphic studies of schizophrenia, however, are marked by ambiguous findings, which makes their interpretation difficult. This fact can be accounted for by methodological differences, racial and ethnic peculiarities of the examined groups and the form and course of the schizophrenic disorder.[8] A novel approach to exploring the potential role of prenatal factors in schizophrenia etiology is by assessing the fluctuating asymmetry in dermatoglyphic traits.[9,10] Fluctuating asymmetry was originally described by Ludwig (1932) as a sign of developmental stability in various species including man.[11] In fluctuating asymmetry the differences between corresponding right- and left-sided structures show a distribution different from the normal one in a population sample, which is approximately zero.

Increased asymmetry in homologous structures or high level of fluctuating asymmetry indicates a decreased capacity of the body to buffer adverse exogenous effects[11] and has an essential value for

Correspondence and reprint request to: St. Sivkov, Department of Anatomy, Medical University Plovdiv 15A Vassil Aprilov St., 4002 Plovdiv, Bulgaria
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the identification of prenatal factors.[5,10,11] High level of fluctuating asymmetry is considered a sign of disturbed neurodevelopment at the time of dermal ridge formation – prenatal months 3 and 4.[12] Fluctuating asymmetry is not related to sex[13] or certain hemispheric dominance expressed by functional asymmetry in the limbs – right- or left-handedness.[14]

The aim of the present study was to establish the validity of fluctuating asymmetry in dermatoglyphic traits as a sign of prenatal injury of schizophrenic patients.

MATERIAL AND METHODS

Subjects

The subjects for this study were 76 schizophrenic inpatients (43 men, 33 women) consecutively admitted to the Clinic of Psychiatry in Plovdiv. Their mean age was 31.47 ± 1.24 years, mean duration of illness 6.86 ± 0.81 years, mean number of hospitalizations 4.22 ± 0.54. The patients satisfied the DSM-IV criteria for schizophrenia (American Psychiatric Association, 1994)[15] on the basis of the case records' review, semistructured interview (conducted by V.A., the present study psychiatrist) based on a checklist of items from DSM-IV and information obtained from relatives in order to enhance the validity of the diagnosis. Potential subjects were excluded if they had a history of drug or alcohol abuse, identifiable neurological disorder (seizure disorder, head injury, multiple sclerosis etc.), any signs of mental retardation or somatic disorder with neurological components.

The normal comparison group comprised 82 subjects (42 men, 40 women) of Bulgarian origin and a mean age of 39.24 ± 1.19 years. Individuals were excluded if their parental or grandparental ethnic group was other than Bulgarian. The groups of men and women were comparable in their socioeconomic background. Normality was defined as the absence of a major axis I or axis II disorder according to DSM-IV.[15] They satisfied exclusion criteria similar to those applied to the patients. In addition, to better separate the controls from the schizophrenic group, potential normal controls were excluded if they had a first-degree relative with a history of a psychotic disorder, major affective disorder or suicide.

The study was approved by the local Ethics Committee and all subjects gave written informed consent to participate.

Dactyloscopy

A set of dermatoglyphic traits with low interracial variability and high diagnostic value was examined.[8] Rolled fingerprints were obtained passively using an ink technique and were read with light (6D) magnification.

Finger ridge patterns were read using the method of Cummins and Midlo.[16] According to the number of triradii the patterns were defined as arches, loops and whorls[17,18] (Fig. 1). The ridge count was assessed by the number of ridges crossed or touched by the line connecting the center of the triradius with the center of the corresponding pattern (Fig. 2).

The fingerprint pattern and finger-ridge count on homologous fingers were used for evaluation of the level of fluctuating asymmetry as an indicator of the organism's ability to buffer adverse effects during the prenatal period.[11]

Arch Loop Whorl

Figure 1. Fingerprint patterns.

Figure 2. Measurement of finger ridge count.

The level of fluctuating asymmetry in quantitative variables was determined by the formula $1-r^2$.[14] The square of the correlation coefficient (r^2) is a measure of the common variance of the examined events and $1-r^2$, known also as the coefficient of indetermination, characterizes the absence of correlation between the events and represents the fluctuating asymmetry.

In qualitative variables the level of fluctuating asymmetry was determined by the degree of pattern discordance in homologous fingerprints.

The data were analyzed using analysis of variance and correlation analysis. The differences were assumed to be statistically significant at $p<0.05$.

RESULTS

Fluctuating asymmetry in fingerprints

Finger ridge count. The correlation coefficients for the homologous fingers of the right and left hand in the schizophrenic patients and control males and females, the measures of the fluctuating asymmetry ($1-r^2$) derived from the correlation coefficients and the differences between the schizophrenic patients and control subjects are shown in Table 1 and Table 2.

Schizophrenic patients showed weaker correlation in the finger ridge count than the control subjects in three of the five homologous fingers and in the total ridge count between right and left hands. The differences were greater, yet failing to reach statistical significance, for the first fingers in the males and third fingers in the females.

Schizophrenic patients showed higher levels of fluctuating asymmetry than the control subjects in three of the five homologous fingers and in the total ridge count between right and left hands. The highest degree of fluctuating asymmetry was

Table 1. Correlation coefficients (rh) for finger ridge count on homologous fingers in schizophrenic patients and control subjects

Fingers	Schizophrenia	Controls	Difference between r
Males			
I	0.6596	0.7730	**p < 0.05**
II	0.7615	0.7999	p > 0.05
III	0.8832	0.8473	p > 0.05
IV	0.8657	0.8191	p > 0.05
V	0.7948	0.8127	p > 0.05
Females			
I	0.7292	0.7888	p > 0.05
II	0.6807	0.5716	p > 0.05
III	0.6542	0.8541	**p < 0.05**
IV	0.8347	0.8748	p > 0.05
V	0.8541	0.7853	p > 0.05

Table 2. Fluctuating asymmetry ($1-r^2$) in finger ridge count on homologous fingers in schizophrenic patients and control subjects

Fingers	Scizophrenia (S)	Controls (C)	Difference (S-C)
Males			
I	0.5649	0.4025	+ **0.1624**
II	0.4201	0.3602	+ 0.0599
III	0.2199	0.2821	- 0.0622
IV	0.2506	0.3201	- 0.0695
V	0.3683	0.3395	+ 0.0288
Females			
I	0.4683	0.3778	+ 0.0905
II	0.5366	0.6733	- 0.1367
III	0.5720	0.2705	+ **0.3015**
IV	0.3033	0.2347	+ 0.0686
V	0.2705	0.3833	- 0.1128

Table 3. Concordance of fingerprint patterns on homologous fingers in schizophrenic patients and control subjects by gender

	Males						Females					
	Schizophrenia (n=38)		Controls (n=36)		Statistical significance		Schizophrenia (n=30)		Controls (n=33)		Statistical significance	
Finger	Mean	SD	Mean	SD	t	p	Mean	SD	Mean	SD	t	p
I	0.79	0.43	0.64	0.49	-1.414	.162	0.83	0.38	0.88	0.34	0.459	.648
II	0.73	0.45	0.63	0.49	-1.299	.199	0.40	0.50	0.66	0.48	1.529	.132
III	0.76	0.43	0.68	0.47	-0.812	.419	0.53	0.51	0.84	0.37	2.403	.020
IV	0.84	0.37	0.74	0.45	-1.096	.277	0.77	0.43	0.69	0.47	-0.690	.493
V	0.79	0.41	0.82	0.39	0.360	.720	0.80	0.41	0.94	0.25	0.793	.431

found for the first fingers in the males and third fingers in the females.

Finger ridge pattern. Finger ridge patterns are not morphometric characteristics, but tend to be identical on homologous fingers. That is why the degree of pattern discordance can be used as an indicator of fluctuating asymmetry.

Table 3 shows the gender distribution of the finger ridge patterns on homologous fingers in schizophrenic patients and control subjects.

As a whole, the schizophrenic patients showed higher degree of discordance in the fingerprints on homologous fingers than the control subjects. There were, however, clearly expressed sex-related differences between the patients and control subjects. In the males, the differences between the homologous fingers did not reach statistical significance ($p > 0.05$). The schizophrenic patients showed higher level of discordance than the controls only between the fifth fingers.

The schizophrenic females showed higher level of discordance in the fingerprint patterns on homologous fingers than the control females. The differences were greater between the second and third fingers, but reached statistical significance only between the third fingers ($p < 0.05$). The other homologous fingers showed almost identical level of concordance both in the patients and control subjects.

The findings of the comparative dermatoglyphic characteristic of schizophrenic patients suggested the following general conclusions with respect to fluctuating asymmetry:

• There is a high degree of fluctuating asymme-

try in the finger ridge count, most evident between the first fingers in males and between the third fingers in females.

• Female patients show higher degree of discordance in the fingerprint patterns on homologous fingers especially well expressed between the second and third fingers.

DISCUSSION

The high levels of fluctuating asymmetry indicate a reduced capacity to neutralize dysontogenic influences. In our sample, the schizophrenic subjects showed increased fluctuating asymmetry in the total finger ridge count. In spite of the evidence of absence of sexual dimorphism in the fluctuating asymmetry, the schizophrenic females in the present study exhibited greater absolute differences in the total ridge count and higher level of fluctuating asymmetry than the control females. In the males, despite the tendency of more expressed asymmetry in schizophrenic subjects, the differences failed to reach statistical significance. Possible explanation of the differences could be found in the sex-related characteristics in the course of schizophrenia.[19] Apparently, more powerful peristaltic impact or higher level of cumulative genetic effect is necessary so that schizophrenia can occur in females. In this direction lies the explanation of the sex-related characteristics in the fluctuating asymmetry. The latter is known to occur more clearly in homozygotic subjects and correlates negatively with heterozygosity. This could imply higher level of homozygosity in schizophrenic females, which correlates with lower resistance to external influences. On the other hand, despite the presence of epigenetic effect, the fluctuating (and directional) asymmetry is under significant genetic control, which is more expressed in males.

The evidence of increased fluctuating asymmetry in the examined dermatoglyphic traits is consistent with the etiological theories of schizophrenia development from the field of genetics, prenatal adverse exposure and dysontogenetic processes.[4]

According to the polygenic theory put forward by Gottesman and Shields (1967) and elaborated by McGue et al. (1983), a threshold in the polygenic system has to be crossed to develop schizophrenic disorder.[20] Alternatively, the increased number of genes in the polygenic system may reduce the general buffering capacity of the organism to neutralize the adverse ontogenetic effects, caused by a particular gene in a particular organ.

It is quite possible that exogenous adversities can have particular contribution to the increase in the fluctuating asymmetry, which disrupts the otherwise normal buffering system of the body. However, to date there has been no sufficiently solid direct evidence of relations between intrauterine insults and consequent development of schizophrenia.

As an addition to the traditional methods of dermatoglyphic examination, fluctuating asymmetry appears to be a promising method for studying schizophrenia. Establishment of more sensitive indices of fluctuating asymmetry could define more precisely the time of possible maldevelopment. Family and twin studies using markers of fluctuating asymmetry could contribute to the elucidation of the problem of genetic heterogeneity of the disease. And last but not least, fluctuating asymmetry could help in determining the relation between prenatal exogenous influences and structural alterations of brain.

CONCLUSIONS

Examination of fluctuating asymmetry in dermatoglyphic indices appears to be a promising method for studying the neurodevelopmental hypothesis of schizophrenic diseases. In a constellation with other specific morphogenetic variables its application could verify the presence and period of occurrence of prenatal exogenous influences with aberrant effect on the processes of neurolation.

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ФЛЮКТУИРУЮЩАЯ АСИММЕТРИЯ В ДЕРМАТОГЛИФИЧЕСКИХ ПРИЗНАКАХ У БОЛЬНЫХ ШИЗОФРЕНИЕЙ

С. Сивков, В. Акабалиев, К. Акабалиева

РЕЗЮМЕ

Цель: *Настоящее исследование ставит себе целью установить достоверность флюктуирующей дерматоглифической асимметрии как признак пренатального повреждения у пациентов, больных шизофренией.*

Материал и методы: *Авторы обследовали 76 госпитализированных больных шизофренией (43 мужчины и 33 женщины; средний возраст – 31.47; покрывают критерии ДСМ-IV для диагноза "шизофрения") и 82 психически здоровых лиц болгарского происхождения (средний возраст – 39.24). По типографическому методу исследованы пальцевые папилярные признаки. Изображения расшифрованы по методике Cummins, Midlo.*

Результаты: *В целом больные шизофренией показывают более высокую степень дискордантности в папилярных изображениях и хребетного числа гомоложных пальцев по сравнению с лицами контрольной группы. Налицо ярко выраженные полово зависимые различия между больными шизофренией и лицами контрольной группы.*

Заключение: *Флюктуирующая асимметрия представляет обещающий метод исследования шизофрении, который смог бы способствовать установлению взаимосвязи между пренатальными экзогенными воздействиями и структурными изменениями в мозге.*

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COMPUTERTOMOGRAPHIC FINDINGS OF STRUCTURAL BRAIN ANOMALIES IN SCHIZOPHRENIC PATIENTS IN SUPPORT OF THE NEURODEVELOPMENTAL HYPOTHESIS OF THE DISORDER

Vessela Georgieva, Stefan T. Sivkov[1], Valentin H. Akabaliev[2], Yvetta A. Koeva[1], Roumiana Y. Angelova[3], Tania I. Deneva[3]

Department of Radiology, [1]Department of Anatomy, Histology and Embryology, [2]Department of Psychiatry and Medical Psychology, [3]Department of Clinical Laboratory, Medical University Plovdiv

ABSTRACT

Aim: The aim of the study was to make a comparative CT examination of schizophrenic patients and find lifetime criteria for recognition of brain changes in schizophrenia.

Material and methods: Twenty-two schizophrenic inpatients (mean age 32.86±2.65 yrs) satisfying the DSM-IV criteria for schizophrenia were examined. The control group comprised 27 clinically healthy subjects (16 men, 11 women, mean age 46.44±2.32 yrs) all of Bulgarian ancestry. All subjects underwent CT examination without venous enhancement at an examination angle of + 15°-20° in relation to the orbitomeatal line. Cortical atrophy was assessed according to criteria determining the external and internal liquor spaces (after Meese and Groome).

Results: There is a consistent low-grade enlargement of the brain ventricles. The variables have increased values (decreased for CMI) in the schizophrenic patients compared with the controls. The patients show moderately increased width of the lateral sulcus and brain convexity sulci.

Conclusion: The brain tissue loss and enlarged extracerebral space suggest that the observed evidence of cortical loss in schizophrenic patients reflects a pathological process operating before completion of the brain growth.

Key words: *schizophrenia, computer tomography, brain atrophy, neurodevelopment*

INTRODUCTION

Despite the evident success in treatment, schizophrenia still remains a puzzle for medicine and challenge for the clinical and fundamental studies. Modern science provides new data of impairment in the brain structure and function of schizophrenic patients.[1] Local aberrations affecting morphology, metabolism, neurotransmitters and electrophysiology of brain have been established.[2] The pathological changes most frequently found include reduced brain mass, enlarged brain ventricles, cortical atrophy of the anterior hippocampus, parahippocampal and superior temporal gyrus of the left hemisphere, disturbed cortical cytoarchitectonics.[3] Schizophrenia is a product of morphological changes in brain. The primary vulnerability to schizophrenia occurs in the early prenatal stages of brain development. A relatively static agent that influences the processes of neurogenesis works within this limited period of time. The effects remain relatively latent long after the activity of the agent has subsided.[4]

Imaging studies of brain of schizophrenic patients have established enlarged lateral ventricles, considerable total loss of brain tissue, reduced brain grey matter, increased external liquor space, marked reduction of brain tissue in the frontal lobes, orbitofrontal and mesiofrontal cortex.[5,6]

The aim of the study was to make a comparative CT examination of schizophrenic patients and find lifetime criteria for recognition of brain changes in schizophrenia.

PATIENTS AND METHODS

Subjects

The patients for this study were 22 (11 men, 11 women) schizophrenic inpatients of mean age

Correspondence and reprint request to: V. Georgieva, Department of Radiology, Medical University Plovdiv 15A Vassil Aprilov St., 4002 Plovdiv, Bulgaria
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32.86 ± 2.65 years, who satisfied DSM-IV criteria for schizophrenia.[7] All patients were of Bulgarian ancestry. According to the form of schizophrenia the following subgroups were identified: paranoid, disorganized, catatonic, simple and non-differentiated form. Patients with diagnoses from the "schizophrenic spectrum" and history of organic psychotic disorder, severe neurological disorder, drug or alcohol abuse were excluded from the study.

The control group comprised 27 clinically healthy subjects (16 men, 11 women) of mean age 46.44 ± 2.32 years and Bulgarian ancestry. Subjects with a history of psychotic disorder, organic psychotic disorder, severe neurological disorder, drug or alcohol abuse were excluded from the study. To reduce the potential risk of schizophrenia development the control subjects selected were about the age of 40.

The study was approved by the local Ethics Committee and all subjects gave their written informed consent to participate.

CT examination

CT examination without venous enhancement at an angle of examination + 15°-20° in relation to the orbitomeatal line was performed. The CT images were assessed by a radiologist, who was blind to the diagnosis, the demographic and symptom status of the patients. Cortical atrophy was assessed according to criteria determining the external and internal liquor spaces (after Meese and Groome) (Figs 1,2).[8]

Criteria for determining the external liquor spaces

- Width of the brain convexity sulci.
- Width of the lateral sulcus.

Criteria for determining the internal liquor spaces

- Cella media index (CMI) – the ratio between the greatest bitemporal distance of the skull (D) and the greatest distance between the lateral ventricles at the level of cella media (E).
- Frontal horns index (FHI) – the ratio between the greatest external distance of the skull (A) and the greatest distance between the frontal horns (B).
- Huckmann digit (HZ) – the sum of the greatest distance between the frontal horns (B) and the distance between the frontal horns at the level of caudate nucleus (C).

Statistical analysis

The data were analysed with Mann-Whitney U test using SPSS 11.0. P values < 0.05 were considered statistically significant.

RESULTS

The data showed differences in the variables between patients and controls. There was a consistent low-grade enlargement of the brain ventricles (Figs 3, 4). The variables had increased values (decreased for CMI) in the schizophrenic patients compared with the controls (Table 1). The patients had moderately

Figure 1. Brain atrophy indexes (after Meese W. and Groome T.)

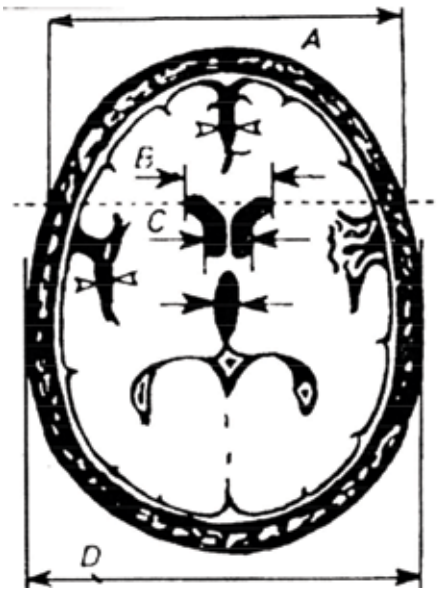

Figure 2. Brain atrophy indexes (after Meese W. and Groome T.)

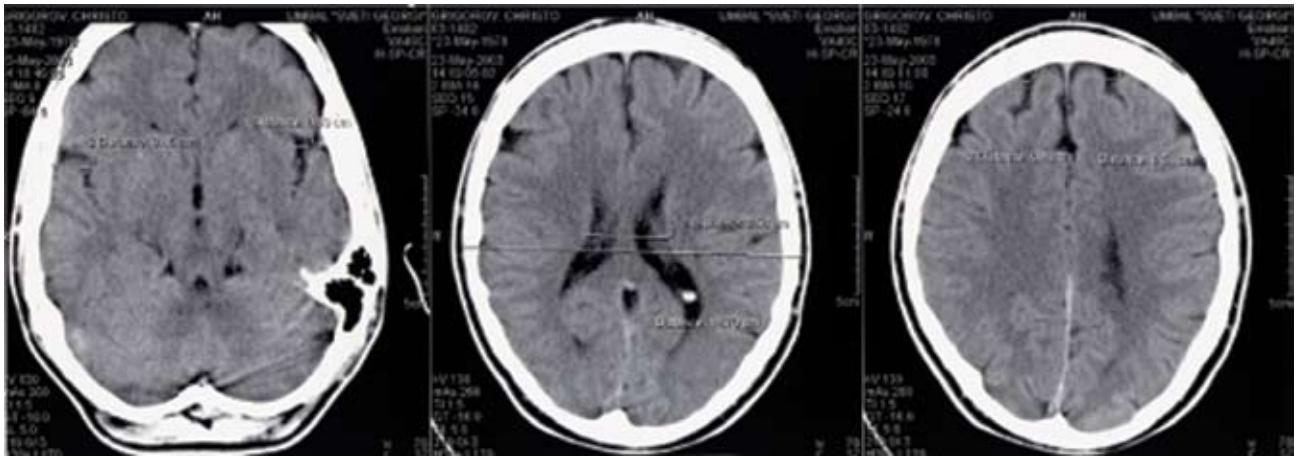

Figure 3. A 26-year-old man with one-year duration of disease. Moderately increased lateral ventricles, decreased anterior horn index.

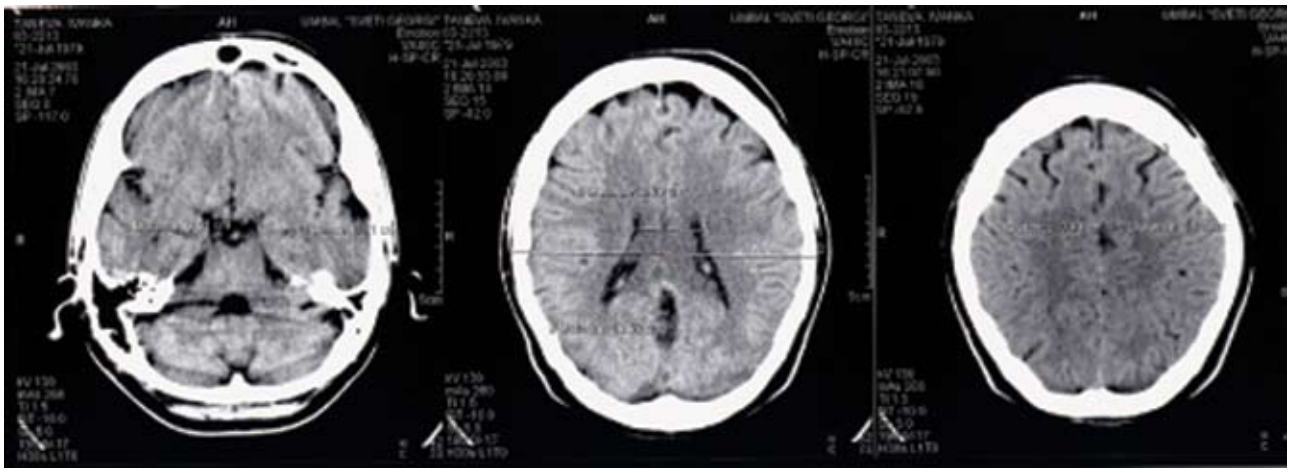

Figure 4. A 23-year-old woman with newly diagnosed schizophrenia. Enlarged lateral ventricles, increased Huckmann digit.

Table 1. External and internal liquor spaces in schizophrenic patients and healthy controls

Variable	Patients		Controls		Statistics
	$\bar{x}$	S_x	$\bar{x}$	S_x	P
Sulci width	4.55	0.25	4.20	0.24	> 0.05
Lateral sulcus width	5.05	0.36	4.69	0.21	> 0.05
Anterior horn index	3.27	0.13	4.08	0.20	< 0.05
Huckmann digit	5.10	0.13	4.57	0.14	< 0.05
Cella media index	4.46	0.08	4.54	0.08	> 0.05

increased width of the lateral sulcus (5.04 ± 0.34 vs. 4.69 ± 0.21, P > 0.05) and brain convexity sulci (4.55 ± 0.25 vs. 4.20 ± 0.21, P > 0.05). The cella media index was slightly decreased in the patients compared with the controls (4.46 ± 0.08 vs. 4.54 ± 0.08, P > 0.05). The frontal horn index showed mild to moderate decrease in the schizophrenic patients compared with that of the controls (3.27 ± 0.13 vs. 4.08 ± 0.20, P < 0.01). The Huckmann digit was slightly increased in the patients compared

with that in controls (5.10 ± 0.13 vs. 4.57 ± 0.14, $P < 0.05$).

DISCUSSION

Our findings show an overall loss of brain tissue or gray matter and increase in the extracerebral (sulcal) and ventricular space, in accordance with other imaging studies of schizophrenic patients using quantitative measurements.[9-11] The reduced brain-ventricle ratio is mainly at the expense of the internal liquor spaces. This could be due to the clearly identifiable ventricle images unlike the ill-defined variables on the brain surface (lateral sulcus and cerebral sulci). The presence of objective data for brain changes at the very onset of the disease suggests that the observed evidence of cortical loss in schizophrenic patients reflects a pathological process operating before completion of brain growth. The possible relevance of these findings of intracranial volume increase and brain volume loss to neurodevelopmental hypothesis of schizophrenia is based on some fundamental principles of normal neurodevelopment. The increase of the intracranial cavity is irreversible after fusion of the skull sutures. Hence, after brain growth reaches its maximum, intracranial size remains constant and any loss of brain volume leads to an equivalent increase in extracerebral and intraventricular CSF volume. It could be assumed that the generalized decrease in the brain volume can be attributed to a process that affects the whole cerebrum over a time period that goes beyond the early stages of brain development. Technical advance in the imaging examinations will probably provide more reliable data for the specific morphological changes. More reliable data for specific morphological alterations would be obtained from magnetic resonance imaging because of the higher image resolution. More strict selection criteria and the considerably high cost of the examination however limit the extensive use of magnetic resonance imaging for evaluation of great number of patients.

Although the interpretation of the results should be done carefully because of the small sample of schizophrenic patients, the present findings are consistent with the hypothesis that morphological alterations in brain are present in a subpopulation of schizophrenic patients, which correspond to specific CT and clinical signs.

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КОМПЬЮТЕРТОМОГРАФИЧЕСКИЕ ИЗМЕНЕНИЯ В МОЗГЕ БОЛЬНЫХ ШИЗОФРЕНИЕЙ

В. Георгиева, С. Сивков, И. Коева, Р. Ангелова, Т. Денева

РЕЗЮМЕ

Цель: *Целью работы является сравнительное компьютертомографическое исследование пациентов, больных шизофренией, а также и найти критерии (прижизненно) для объективизирования изменений в мозге при шизофрении.*

Пациенты и методы: *Обследовано 22 больных шизофренией (средний возраст 32.86 ± 2.65 г.), покрывающих критерии болезни по DSM-IV. Контрольная группа включала 27 клинически здоровых лиц болгарского происхождения (средний возраст 46.44 ± 2.32). Всем пациентам проведено CT- исследование без венозного усиливания при угле исследования +15º - 20º по отношению к орбитомеатальной равнине. Мозговая атрофия оценена по параметрам, определяющим наружные и внутренние ликворные пространства (по Meese и Groote).*

Результаты: *Данные показывают консистентно легкостепенное увеличение мозговых вентрикулов, при чем параметры имеют более высокие стоимости (уменьшены - cella media index) у больных шизофренией по отношению к лицам контрольной группы. У больных наблюдается умеренное расширение sulcus lateralis и борозды по конвексной поверхности мозга.*

Заключение: *Потеря мозговой ткани и повышение экстрацеребрального пространства предполагают, что наблюдаемые данные о потере коры у больных шизофренией отражают патологический процесс, действующий до завершения роста мозга.*

ASSOCIATION BETWEEN ANGIOGRAPHIC CORONARY STENOSIS MORPHOLOGY AND ACUTE CORONARY SYNDROME MANIFESTATION IN PATIENTS WITH ISCHEMIC HEART DISEASE

Ivan H. Manukov, Julia B. Djorgova, Atanas B. Djurdjev,[1] Maria P. Tokmakova[1], Lyudmila V. Kitova[1], Okan I. Aliman[1]

Clinic of Invasive Cardiology, St George's University Hospital, [1]Clinic of Cardiology, Medical University, Plovdiv

ABSTRACT

Aim: Patients with acute coronary syndrome (ACS) often show complex morphology of coronary stenosis at angiography. In the present study we evaluated the association between different clinical forms of manifestation of acute coronary syndrome and the angiographic morphological patterns of coronary stenosis.

Patients and methods: A total of 112 patients with angiographically verified single vessel coronary artery disease were divided into two groups: a control group of 44 patients with simple coronary stenosis at angiography and a study group of 66 patients with complex coronary stenosis. Angiographic analysis was performed using a modified Ambrose classification. The two groups were compared according to the manifestation and distribution of the acute coronary syndrome based on Braunwald classification.

Results: There were no statistically significant differences between the mean values of stenosis severity in the group with simple stenosis (79.8% ± 10.7%) and the group with complex stenosis (82.7% ± 8.2%) (P > 0.05). The incidence of current acute coronary syndrome – unstable angina or myocardial infarction – was higher in the group with complex stenosis (30.00% ± 8.37% vs. 52.00% ± 7.07%, P < 0.05). Patients with previous ACS were prevailing in the group with simple stenosis (70.00% ± 8.37% vs. 48.00% ± 7.07%, P < 0.05).

Conclusion: Complex coronary stenosis is associated with higher prevalence of acute coronary syndrome in acute clinical stage while simple coronary stenosis is associated with higher prevalence of previous acute coronary syndrome. A possible metamorphosis of coronary stenoses is taken into consideration.

Key words: *acute coronary syndrome, complex stenosis, Ambrose*

INTRODUCTION

The repeated angiographic diagnosis of stable stenocardia patients with acute myocardial infarction and a control group with stable stenocardia without clinical changes revealed the prevalence of myocardial infarctions on the basis of a previous mild stenosis - below 50%.[1] Muller et al.[2] defined them as "vulnerable" atherosclerotic plaques characterized by a large burden of extra cellular lipids and detritus in their core, a thin fibrous cap complicated with a rupture, intramural hematoma and a current or already dissolved intra-luminal thrombus.[3] Complex stenoses are the angiographic equivalent of this type of complicated plaques.[4] The aim of this study was to determine the association between coronary stenosis angiographic morphology and the manifestation of acute coronary syndrome in patients with heart ischemic disease.

MATERIALS AND METHODS

A total of 112 patients with angiographically verified single vessel coronary artery disease (using selective coronarography) were divided into two groups: a control group of 46 patients with simple coronary stenosis and a study group of 66 patients with complex coronary stenosis.

Coronary stenosis was defined as "simple" or "complex" using a modified Ambrose classification. Simple stenosis (of concentric or eccentric type) is characterized by smooth edges while complex

Correspondence and reprint request to: Iv. Manukov, Clinic of Invasive Cardiology, University Hospital "St. George", Medical University, Plovdiv
15A Vassil Aprilov St., 4002 Plovdiv, Bulgaria
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stenosis is characterized by irregular outlines, overhanging and sharply outlined edges, ectasia, asymmetric narrowing, multiple roughnesses and a saw-like profile suggesting ulceration. We used Braunwald classification of unstable angina pectoris when one of the following criteria was available:

- New onset angina, at least class III according to the Canadian Cardiovascular Society (CCS).
- Accelerated angina with increased severity to a higher CCS class, at least class III.
- Angina at rest lasting more than 20 minutes usually.

Variant angina pectoris was considered as a particular form.

Myocardial infarction with/without Q wave was diagnosed according to the definition of the European Society of Cardiology and the American College of Cardiology (ESC/ACC).[5]

Data were analysed using the analysis of variance, the alternative and non parametric analyses with the help of software based on the MS Windows 98 operation system. Differences were considered statistically significant and confirming the alternative hypothesis if greater than the critical value of *u* at a = 0.05.

RESULTS

The mean age of the patients was 54 ± 8.4 years (52.8 ± 8.4 years for the control group and 54.9 ± 9.1 years for the complex stenosis group), P > 0.05. In the complex stenisos group 57 of the patients were men (86.36% ± 4.22%) and 9 were women (13.64% ± 4.22%), while in the simple stenosis group 37 of the patients were men (80.43% ± 5.85%) and 9 were women (19.57% ± 5.85%). The sex distribution in the two groups was similar, P > 0.05.

Both groups were matched for anthropometric characteristics (Table 1).

There was no significant difference between the groups in the main risk factors for the development of ischemic heart disease (Table 2).

The mean values of stenosis severity in the group with simple coronary stenosis (79.8% ± 10.7%) and in the group with complex coronary stenosis (82.7% ± 8.2%) did not differ significantly at angiography (P > 0.05).

The within-group analysis showed that the relative quota of patients with current or previous acute coronary syndrome (ACS) was higher compared to the patients without ACS (u = 3.06, P < 0.01 for the control group while u = 6.92, P < 0.001 for the complex stenosis group) (Table 3). That proved to be an important clinical manifestation of patients with a single vessel disorder and a tendency of higher incidence was observed in the complex stenosis group but the difference failed to reach statistical significance (P > 0.05, χ^2 = 1.48).

Of the 80 patients with acute coronary syndrome 35 (43.75% ± 5.55%) had current (new onset) ACS at admission and 45 (56.25% ± 5.55%) had previous ACS more than a month before (9.6 ± 14.2 months at an average), u = 1.58.

There was a statistically significant difference between the two groups when compared on the indicator "previous or current ACS"; the relative quota of patients with previous ACS was greater in the simple stenosis group (70.00% ± 8.37% vs. 48.00% ± 7.07%, P < 0.05) (Table 3). Current ACS was prevalent in the group of patients with complex stenosis (30.00% ± 8.37% vs. 52.00% ± 7.07%, P < 0.05) and the difference between the

Table 1. Mean age and the basic anthropometric characteristics

	Simple stenosis		Complex stenosis			
Indicator	$\bar{x} \pm S_x$	$S_{\bar{x}}$	$\bar{x} \pm S_x$	$S_{\bar{x}}$	u	P
Age (years)	52.8 ± 8.4	1.24	54.9 ± 9.1	1.12	1.26	> 0.05
Height (meters)	1.70 ± 0.07	0.01	1.70 ± 0.08	0.01	0.00	> 0.05
Weight (kilograms)	78.3 ± 11.9	1.75	81.1 ± 9.7	1.19	1.32	> 0.05
Body surface (m^2)	1.88 ± 0.20	0.03	1.92 ± 0.15	0.02	1.15	> 0.05
BMI	27.0 ± 3.4	0.50	27.9 ± 2.6	0.32	1.51	> 0.05

BMI - body mass index

Table 2. Distribution of the general risk factors

Risk factors	Simple stenosis			Complex stenosis				
	N	%	Sp	N	%	Sp	u	P
AH	22	47.83	7.36%	42	63.64	5.92%	1.67	> 0.05
Diabetes mellitus	5	10.87	4.59%	3	4.55	2.55%	1.22	> 0.05
Smoking	22	47.83	7.36%	32	48.48	6.15%	0.07	> 0.05
Heredity	10	21.74	6.08%	8	12.12	4.01%	1.32	> 0.05
Higher levels of total cholesterol	8	17.39	5.59%	15	22.73	5.16%	0.70	> 0.05
Higher levels of triglycerides	14	30.43	6.78%	11	16.67	4.59%	1.67	> 0.05
Lower levels of HDL	2	4.35	2.99%	6	9.09	3.54%	1.04	> 0.05

Note: The sum of values exceeds 100% as some of the patients were risky for several factors.
AH - arterial hypertension, HDL - high density lipoproteins

two groups was statistically significant. At a level of significance above 95.0% the null hypothesis can be rejected and the alternative hypothesis H_1 can be accepted which suggested prevalence of the relative part of new onset ACS in single vessel patients with complex coronary stenosis. Vice versa – previous ACS prevailed in single vessel patients with simple coronary stenosis.

Within the control group with ACS, 5 of the patients had had the syndrome more than once (16.67% ± 6.80%). The cases in the group with complex stenosis were 20 (40.00% ± 6,93%). At u = 2.40 and P < 0.02 that difference was considered statistically significant, and the H_1 hypothesis was accepted for a higher percentage of repeated ACS in single vessel patients with complex coronary stenosis, at level of significance above 98.0%.

We had 72 patients with myocardial infarction (MI) in different stages (64.29% ± 4.53%) but there was no significant difference between the two groups concerning MI incidence (P > 0.05, χ^2 = 0.40) (Table 4).

Of the 72 patients with MI 50 patients had chronic MI (69.44% ± 5.43%), while 22 patients had acute MI (30.56 ± 5.435, u = 5.05). The number of patients with chronic myocardial infarction was significantly higher in the group with simple stenosis (82.14% ± 7.24% vs. 61.36% ± 7.34%, respectively P < 0.05), while the percentage of patients with acute myocardial infarction was significantly higher in the group with complex stenosis (17.86% ± 7.24% vs. 38.64% ± 7.34%,

Table 3. Distribution of the patients by ACS

	Simple stenosis			Complex stenosis					Total	
	N	%	Sp	N	%	Sp	u	P	N	%
ACS	30	65.22	7.02%	50	75.76	5.27%	1.21	> 0.05	80	71.43
Without ACS	16	34.78	7.02%	16	24.24	5.27%	1.21	> 0.05	32	28.57
Total	46	100		66	100				112	100
Previous ACS	21	70.00	8.37%	24	48.00	7.07%	2.01	< 0.05	45	56.25
Current ACS	9	30.00	8.37%	26	52.00	7.07%	2.01	< 0.05	35	43.75
Total	30	100.0		50	100.0				80	100

ACS – acute coronary syndrome

respectively, P < 0.05). At probability above 0.95 we accepted the H_1 hypothesis of a higher relative part of acute MI in single vessel patients with complex coronary stenosis and respectively, a higher relative part of chronic MI in patients with simple coronary stenosis.

Thirty seven of the MI patients had MI with Q wave (51.39 % ± 5.89%), while 35 had MI without Q wave (48.61% ± 5.89%) (u = 0.34).

We did not find any significant differences regarding the abnormal Q wave (u = 0.63, P > 0.1) and both groups showed identical distribution and were fully compatible (Table 5).

Thirty three of the 80 patients with ACS had unstable angina pectoris or variant angina (41.25% ± 5.50%). Only 2 of them had a history of more than one month (6.06% ± 4.15%) (variant angina from the complex stenosis group). The remaining 31 patients had an acute disorder (93.94% ± 4.15%). We found significant prevalence of UAP cases in the group with complex stenosis and statistically significant difference between the groups at u = 2.74 and P < 0.01 (Table 6). At a level of significance above 99.0% the alternative hypothesis H_1 might be accepted of a higher relative part of UAP in single vessel patients with complex coronary stenosis.

Table 7 shows the distribution of the patients by UAP type indicator in the two groups.

DISCUSSION

The complex coronary stenoses are an angiographic equivalent of complicated "vulnerable" plaques – atherosclerotic plaques with a large central core of extra-cellular lipids and cellular detritus and a thin fibrous cap susceptible to fissure formation and ulceration.[6] Small fissures in the plaques provoke intramural thrombus formation. Along with smooth muscle reparation and collagen accumulation this leads to thrombus enlargement. Because of broken local balance between thrombosis and spontaneous thrombolysis, tPA/ PAI-1 discrepancy and some tissue factors, larger ruptures of the unstable plaques may cause a combination of intramural thrombus above the site of rupture with coronary occlusion of different degrees.[7] Acute coronary syndrome is

Table 4. Distribution of patients by myocardial infarction and stage of disease

MI	Simple stenosis			Complex stenosis					Total	
	N	%	Sp	N	%	Sp	u	P	N	%
With MI	28	60.87	7.19%	44	66.67	5.80%	0.63	> 0.05	72	64.29
Without MI	18	39.13	7.19%	22	33.33	5.80%	0.63	>0.05	40	35.71
Total	46	100		66	100				112	100
Chronic stage	23	82.14	7.24%	27	61.36	7.34%	2.01	< 0.05	50	69.44
Acute stage	5	17.86	7.24%	17	38.64	7.34%	2.01	< 0.05	22	30.56
Total	28	100		44	100				72	100.0

MI – myocardial infarction

Table 5. Distribution of patients by type of myocardial infarction (with or without Q wave)

MI type	Simple stenosis			Complex stenosis					Total	
	N	%	Sp	N	%	Sp	u	P	N	%
With Q wave	14	50.0	9.45%	23	52.3	7.53%	0.19	> 0.05	37	51.39
Without Q wave	14	50.0	9.45%	21	47.7	7.53%	0.19	> 0.05	35	48.61
Total	28	100		44	100				72	100

the clinical manifestation of the process described above.[8-10]

CONCLUSIONS

Acute coronary syndromes – previous or new onset - represent an important clinical manifestation of the ischemic heart disease in single vessel coronary artery patients. They tend to prevail in the group with complex stenosis. ACS in acute clinical stage is more frequent in single vessel patients with complex coronary stenosis while previous ACS is more frequent in single vessel patients with simple coronary stenosis.

In single vessel patients with complex coronary stenosis the incidence of repeated ACS is higher.

Acute myocardial infarction (MI) has a higher prevalence in patients with complex coronary stenosis while chronic MI has a higher prevalence in patients with simple coronary stenosis.

Unstable angina pectoris (UAP) has a higher prevalence in single vessel patients with complex coronary stenosis.

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Table 6. Distribution of patients by unstable angina pectoris

UAP	Simple stenosis			Complex stenosis					Total	
	N	%	Sp	N	%	Sp	u	P	N	%
With UAP	7	23.33	7.72%	26	52.00	7.07%	2.74	< 0.01	33	41.25
Without UAP	23	76.77	7.72%	24	48.00	7.07%	2.74	< 0.01	47	58.75
Total	**30**	**100**		**50**	**100**				**80**	**100**

UAP - unstable angina pectoris

Table 7. Distribution of patients with unstable angina pectoris (Braunwald)

Type of angina	Simple stenosis	Complex stenosis	Total
	N	N	N
I	2	4	6
II	1	4	5
III	2	8	10
Variant Angina	2	10	12
Total	**7**	**26**	**33**

UAP - unstable angina pectoris

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СВЯЗЬ МЕЖДУ АНГИОГРАФИЧЕСКОЙ МОРФОЛОГИЕЙ КОРОНАРНОГО СТЕНОЗА И ПРОЯВЛЕНИЕМ ОСТРОГО КОРОНАРНОГО СИНДРОМА У БОЛЬНЫХ С ИШЕМИЧЕСКОЙ БОЛЕЗНЬЮ СЕРДЦА

И. Мануков, Ю. Джоргова, А. Джурджев, М. Токмакова, Л. Китова, О. Алиман

РЕЗЮМЕ

***Цель**: У большинства пациентов с ос-трым коронарным синдромом (ОКС) устанавливается наличие коронарного стеноза с комплексной ангиографической морфологией. В настоящей работе исследован тип кли-нического проявления ОКС у больных с различной ангиографической морфологией коронарного стеноза.*

***Пациенты и методы**: 112 больных с доказанной инвазивно коронарной болезнью обычного ангиографического типа разде-лены на две группы: контрольную (44 больных) – с коронарным стенозом обычного ангиографического типа и экспериментальную (66 пациентов) с комплексным коронарным стенозом. Для ангиографического разделения использована модифицированная классификация Ambrose. Обе группы сравниваются в за-висимости от проявления и распределения острого коронарного синдрома.*

***Результаты**: Средние стоимости тяжести стеноза в группе с обычным стенозом (79.8% ± 10.7%) и в группе с комплексным стенозом (82.7 ± 8.2%) не различаются статистически достоверно (P > 0.05). Процент случаев с настоящим ОКС – (нестабильная ангина или инфаркт миокарда) более высокий в группе с комплексным стенозом (30.00% ± 8.37% к 52.00% ± 7.07%; P < 0.05). Случаи с уже перенесенным ОКС преобладают в группе больных с обычным стенозом (70.00% ± 8.37% к 48.00% ± 7.07%; P < 0.05).*

***Выводы**: Коронарные стенозы с комплекс-ной морфологией связаны с превалированием острых коронарных синдромов в острой клинической фазе, в то время как при обычном коронарном стенозе преобладают случаи с перенесенным острым коронарным синдромом. Во внимание принимается возможная мета-морфоза коронарных стенозов.*

DIRECT RUPTURE OF SPLENIC HYDATID CYST WITHOUT VOLUME LOSS: REPORT OF A UNIQUE CASE

Dimitrios Passomenos, Dimitrios Filippou[1], Paraskevi Makrodimitri, Efthalia Karzi, Spiros Rizos, Panagiotis Skandalakis[2], Constantinos Simopoulos[3]

Department of Computed Tomography, [1]Department of Surgery, Tzaneion Hospital, Piraeus, Greece; [2]II Propaedeutic Department of Surgery, Medical School, University of Athens, Laiko Hospital, Athens, Greece; [3]Department of Surgery, II Department of Surgery, Medical School, University of Thrace, Alexandroupolis, Greece

ABSTRACT

We present a rare case of primary spleen Echinococcus cyst with rupture evidence due to trauma. The cyst presented with an echogenic pattern at ultrasound examination but subsequent CT study revealed calcification of the cyst wall and signs of rupture. Serological tests were positive for echinococcus and eosinophilia defined. Volume changes of hydatid cyst were not discernible in the imaging studies despite evidence of rupture. Both imaging methods may suggest the diagnosis unless typical diagnostic features are present. Hydatid disease was confirmed histologically on the specimen after surgical excision of the cyst.

Key words: *hydatid cyst, spleen, computed tomography*

INTRODUCTION

Hydatid disease of spleen is quite a rare entity. Reported incidence is less than 2% of all hydatid disease cases. In all reported cases rupture of a splenic hydatid disease cyst is followed by reduction of the cyst volume.

CASE REPORT

A 45-year-old male farmer from a rural area was admitted to our hospital after having sustained a car accident. The patient complained of aggravated left upper quadrant pain and had difficulty in walking and standing on his left foot. He had been an otherwise healthy individual but the last 3 years he had been complaining of discomfort and fullness of abdomen unrelated to physical activities. He also reported that his brother had had a history of liver echinococcosis 7 years before. At admission the patient presented with mild pyrexia.

The chest and spine radiographs did not reveal any pathology while pelvic plain radiograph revealed a left hip fracture. Blood test analysis was normal with mild anemia and subsequent leucocytosis with eosinophil prevalence. Ultrasound study revealed an echogenic mass with no air-fluid levels at the anatomical site of the spleen. The diameter of the mass was about 8 cm. A perisplenic fluid collection was also diagnosed on ultrasound examination. A CT study after IV contrast administration revealed a large cystic lesion with a partially calcified wall in an enlarged spleen. Some more cystic lesions were also diagnosed in the left subdiaphragmatic space (Fig. 1). No contrast enhancement of the cyst was seen. The measuring attenuation values of the cyst content were higher than those of water, while its lateral wall side was discontinuous allowing fluid to spread (Fig. 2). The patient's history and the radiological findings suggested that the lesion may be a hydatid cyst. Further on, serologic tests, such as enzyme-linked immunoassay (ELISA), proved positive for echinococcus. Chemoprophylaxis with albendazole for 2 months was administered to the patient who finally received. During the operation the fluid of the cyst was aspirated and all prophylactic measures were taken in order to avoid dissemination of the disease and contamination of the abdomen. In the aspirated cyst material some scolices and broken hooklets, typical of echinococcus granulosus, were detected. The left hip fracture has been treated conservatively. Five months after the initial surgery the patient had no symptoms and the CT study suggested no disease although high echinococcal titer was persistently found. No recurrence of disease was detected nor any other site of involvement in the peritoneal cavity was found.

Correspondence and reprint request to: D. Passomenos, Department of Computed Tomography, Tzaneion Hospital 18536 Piraeus, Greece

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Figure 1. CT axial image at the level of diaphragmatic domes. Multiple smaller sized, lobulated fluid collections at left subdiaphragmatic area are present.

Figure 2. CT axial image following IV contrast administration. A large cyst in spleen can be seen with partial wall calcification and discontinuous lateral wall contour. Lobulated fluid around the spleen space and a smaller fluid area on the posterior surface of spleen are also seen.

DISCUSSION

Hydatid disease (HD) is a parasitic infestation caused by the larva of echinococcus, more frequently encountered in endemic areas such as Mediterranean countries, South America, Africa and Middle East.[1,2] The disease primarily affects the liver via the portal circulation (70% of cases) with typical imaging findings and the lungs (30%). Other less common sites of the disease such as the bones, muscles or vessels have also been reported.[3-5]

An enlarged liver hydatid cyst may rupture into the peritoneal cavity which may cause spillage in the abdomen and anaphylactic reactions. Hence, involvement of any peritoneal organ may be secondary in most cases. Primary involvement of spleen is a rather rare event (4%).

The clinical symptoms vary and may be non-characteristic such as abdominal pain or discomfort, spleen enlargement or fever.

A hydatid cyst usually has a smooth, sharply delineated wall and hypo-attenuated fluid content (unless infected); it is initially unilocular. With time, cyst enlarges and becomes multilocular. Multiple daughter cysts contained in a larger mother cyst show a characteristic configuration. The contained fluid may be clear or subpale with attenuation values similar to those of water (3-30 HU). Presence of sand, intracystic debris or inflammatory material may increase measured fluid density.[6] Wall calcification (arcuate or ring-like) is seen in about 20-30% of cases in plain films, reflecting pericyst calcification. Cyst wall displays an outer (pericyst) and an inner (endocyst) layer. Cyst rupture may be 'direct' or 'contained', with both endocyst and pericyst ruptured or only endocyst rupture occurring, respectively.[7]

Rupture into the peritoneal space may induce signs of irritation and subsequent hypersensitivity reactions, while in cases of contained rupture (regardless of pericyst-endocyst congruity) spillage of scolices does not occur. Volume loss suggests rupture of cyst occurs most often into the peritoneal space[8] but - to our opinion - this is not a common feature in every case of cyst rupture. In this case, although there was discontinuation of cyst wall (rupture), volume loss of cyst was not discernible on imaging studies. As far as we are concerned, this is the first case of spleen echinococcosis direct rupture with no evidence of volume loss, contrasting previously reported cases in medical literature. To our opinion, cyst location is related with the free spillage and therefore the volume loss. The clinical symptoms and signs in such a case are not efficient to establish the diagnosis. Imaging features combined with the blood test and the appropriate patient's history may set the appropriate diagnosis. Diagnostic features may be presence of daughter cysts and wall calcification (if any).

In conclusion, the possibility of HD should always be kept in mind when referring an individual with a calcified cyst of spleen, especially from an endemic area regardless of the presenting symptomatology.

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Dental Investigations

PREVALENCE OF DENTAL FLUOROSIS AMONG 4-TO 14-YEAR-OLD CHILDREN FROM THE TOWN OF DIMITROVGRAD (BULGARIA)

Maria P. Kukleva, Alexandra V. Isheva, Veselina K. Kondeva, Mariana M. Dimitrova, Svetla G. Petrova

Department of Pediatric Dentistry, Faculty of Dental Medicine, Medical University, Plovdiv

ABSTRACT

Introduction: There has been no study on the prevalence of dental fluorosis in Bulgaria of today where people have free access to some fluoride-containing products.

Aim: The aim of the study was to investigate the prevalence of dental fluorosis among children 4 to 14 years old from the town of Dimitrovgrad, where due to unsatisfactory qualities of tap water people consume bottled water including such with fluoride levels higher than 1.5 mg/l.

Patients and methods: The study included 1504 randomly selected children. We analysed subjects with dental fluorosis according to Dean's modified criteria. The following severity levels were defined: 0 – normal; 0.5 – suspicious; 1 – very mild; 2 – mild; 3 – moderate; 4 – severe. Data were analyzed separately for the different types of dentitions.

Results: Results showed that 54.52% of all children included in the study had dental fluorosis in different degrees. Primary teeth were affected by dental fluorosis less frequently than permanent teeth (P < 0.001). In mixed dentition cases 41.41% of the children had fluorosis of permanent teeth only, 1.64% had dental fluorosis of primary teeth only and 12.50% had both their primary and permanent teeth affected. The proportion of individuals with the lowest degree of severity – 0.5, was the greatest both for the primary and permanent teeth. Comparison with the proportions of children with more severe degrees of fluorosis revealed significant differences (P < 0.001).

Conclusion: The results of the study showed excessive fluoride intake during tooth development and suggested a need for further research of risk factors.

Key words: *dental fluorosis, prevalence, Dean's criteria*

INTRODUCTION

By the middle eighties of last century the prevalence of dental caries among children in North America had decreased significantly, which was attributed mainly to fluoride prophylaxis.[1,2] A similar trend was observed in the developed European countries.[3] Soon afterwards the prevalence of dental fluorosis was reported to be increased in a number of studies. It was suggested that improper fluoride intake during the first years of life, swallowing fluoride toothpastes, eating food and drinking water with high fluoride content, the use of fluoride-containing dental products are possible risk factors for development of dental fluorosis.[4-7] As a result of these studies new recommendations for fluoride prophylaxis of dental caries were put forward[8-10], aimed mainly at reducing the additional fluoride intake. In recent years, studies reporting presence of dental fluorosis in various countries around the world have increased in number.[11-17] In Bulgaria, the fluoride prophylaxis of dental caries is implemented through fluoride tablets, drops, fluoridated milk, and local application of fluoride toothpastes and fluoride-containing dental products (varnishes, gels, solutions, and sealants). In addition, bottled water with fluoride concentration greater than the optimal is commercially available. According to data reported by the Union of Producers of Non-alcoholic Drinks the consumption of non-alcoholic drinks in Bulgaria in 1990 amounted to 8.1 liters per person, and in 2001 – to 30.6 liters.[18] A steady increasing tendency in the consumption of such

Correspondence and reprint request to: M. Kukleva, Department of Pediatric Dentistry, Faculty of Dental Medicine, Medical University, Plovdiv
15A Vassil Aprilov St., 4002 Plovdiv, Bulgaria
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drinks has been observed, which is due largely to the bottled water intake including water with fluoride concentrations higher than 1.5 mg/l. The ignorance of this fact by dentists and the parents' insufficient information about the toxic effects of fluoride on dental tissues can lead to the extremely high risk of overdose as a result of consumption of such mineral water and the application of other fluoride-containing prophylactic products. Most dentists in Bulgaria have never encountered dental fluorosis in their practice and are little prepared to diagnose and treat it. The predominant number of cases of dental fluorosis comes from the towns of Dimitrovgrad, Haskovo, and Harmanli. Case histories revealed that people there were used to consuming types of mineral water, including such with fluoride levels higher than 1.5 mg/l, due to the inadequate quality of tap water.

To date, there has been no study in Bulgaria on dental fluorosis, as there has been no assessment conducted of the risk factors arising from the currently conducted fluoride prophylaxis and the free access of the population to fluoridated products. Therefore, we set up a research project to study the prevalence, risk factors and treatment of dental fluorosis, and the present study is a part of this larger project.

AIM

To investigate the prevalence of dental fluorosis among children aged 4 to 14 years from the town of Dimitrovgrad.

PATIENTS AND METHODS

The investigation included 1504 randomly selected children aged 4 to 14 years. Children who had not lived in Dimitrovgrad until they became 7 years of age were excluded from the study. The involvement of permanent teeth by dental fluorosis was evaluated in 1158 children with permanent or mixed dentition aged 7 to 14 years, while the involvement of primary teeth was assessed in 1074 children with primary or mixed dentition aged 4 to 14 years. A subject with dental fluorosis represented the unit of analysis. Dental fluorosis was diagnosed using Dean's modified criteria.[19] The following severity levels were defined: 0 – normal; 0.5 – suspicious; 1 – very mild; 2 – mild; 3 – moderate; 4 – severe.

The "one person, one disease" approach common in medical epidemiology was utilized in evaluating dental fluorosis. The two worst affected teeth were used to rate the degree of dental fluorosis severity in the individual. If the degrees of involvement of the two teeth were different, the lower score was taken. The proportions of children with dental fluorosis were calculated for the whole group, as well as for the different degrees of severity and the different age groups. The prevalence of dental fluorosis in children with mixed dentitions was estimated for primary teeth only, for permanent teeth only and for both types of teeth together. Descriptive statistics and t-criterion were used for data analysis. A p value less than 0.05 was considered statistically significant.

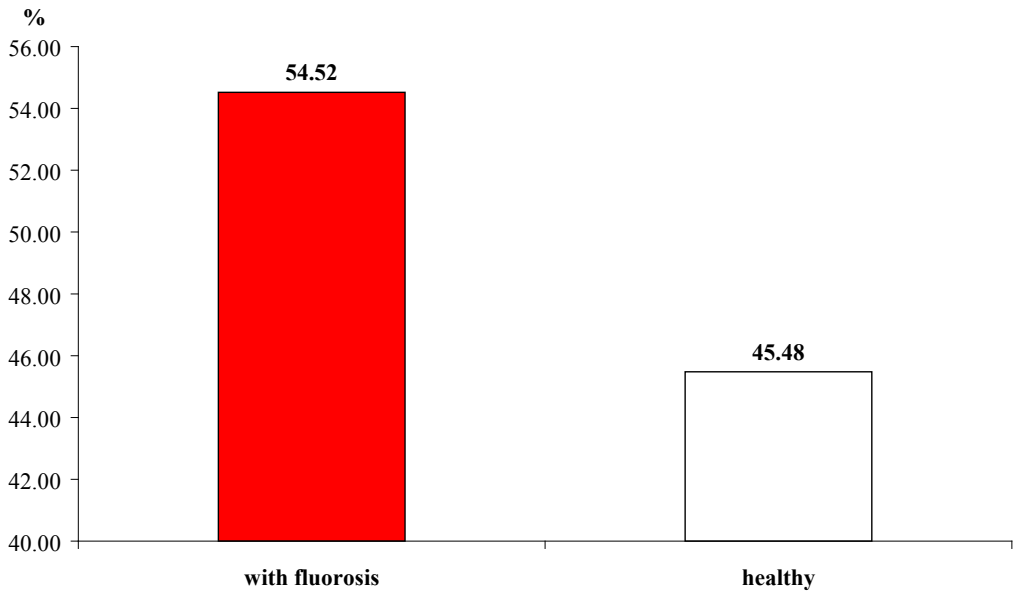

Figure 1. Prevalence of dental fluorosis among children aged 4 to 14 years from the town of Dimitrovgrad.

RESULTS

The results of the study showed that 54.52% of the examined children had dental fluorosis (Fig. 1).

The lowest prevalence was found at the age of 4 (40.52%), and 5 years (40.57%). Children at this age had primary dentition. The prevalence of dental fluorosis was higher in the remaining age groups of children with mixed and permanent dentitions. The comparison of prevalences among 4-year-old children with the other age groups revealed statistically significant differences, except for 6-,

Table 1. Prevalence of dental fluorosis among children from the town of Dimitrovgrad according to age groups

Age	Total number of examined children	Children with dental fluorosis			t	Pt
		Number	%	Sp		
4	116	47	40.52	4.56	_	_
5	106	43	40.57	4.78	0.01	> 0.05
6	124	57	45.97	5.94	0.73	> 0.05
7	143	88	61.54	4.07	3.44	< 0.001
8	149	92	61.74	3.98	3.51	< 0.001
9	155	77	49.68	4.02	1.51	> 0.05
10	156	81	51.92	4.00	1.88	> 0.05
11	127	80	61.99	4.31	3.42	< 0.001
12	137	89	64.96	4.08	3.99	< 0.001
13	147	90	61.22	4.02	3.40	< 0.001
14	144	76	52.78	4.16	1.99	< 0.05
Total	**1504**	**820**	**54.52**	**1.28**	_	_

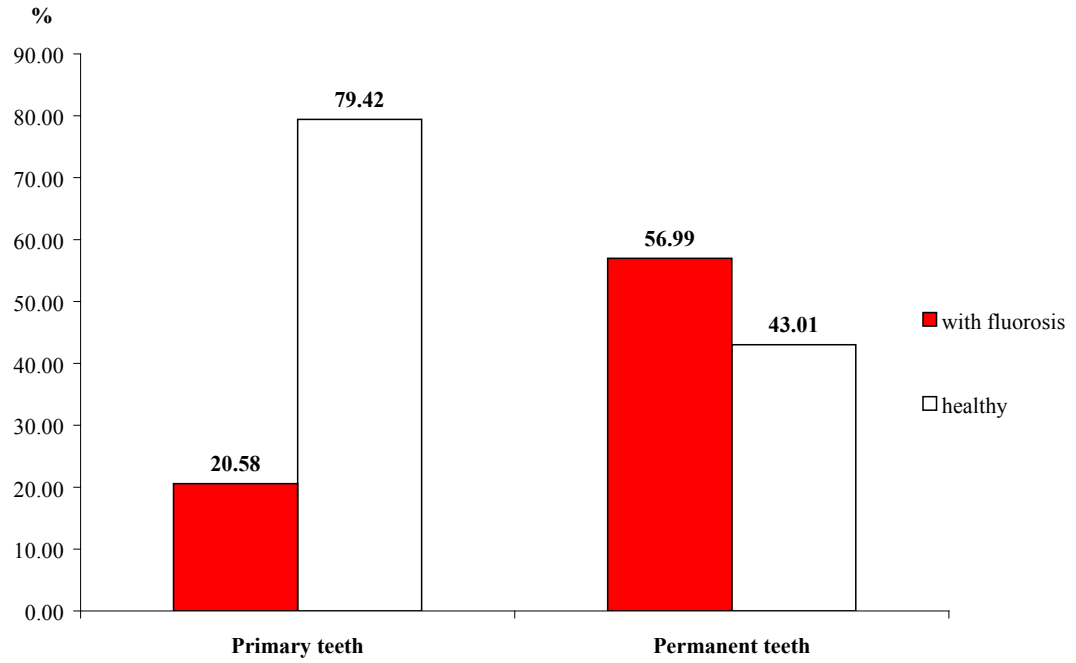

Figure 2. Prevalence of dental fluorosis among children from 4 to 14 years of age from the town of Dimitrovgrad according to the type of dentition.

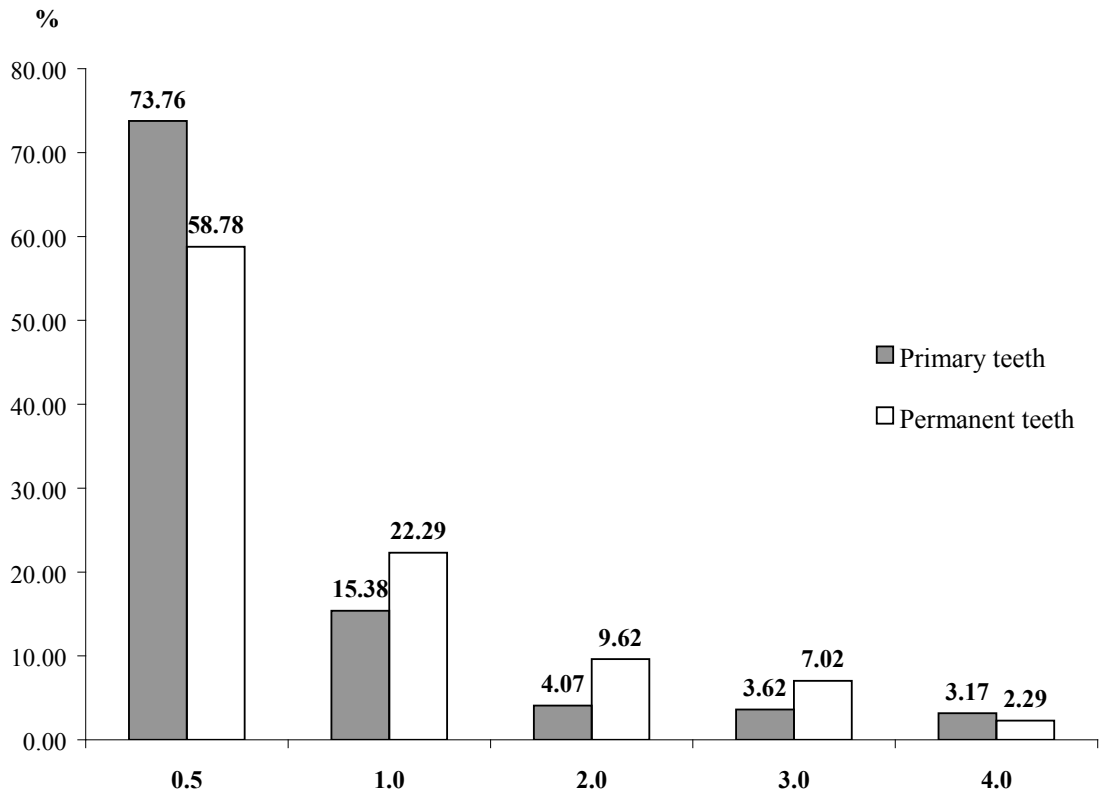

Figure 3. Distribution by severity of dental fluorosis among children from the town of Dimitrovgrad.

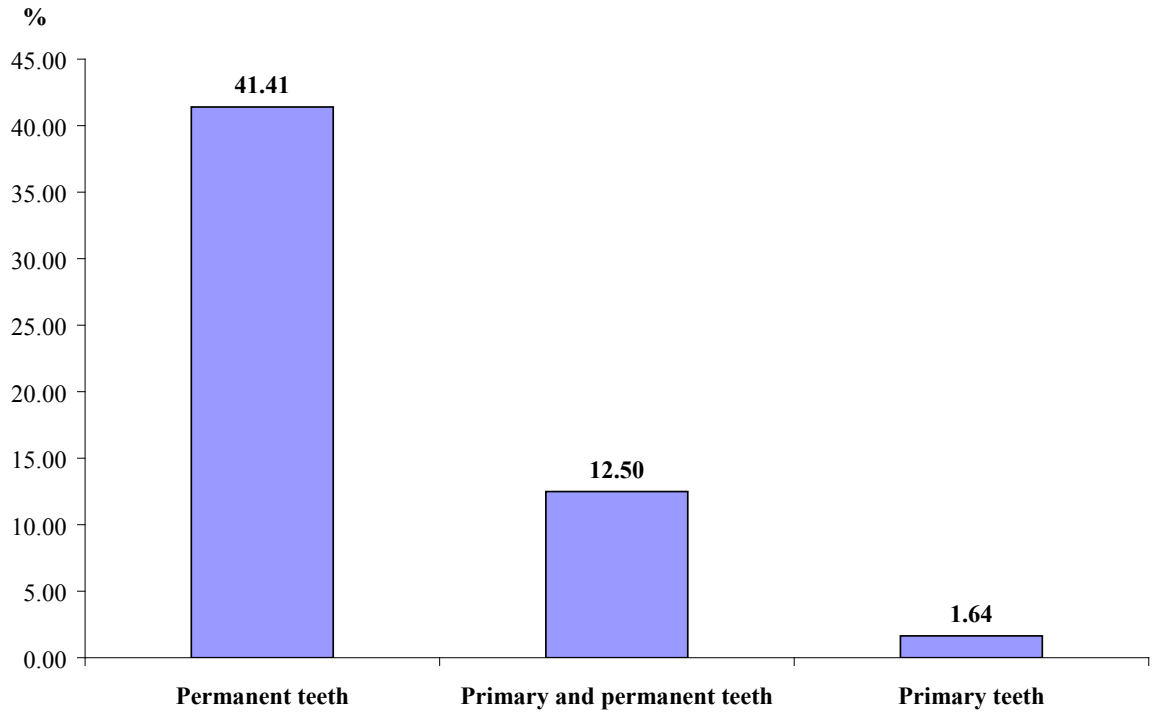

Figure 4. Prevalence of dental fluorosis among children with mixed dentition from the town of Dimitrovgrad.

9- and 10-year-old groups (Table 1). Our results demonstrated that primary teeth were less affected by dental fluorosis (Fig. 2). The prevalence of dental fluorosis among children with primary dentition was significantly lower than among children with permanent teeth ($P < 0.0001$). This result should

be taken into account while investigating the risk factors for development of dental fluorosis in the studied populations.

The distribution by severity of dental fluorosis showed that children with the lowest degree of severity – 0.5, comprised the largest group (Fig. 3). The number of affected persons decreased with the increase of severity. The comparison of the proportion of children with the lowest degree of dental fluorosis – 0.5, with the proportions of children with more severe involvement – 1, 2, 3, 4, showed significant differences for both types of dentitions ($P < 0.001$). Some authors reject the diagnosis of the least severe degree when the prevalence of dental fluorosis is low. In our opinion, differential diagnostic errors were minimized in this study considering the high prevalence of dental fluorosis. Hence, the diagnosis of the lowest degree of severity 0.5 gave a better idea of the chronic toxic effect of fluoride in the studied population.

Among children with mixed dentition, 1.64% had fluorosis of primary teeth only, 41.41% demonstrated fluorosis of permanent teeth only and 12.5% had fluorosis of both types of teeth (Fig. 4). The differences between the proportion of children with fluorosis of permanent teeth only on one hand and the proportions of children with fluorosis of primary teeth only or of both types of teeth, on the other hand, were statistically significant ($P < 0.001$). These results confirmed again less involvement of primary compared to permanent teeth.

DISCUSSION

At present different systems for classification of dental fluorosis are used. The choice of an index to be employed depends in the first place on the aim of the investigation. We chose Dean's index because it had been designed specifically for epidemiologic studies. The individual represents the unit of measurement which permits an evaluation of the prevalence in the studied population.

A brief literature review showed that most epidemiologic studies utilizing Dean's index did not distinguish between involvement of primary and permanent teeth. Our research group believed that a differential approach to the analysis of the results was important in order to identify risk factors and the duration of their influence. Our findings in children with primary teeth aged 4 and 5 years, as well as in individuals with mixed dentitions convinced us of the appropriateness of our approach.

The chronic toxic effect of fluoride on primary and permanent tooth germs is exerted during different time periods which do not coincide entirely. The damage to primary tooth germs starts during the second half of pregnancy due to higher than optimal daily intake of fluoride by the mother. In permanent teeth germs, injuries result from overdoses of fluoride during the first five years after birth. Different groups of permanent teeth or of the dentition as a whole can be affected depending on the duration of intake. The toxic effects may vary in intensity and the clinical manifestations of dental fluorosis may vary in severity in the different groups of teeth depending on the quantity of fluorides taken. These facts have been known for a long time and have been well investigated. They are taken into consideration by Dean's index where the severity evaluation is based on the two worst affected teeth.

In the present study we also found children with different degrees of severity of dental fluorosis in the different groups of permanent teeth. There were children in whom the first permanent molars and the frontal teeth were not affected, while the premolars and second molars demonstrated dental fluorosis. In other children only the anterior teeth and first molars were involved. These results proved that the daily intake of fluoride by these children was not constant and varied during the period of formation of the different groups of teeth. The presence of dental fluorosis in the primary dentition attested to excessive fluoride intake during pregnancy. The lesser relative share of children with fluorosis of primary teeth only and the larger relative share of children with fluorosis of permanent teeth only, as found in the group with mixed dentition, supported the view about the maternal mechanisms of protection of the fetus. The presence of children with fluorosis of both types of teeth confirmed the prolonged influence of incorrect fluoride intake as a risk factor.

The prevalence of dental fluorosis shows wide variations which depend on both the types of risk factors and the duration of their influence. Analysis of 55 publications in Medline from 1980 till 2000 showed that the prevalence of dental fluorosis ranged from 16.7% to 32.2% depending on the fluoride content of drinking water. Moreover, an increase was found compared to data from the 1940s. Juxtaposing our results with those studies showed high prevalence of dental fluorosis among

the children from the town of Dimitrovgrad. The results clearly indicate a need of in-depth investigation of the risk factors and development of a program to eliminate their influence.

CONCLUSIONS

The prevalence of dental fluorosis in the studied population group was high – 54.52%. It proved the excessive use of fluoride during the period of tooth development. Permanent teeth were affected more extensively, the difference being statistically significant ($P < 0.001$). The lower severity degrees of dental fluorosis predominated and this held true for both types of dentitions. The high prevalence of dental fluorosis among the children aged 4 to 14 years from the town of Dimitrovgrad imparts social significance to the problem and makes further studies necessary to clarify the risk factors.

Our investigation showed that the contemporary way of life allowing people to have free access to fluoride-containing products, carries a risk of excessive fluoride ingestion. Dentists should take this fact into consideration when they develop individual prophylactic programs prescribing additional fluoride intake.

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ЭПИДЕМИЧНОСТЬ ЗУБНОГО ФЛЮОРОЗА У ДЕТЕЙ В ВОЗРАСТЕ 4-14 ЛЕТ (ГОРОД ДИМИТРОВГРАД, БОЛГАРИЯ)

М. Куклева, А. Ишева, В. Кондева, М. Димитрова, С. Петрова

РЕЗЮМЕ

Введение: *В Болгарии отсутствуют данные об эпидемичности зубного флюороза в совре-менных условиях жизни, при которых у людей имеется свободный доступ к некоторым флюорсодержащим средствам.*

Цель: *Работа ставит себе целью исследовать эпидемичность зубного флюороза у детей в возрасте от 4 до 14 лет из Димитровграда, где из-за неудовлетворительных качеств питьевой воды население использует бутылочную воду, в том числе воду, содержащую флюор (более чем 1.5 мг/л).*

Пациенты и методы: *Обследовали 1504 случайно подобранных детей. В качестве единицы наблюдения определено лицо с зубным флюорозом согласно модифицированным критериям Dean. Определяются следующие степени тяжести: 0 – норма, 0.5 – сомнительная, 1.0 – очень легкая, 2.0 – легкая, 3.0 – умеренная, 4.0 – тяжелая. Анализ данных делается дифференцированно в зависимости от прикуса.*

Результаты: *Результаты показывают, что у 54.52% всех обследованных детей наблюдаются различные степени зубного флюороза. Временные зубы затрагиваются флюорозом в меньшей степени по сравнению с постоянными ($p < 0.001$). У детей со смешанным прикусом в 41.41% случаев налицо зубной флюороз только на постоянных зубах. У 1.64% детей имеется зубной флюороз только на временных зубах. У 12.50% детей обнаруживается зубной флюороз, затрагивающий и временные, и постоянные зубы. Самый высокий это процент лиц с самой низкой тяжестью – 0.5, и это относится как к временным, так и к постоянным зубам. Сопоставление с более высокими степенями тяжести по-казывает статистическую значимость разниц ($p < 0.001$).*

Заключение: *Результаты исследования доказы-вают повышенную дозировку флюора во время развития зубов и необходимость в дополнительных исследованиях с целью установления факторов риска.*

EFFECT OF NON-SURGICAL PERIODONTAL THERAPY ON PATIENTS WITH TYPE 2 DIABETES MELLITUS

Fu Yun, Elena I. Firkova[1], Lin Jun-Qi[2], Hong Xun[3]

Department of Periodontal and Oral Diseases, Guanghua College of Stomatology, Sun Yat-sen University, Guangzhou, China, [1]Department of Periodontology and Oral Diseases, Faculty of Dentistry, Medical University - Plovdiv, Bulgaria, [2]Endondontic Department, Guanghua College of Stomatology, [3]Center of Experimental Animal Models, Pharmaceutical College, Sun Ya Sen University, Guangzhou, China

ABSTRACT

The AIM of this study was to evaluate the effect of non-surgical therapy on clinical variables and glycemic control on type 2 diabetics with chronic periodontitis.

PATIENTS AND METHODS: Forty six type 2 diabetics with chronic periodontitis were randomized into two groups (group A and group B). Treatment included scaling and root planning for group A plus systematic use of doxycycline in both groups. Assessment was made prior to and 16 weeks following the therapy.

RESULTS: Analysis of data showed that both groups had clinical and glycated hemoglobin (HbA1c) improvement after the treatment. Group A had a statistically significant reduction of plaque index and bleeding on probing scores compared with controls (P<0.05) at 16 weeks.

CONCLUSION: These results suggest that non-surgical therapy is of value in maintaining periodontal health and may be beneficial in reducing blood glucose level in type 2 diabetics with chronic periodontitis.

Key words: *chronic periodontitis, type 2 diabetes, debridement, glycated hemoglobin A1c.*

INTRODUCTION

Diabetes mellitus (DM) is one of the most frequent metabolic disorders with an estimated prevalence of at least 15% in the adult world population.[1] It is associated with a wide range of complications, such as retinopathy, neuropathy, nephropathy, micro- and macro-vascular disease and altered wound healing.[2] Periodontitis is the most prevalent dental disease in diabetics. It was estimated that mild and moderate to severe periodontitis affects 22% and 13% of adults, respectively.[3,4] Epidemiological studies have shown diabetes to be a risk factor for periodontitis and periodontitis has been found to be "the sixth complication of DM".[5,6] Both hyper- and hypoglycemia may directly impair the wound healing ability of periodontal connective tissues by affecting the cellular homeostatic system and phagocytotic activity. Periodontitis is usually more frequent and severe in uncontrolled diabetic individuals than individuals without diabetes or individuals with well controlled diabetes.[9-11] Moreover, DM and periodontitis seem to interact in a bidirectional manner.[13] At present, there is strong evidence to suggest that the incidence and severity of periodontitis is influenced in part by DM[14] and periodontal therapy exert a beneficial effect on the level of blood glucose control[15].

DM encompasses a heterogeneous group of disorders with altered or impaired lipid and carbohydrate metabolism.[15] Hyperglycemia progressively glycates body proteins, forming advanced glycated end products (AGE). This in turn may stimulate phagocytes to release inflammatory cytokines such as tumor necrosis factor-α (TNF-α) and interleukin-6 (IL-6) that play a central role in diabetic complications and impair the normal formation of extracellular matrix components.[16] These alterations have adverse effect on periodontal tissues, especially on the collagen stability and the vascular integrity, enhancing the susceptibility to tissue destruction. It has been proposed that AGE mediating responses to periodontal infections may be synergistic in producing more severe periodontitis in subjects with diabetes.[13]

Type 2 DM usually becomes clinically appar-

Correspondence and reprint request to: Fu Yun, Department of Periodontal and Oral Diseases, Guanghua College of Stomatology, Sun Yat-sen Univesity, Guangzhou, China

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ent after the age of 40 years. There is increased production of glucose by the liver in the fasting stable, altered cell receptors for insulin as well as resistance to insulin's action in target (liver, muscle, fat) tissue. Moreover, in the initial phases of the disease, there is an increased secretion of insulin by the pancreas.

Most studies showed that non-surgical periodontal therapy such as scaling and root planning in type 2 diabetics was associated with a small reduction in glycated hemoglobin levels, which in turn improved the metabolic control of diabetes. As a result it has been suggested that effective control of periodontal infection in diabetics may reduce the level of advanced glycation in serum to normal levels.[19]

There have been numerous studies that have found a positive correlation between poor glycemic control in DM patients and advanced periodontitis. Conversely, limited data is available that may suggest that periodontitis affects glycemic control in individuals with type 2 DM.

The purpose of this study was to compare the changes in the clinical variables and glycemic level of type 2 DM patients after undergoing non-surgical periodontal therapy with a control group with type 2 DM not receiving periodontal treatment.

PATIENTS AND METHODS

Subjects were chosen with newly diagnosed type 2 DM who responded to the physician's referral to examine their periodontal condition in the periodontal department of Guanghua College of Stomatology, Sun Yat-sen University, China. The inclusion criteria were: (1) no history of another major illness and no antibiotics or other medications received for at least three previous months; (2) at least 14 standing teeth, pocket probing depth was ≥ 5 mm, but < 8 mm in at least one site in 4 teeth in at least 2 different quadrants; (3) bleeding and/or suppuration on probing; (4) no periodontal treatment for 6 months prior to baseline examination; (5) female subjects should not be pregnant or breast-feeding. Patient fulfilling the inclusion criteria were informed about the study and they signed the informed consent.

Forty six patients with chronic periodontitis were recruited. They were randomly divided into two groups: group A: non-surgical periodontal treatment; group B: temporarily without periodontal treatment.

Periodontal assessment was carried out of each standing tooth on four sites. The following parameters were evaluated: probing pocket depth (PPD), clinical attachment level (CAL), plaque index (PI), and bleeding on probing (BOP). Glucose control was determined by HbA1c, and measured by high pressure liquid chromatography.

Subjects were treated in a schedule of five 60-minute appointments on a weekly basis. At the first appointment, intensive oral health instruction and supragingival scaling and polishing were performed. In the following up appointment, oral hygiene was monitored and reinforced. Subgingival scaling and root planning was carried out under topical anesthesia on a quadrant-by-quadrant basis.

Subjects in group A were reassessed 8 weeks after periodontal treatment. Group B acted as a non-treatment control group for the duration of the study, and then they were treated in exactly the same manner as described for group A.

Each subject was prescribed doxycycline in a dose of 100 mg/day for 14 days.

STATISTICAL ANALYSIS

Statistical analysis was performed using one way analysis of variance (ANOVA) for dental variables and HbA1c values. A value of $P < 0.05$ was considered statistically significant. The results are expressed as mean and standard deviations.

RESULTS

Tables 1 and 2 present the basic demographics of the patients in the two groups at baseline. Table 3 shows the mean values of the periodontal parameters at baseline and 16 weeks after the periodontal treatment. These groups were well matched for initial HbA1c level, oral hypoglycemic medication, the proportion of patients prescribed diet control, age, and clinical variables, such as PI, BOP, PPD, CAL.

The test group did have a significant improvement compared with the control group at 16 weeks ($P<0.05$) for PI and BOP (Table 3). There was no statistically significantly difference in PPD and CAL between the two groups ($P>0.05$). However, these indices tended to be lower in the first group, where the patients underwent periodontal treatment.

Glycated hemoglobin measurement (HbA1c) assessed at baseline and 16 weeks later showed a numerical improvement for both groups (Table 3). However, there was no significant difference between the two groups.

Table 1. Subjects' demographics at baseline (gender)

Group \ Gender	Male			Female			All	
	n	%	Sp	n	%	Sp	n	%
A	10	43.48	10.10	13	56.52	10.10	23	100.0
B	12	52.17	10.20	11	47.82	10.20	23	100.0
All	22	47.83	7.37	24	52.17	7.37	46	100.0

Table 2. Subjects' demographics at baseline (age)

Group \ Age	n	$\bar{x} \pm S\bar{x}$	*Sx*	t	Pt	u	Pu
A	23	53.41 ± 2.42	11.60	4.60	< 0.001	0.47 > 0.05	
B	23	55.10 ± 2.64	12.71	4.33	< 0.001		

Table 3. Mean values of periodontal parameters and HbA1c level before and after treatment

Parameters	Group	n	Time of the trial						u*	Pu*	u**	Pu**
			before		after		t	Pt				
			$\bar{x} \pm S\bar{x}$	*Sx*	$\bar{x} \pm S\bar{x}$	*Sx*						
Plaque	A	23	3.23 ± 0.18	0.86	1.57 ± 0.09	0.43	8.3	<0.001	0.23	>0.05	9.42	<0.001
	B	23	3.16 ± 0.12	0.58	3.04 ± 0.13	0.62	0.67	>0.05				
Bleeding	A	23	71.5 ± 5.26	25.25	42.47 ± 2.59	14.43	4.45	<0.001	0.11	>0.05	3.39	>0.05
	B	23	70.69 ± 6.77	32.50	58.18 ± 3.85	18.48	1.61	>0.05				
PPD	A	23	3.86 ± 0.13	0.62	3.42 ± 0.24	1.15	1.62	>0.05	0.39	>0.05	0.76	>0.05
	B	23	3.77 ± 0.20	0.96	3.61 ± 0.07	0.34	0.76	>0.05				
CAL	A	23	4.30 ± 0.18	0.86	4.11 ± 0.12	0.58	0.86	>0.05	0.17	>0.05	1.14	>0.05
	B	23	4.25 ± 0.23	1.10	4.20 ± 0.78	3.74	0.06	>0.05				
HbA1	A	23	8.26 ± 0.31	1.49	7.49 ± 0.28	1.34	1.85	>0.05	0.07	>0.05	0.34	>0.05
	B	23	8.22 ± 0.45	2.16	7.64 ± 0.36	1.73	1.01	>0.05				

DISCUSSION

It has been well documented that periodontal treatment in diabetics has a positive effect in relation to the improvement of periodontal health. Moreover, many controlled clinical trials evaluating the effects of the periodontal therapy in diabetics indicate that most of them exhibit clinical healing similar to non-diabetics in the short term, and that diabetics and non-diabetics alike exhibit a low frequency of periodontal disease recurrence if plaque is carefully controlled, as in a 3-month maintenance program.

In our study, we evaluated the efficacy of non-surgical periodontal therapy on the periodontal health of diabetics. The analysis showed that for some clinical variables measured, including PI and BOP, there was a significant improvement after professional scaling and root planning. This result is also in agreement with a recent study concerning comparative evaluation of adjunctive oral irrigation in diabetics in which both scaling root planning group and scaling root planning plus subgingival water irrigating group had clinical and systemic improvement 12 weeks after treatment.

There is a lot of conflicting evidence about the effect of periodontal therapy on the glycemic control in patient with diabetes. The results of most studies have shown that controlling periodontal infection improves the metabolic control of diabetes. Stewart et al[26] reported that during a nine-month observation, subjects in periodontal treatment group had 17.1% improvement of the glucose level, which was significant when compared with the control

group. Another study examined the correlation between gingival bleeding and blood glucose level in 9 diabetics with moderate to severe periodontitis. The results indicated that a non-significant decrease in HbA1c from 9.44% to 9.01% was observed in the subjects, that 5 out of 9 subjects had a consistent reduction in bleeding on probing. In these 5 subjects, HbA1c was reduced from 8.7% to 7.8% which was significant.

In the present study, both groups had a small positive improvement for HbA1c values. This coincided with some previous studies which reported that effective control of periodontal infection in patients with diabetes could be of help in reducing the level of the advanced glycation end products in serum. However, the minimal difference between test and control groups did not reach statistical significance. This may be comparable to the results reported by Christgau et al that thorough treatment of moderate to advanced periodontal disease did not improve metabolic control of diabetics 4 months after the treatment procedures. A possible explanation of this is that 16 weeks may be too short an interval to observe a significant change in HbA1c. Short-term improvement due to the non-surgical therapy was undetectable, since initial periodontal healing may take 4-6 weeks.Similarly, no significant difference in HbA1c was observed 4 months after periodontal therapy.

An interesting finding in this study was the slight improvement of clinical parameters in the control group. There are two possible explanations: first, type 2 DM management in the Diabetic Clinic was beneficial to the control of periodontal inflammation; second, probably the systemic use of doxycycline, which can inhibit metalloproteinase activity in addition to its antimicrobial effects, also had a good clinical effect in these patients.

Because of the small number of subjects included in this trial, it is quite likely that this study is insufficient to definitely show the differences which may actually exist. Longer term studies with more patients are necessary to further evaluate the variables involved in the present investigation. However, currently there is no evidence that the diabetics require more thorough and aggressive periodontal therapy than the common therapy applied for non-diabetics with periodontal disease. For those patients, who do not respond well to the initial therapy, it may be appropriate to select an antibiotic based on the results of microbial testing. Doxycycline may be preferred due to its special functions.

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ЭФФЕКТ НЕХИРУРГИЧЕСКОГО ПАРОДОНТАЛЬНОГО ЛЕЧЕНИЯ У ПА-ЦИЕНТОВ С САХАРНЫМ ДИАБЕТОМ ТИПА 2

Fu Yun, Елена Фиркова, Lin Jun-Qi, Hong Xun

РЕЗЮМЕ

***Цель**: Оценить эффект нехирургического пародонтального лечения на клинические параметры и гликемический контроль у паци-ентов с сахарным диабетом и хроническим пародонтитом.*

***Пациенты и методы**: 46 пациентов с диабетом типа 2 и с хроническим пародонтитом на случайном принципе разделены на 2 группы (А и В). Лечебный план включает очищение и заглаживание поверхностей корней только в группе А и систематический прием доксициклина в обеих группах. Контроль клинических и лабораторных параметров проведен до и 16 нед. после лечения.*

***Резултаты**: Улучшение клинических пародонтальных параметров и гликированного гемо-глобина (HbA1c) наблюдается у пациентов обеих групп. В группе А однако наблюдается статистически значимая редукция в стоимостях индекса бляшки и кровоточивости при зонди-ровании (P < 0.05) по сравнению с контрольной группой на 16 нед.*

***Вывод**: Результаты показывают, что нехирургическое пародонтальное лечение оказывает благоприятный эффект по отношению к гликемическому контролю у пациентов с сахарным диабетом типа 2 и с хроническим пародонтитом, выражающимся в редукции уровня гликированного гемоглобина.*

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EFFECTS OF SURGICAL PERIODONTAL THERAPY ON SERUM LEVELS OF TNF-α IN PATIENTS WITH CHRONIC PERIODONTITIS

Fu Yun, Elena I. Firkova[1], Hong Xun[2], Lin Jun-Qi[3]

Department of Periodontal and Oral Diseases, Guanghua College of Stomatology, Sun Yat-sen University, Guangzhou, China; [1]Department of Periodontology and Oral Diseases, Faculty of Dentistry, Medical University-Plovdiv, Bulgaria; [2]Center of Experimental Animal Models, Pharmaceutical College, Sun Yat-Sen University, Guangzhou, China; [3]Endondontic Department, Guanghua College of Stomatology, Sun Yat-sen University, Guangzhou, China

ABSTRACT

Aim: To find if serum levels of tumor necrosis factor-α (TNF-α) can change after a surgical periodontal treatment.

Patients and methods: Moderate to severe periodontitis subjects who received basic periodontal treatment were included in the study and randomly divided into two groups. In group A, periodontal surgery was performed. In group B, supportive periodontal therapy was administered. Clinical parameters including probing pocket depth, plaque index and circulating serum TNF-α levels were assessed at baseline and 16 weeks following treatment.

Results: There was a significant improvement in periodontal health of the surgery group compared with the control subjects, with reduction in plaque index and pocket probing depths. The circulating levels of TNF-α decreased statistically significantly in the group A ($P < 0.05$).

Conclusion: Surgical periodontal therapy can reduce the circulating serum levels of TNF-α.

Key words: *periodontitis, periodontal surgery, tumor necrosis factor-α.*

INTRODUCTION

Periodontitis is initiated by specific bacteria and the local host response to these bacteria includes the recruitment of leukocytes and the subsequent release of inflammatory mediators and cytokines such as interleukin-1 (IL-1), tumor necrosis factor-α (TNF-α), which are thought to play an important role in the pathogenesis of periodontitis. These increased levels of several cytokines are involved in periodontal tissue destruction.[1] It has been proposed that patients with periodontitis may have elevated circulating levels of some inflammatory mediators which can be used to predict an increased risk of cardiovascular and cerebrovascular diseases.[2-5] However, there is still much debate regarding the nature and the degree to which this may happen.[6,7]

TNF-α is a monocyte-derived protein that has a wide range of proinflammatory and immunomodulatory effects on a number of different cell populations. TNF-α can stimulate fibroblasts, including gingival fibroblasts to produce collagenase[8] - an enzyme implicated in the tissue destruction of periodontal disease, and to stimulate bone resorption. For example, TNF-α activates monocytes and stimulates the production of IL-β platelet activating factor and prostaglandins.[9] Monocyte stimulation by lipopolysaccharides enhances the production of TNF-α, which has also been proved to induce collagenase release and bone resorption in vivo and the production of matrix metalloproteinases and PGE2.[8]

AIM

The aim of this study was to find if surgical therapy of moderate to advanced chronic periodontitis could influence circulating serum levels of TNF-α.

PAATIENTS AND METHODS

Subjects were selected from systemically healthy non-smoking adults with moderate to severe peri-

Correspondence and reprint request to: Fu Yun, Department of Periodontal and Oral Diseases, Guanghua College of Stomatology, Sun Yat-sen Univesity, Guangzhou, China
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odontitis, who were referred to the periodontal department of the affiliated hospital of stomatology, Sun Yat-Sen University, China. We choose to examine non-smoking subjects to avoid the confounding effects of this important factor.

Six weeks after initial periodontal therapy, a decision was made at re-assessment, regarding the potential value of a surgical intervention. The need for surgery was decided purely on clinical grounds by the clinician responsible for the treatment. All subjects were instructed about the nature and purpose of the study and informed consent was obtained. The inclusion criteria were: (1) age - between 30 and 60 years; (2) at least 20 standing teeth, none of which had untreated periapical lesions; (3) the presence of at least one tooth with a probing pocket depth (PPD) ≥ 5 mm after completion of initial periodontal therapy, comprising oral health instructions, scaling and root planning.

Subjects were randomly divided into two groups: group A - periodontal surgery; group B - supportive periodontal treatment. Thirty six subjects were initially recruited to the study. However, 3 subjects were lost to follow up. The rest were divided so that group A included 20 subjects and group B included 13 subjects.

Subjects provided 2 ml venous blood samples and underwent a standardized periodontal examination before and 16 weeks following the completion of periodontal treatment.

Periodontal assessment was carried out at standing tooth on four sites. The variables of interest were: probing pocket depth (PPD) and plaque index (PI). The parameters were evaluated before the treatment and 16 weeks after the therapy.

In group A, modified Widman flap procedures were performed in the areas where there were deep pockets. Following reverse incisions, full thickness mucoperiostal flaps were raised, granulation tissue removed, root instrumentation performed and alveolar bone shape corrected. The flaps were replaced to their original position and secured with sutures which were removed 1 week after surgery.

In group B, scaling and root planning was taken in the diseased root surfaces. Subjects were advised to rinse with 0.2% chlorhexidine solution twice daily for 2 weeks.

The serum samples were centrifuged for 10 minutes at 11,000 rpm, separating the cells from the serum, which were then collected in Eppendorf tubes and stored at -70°C until required for analysis. Samples were assayed for TNF-α using a commercially available chemofluoresent ELISA kit which was purchased from R&D System Int. The procedures were according to the kit's instructions. Victor Multilable Counter (manufactured by WALLAC OY 20101. Turku, Finland) was used for chemofluorescent assay.

STATISTICAL METHODS

Baseline demographic data were compared using *t*-tests. Clinical parameters and TNF-α levels at baseline and at 16 weeks were compared within each group using paired *t*-tests.

RESULTS

Thirty three subjects completed the study. The demographic characteristics of the subjects are shown in Table 1.

There were no significant differences between the groups at baseline in terms of age and number of standing teeth (P > 0.05).

Table 2 illustrates the findings for the variables for each group at baseline and 16 weeks following periodontal treatment. It can be seen that there was significant improvement in the periodontal health of the surgical group compared with that of the control subjects, with reduction in plaque and pocket probing depths. At the same time, in group A, the circulating levels of TNF-α decreased significantly.

Table 1. The demographics of the subjects at baseline

Parameters	Group	n	$\bar{x} \pm S\bar{x}$	*Sx*	t	Pt
Age	A	20	45.82 ± 7.30	22.63	0.12	> 0.05
	B	13	46.50 ± 6.83	19.25		
Number of standing teeth	A	20	27.83 ± 3.11	13.90	0.10	> 0.05
	B	13	27.40 ± 2.94	10.58		

Table 2. Mean values of clinical variables and serum TNF-α levels for both groups at baseline and 16 weeks following periodontal treatment

Parameters	Group	n	Time of the trial				t	Pt	u*	Pu*	u**	Pu**
			before		after							
			$\bar{x} \pm S\bar{x}$	Sx	$\bar{x} \pm S\bar{x}$	Sx						
Plaque	A	20	46.52 ± 4.14	18.49	34.16 ± 3.04	13.58	2.41	< 0.05	0.17	> 0.05	1.99	< 0.05
	B	13	47.71 ± 5.90	21.30	42.86 ± 4.89	17.66	0.64	> 0.05				
PPD	A	20	5.24 ± 0.25	1.13	2.41 ± 0.61	0.72	4.29	0.001	0.19	> 0.05	3.24	< 0.001
	B	13	5.16 ± 0.35	1.25	4.58 ± 0.28	1.02	1.29	> 0.05				
TNF-α	A	20	7.16 ± 0.44	1.97	5.33 ± 0.36	1.62	3.22	< 0.001	0.53	> 0.05	3.22	< 0.001

u * - before treatment for both groups
u ** - after treatment for both groups

DISCUSSION

It was proposed that the benefits of repeated subgingival scaling and root planning for the patients with persistent periodontal disease are questionable.[10] The clinical variables, such as pocket probing depths, clinical attachment level, bleeding on probing and plaque scores were not significantly different from baseline values at any time point between the groups with subgingival and supragingival debridement during the supportive periodontal therapy. In our control group, although there was a trend toward PPD and PI reduction, the mean values were not significantly different from baseline and 16 weeks following treatment, which is in agreement with the point above.

The results of many investigations show that the long-term data of non-surgical and surgical therapy were equally effective in establishing gingival health and preventing further loss of attachment[11], but when sites with initial PPD > 6 mm were treated with open flap procedures, there was significantly more attachment gain and PPD reduction. The values of PPD and PI in this study were detected 16 weeks after the basic periodontal treatment. It is reasonable to suppose that the initial PPD values of the subjects involved were more than 6 mm. Consequently, subjects in group A demonstrated more PPD and PI reduction compared with those in the non-surgical group. In this study, the relation between the surgical periodontal treatment and the circulating levels of TNF-α, which plays an important role in the pathogenesis of cardiovascular diseases, was evaluated. It was shown that there was a positive effect of the surgical therapy on this cytokine in patients with moderate to advanced chronic periodontitis. The present study may be compared with that of Ide et al.[12] The authors did not observe a reduction of circulating TNF-α levels following non-surgical periodontal treatment. The conflicting results may be due to different methods of study. First, their reassessment time was 6 weeks and this maybe is insufficient time for any biochemical changes to be established after periodontal disease reduction. Second, Ide et al. performed a non-surgical periodontal treatment. Perhaps they could not eliminate all the diseased sites. As a result, some residual irritants remained and they may have had some bearing on the results obtained.

CONCLUSIONS

Within the limitations of this study, it appears that surgical therapy of moderate to advanced chronic periodontitis may influence circulating serum levels of TNF-α. These findings suggest that extensive periodontal treatment may be helpful in reducing the cardiovascular and the cerebrovascular risk.

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ЭФФЕКТ ПАРОДОНТАЛЬНОГО ХИРУРГИЧЕСКОГО ЛЕЧЕНИЯ НА СЫВОРОТОЧНЫЕ УРОВНИ TNF- У ПАЦИЕНТОВ С ХРОНИЧЕСКИМ ПАРОДОНТИТОМ

Fu Yun, Елена Фиркова, Hong Xun, Lin Jun-Qi

РЕЗЮМЕ

***Цель**: Установить изменяются ли сывороточные уровни опухоль-некротизирующего фактора после хирургического пародонтального лечения.*

***Пациенты и методы**: Исследование охватывает пациентов со средне выраженным до далеко зашедшего хронического пародонтита после проведения основного пародонтального лечения. На случайном принципе пациенты разделены на две группы, при чем пациентам группы А проведена пародонтальная хирургия, а в группе В проведена только поддерживающая пародонтальная терапия. Глубина карманов при зондировании, индекс бляшки и сывороточные уровни TNF- оценены до начала и 16 нед. после лечения.*

***Результаты**: После проведенного хирургического лечения установлено значительное улучшение пародонта у пациентов группы А по сравнению с контрольной группой пациентов. Налицо значительная редукция стоимостей индекса бляшки и глубин при зондировании пародонтальных карманов. Уровни TNF- уменьшаются со статистической значимостью (P < 0.05).*

***Вывод**: Хирургическое пародонтальное лечение редуцирует сывороточные уровни TNF-a.*

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EFFECT OF TIBOLONE ON SEXUAL FUNCTION IN LATE POSTMENOPAUSAL WOMEN

Zdravko A. Kamenov, Maria K. Todorova[1], Vladimir G. Christov

Clinic of Endocrinology, Alexandrov's Hospital, [1] Department of Pathophysiology, Medical University, Sofia, Bulgaria

ABSTRACT

Background: Sexual dysfunction may significantly affect quality of life and marital relations in the postmenopausal period. The aim of the study was to assess the effect of tibolone on climacteric symptoms and sexuality in late postmenopausal but still symptomatic women.

Patients and methods: A six-month prospective study was conducted of two groups of clinically healthy postmenopausal women: a control group (n = 18; mean age 57.8 ± 4.1 yrs; menopause at 49.7 ± 2.5; years of amenorrhea 8.1 ± 4.0 yrs) and a tibolone group (n = 22; mean age 57 ± 4.5 yrs; menopause at 47.7 ± 3.9; years of amenorrhea 9.2 ± 4.6 yrs), who received 2.5 mg tibolone daily for six months. The Kupperman menopausal index (KI) was calculated for both groups at baseline and at six months. Sexual function was assessed by the Female Sexual Function Index (FSFI) questionnaire at the beginning and at the end of the study. The FSFI comprised five main domains: desire, arousal, lubrication, orgasm and pain. Satisfaction and a total score were also recorded.

Results: The results showed that during the observation period KI decreased significantly in the tibolone group (15.7 ± 9.2 vs 11.3 ± 6.8, $p < 0.001$), while in the control group no difference was observed. There was a significant improvement of sexual function in the tibolone group in all domains: desire – from 2.6 ± 1.0 to 3.1 ± 1.0 ($p < 0.001$); arousal – from 2.3 ± 1.8 to 3.4 ± 1.1 ($p < 0.001$); lubrication - 2.6 ± 2.1 and 3.5 ± 1.4 ($p < 0.05$). The ability to reach orgasm increased ($p < 0.001$) and pain and discomfort during and after sexual intercourse significantly decreased ($p < 0.01$). The overall satisfaction and the total score in the treated group changed favourably in a statistically significant manner, while these parameters did not change in the control group.

Conclusions: Treatment with tibolone had a beneficial effect on the climacteric symptoms and sexual function of late postmenopausal women. Moreover, tibolone seems to have an advantage over conventional hormone replacement therapy (HRT) in improving desire and arousal.

Key words: *tibolone, postmenopausal women, sexual function*

INTRODUCTION

Menopausal transition is associated with unfavorable changes such as climacteric symptoms, bone loss, alterations of the anatomical structure, vascular responsiveness, neurological and psychological functions that accompany the aging process. Female sexual dysfunction is a common finding in the postmenopausal period. Although many women do not seek medical help for climacteric complaints, their quality of life and marital status may significantly be affected by mood changes, reduced sexual drive, and sexual dysfunction.[1] Treatment of the hormonal insufficiency with conventional HRT has many beneficial effects: HRT alleviates the climacteric complaints, prevents bone loss and fractures, improves the unfavorable changes of the genito-urinary system.[2] HRT has been found to improve some factors of the sexual function related predominantly to sexual performance as lubrication, pain and orgasm, while desire and arousal remain unaffected.[3] Besides, HRT has undesirable effects on the breast tissue. The inconclusive overall effects of HRT on sexuality may be attributed to the fact that the therapeutic regimens include either estrogen alone or estrogen in combination with progestagens. Androgens play a central role

Correspondence and reprint request to: Z. Kamenov, Clinic of Endocrinology, Alexandrov's Hospital, Department of Pathophysiology, Medical University, Sofia, Bulgaria
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in female sexuality, especially with regard to desire and arousal, which were found to be the most affected domains after the menopause.[3] Menopause is associated with decrease in the levels of circulating androgens, such as dehydroepiandrosterone (DHEA), dehydroepiandrosterone sulfate (DHEA-S), androstendione and testosterone.[4] Some trials report that methyltestosterone taken orally[5] or applied transdermally[6] significantly improve not only sexual desire, but also frequency of intercourse when compared with the effects of both estrogen replacement therapy alone and placebo.

Tibolone is a synthetic tissue-specific steroid [*(7α, 17α)-17-hydroxy-7-methyl-19- norpregn-(10)-en-20-yn-3-one*] combining tissue specific estrogenic, progestagenic and weak androgenic properties.[7] It has been shown to be effective in relieving climacteric symptoms with almost no stimulatory effect on the breast and the endometrium.[8]

The aim of the study was to evaluate the effect of tibolone on sexuality after 6 months of administration in late postmenopausal women who still had climacteric symptoms.

PATIENTS AND METHODS

Late postmenopausal women who still had climacteric symptoms were recruited from patients attending the Clinic of Endocrinology, Medical University, Sofia. Informed consent was obtained from all patients in compliance with the requirements of the local ethical committee. A six-month case-control, prospective, non-randomized study was conducted of two groups of clinically healthy postmenopausal women: a control group (n = 18; mean age 57.8 ± 4.1 yrs; menopause at 49.7 ± 2.5; years of amenorrhea 8.1 ± 4.0 yrs) and a tibolone group (n = 22; mean age 57 ± 4.5 yrs; menopause at 47.7 ± 3.9; years of amenorrhea 9.2 ± 4.6 yrs). The tibolone group received 2.5 mg of tibolone (Livial, Organon, Netherlands) daily for six months. The study sample included women without contraindications to HRT. To evaluate the effect of tibolone on climacteric symptoms the Kupperman menopausal index (KI) was calculated for both groups at baseline and at six months. Sexual function was assessed using the Female Sexual Function Index (FSFI) questionnaire[9] at the beginning and at the end of the study. The FSFI comprised 19 questions, divided into domains: desire, arousal, lubrication, orgasm and pain. Satisfaction and the total score were also recorded. The principal components of the five domains were:

- for desire - frequency and level;
- for arousal – frequency, level, confidence and satisfaction;
- for lubrication – frequency, difficulty, frequency in maintaining and difficulty in maintaining;
- for orgasm – frequency, difficulty and satisfaction;
- for pain – frequency during and following vaginal penetration and level during or following vaginal penetration.

Satisfaction was evaluated taking into consideration the closeness with the partner, with sexual relationship and with overall sex life. Table 1 presents the scale for evaluation of domain scores and sexual function score (FSFI). The control group comprised women who refrained from taking tibolone. They were asked to complete the FSFI and the Kupperman Index questionnaires at the first visit and after 6 months.

Table 1. Evaluation of domain scores and sexual function total score (FSFI)

Domain	Questions	Score range	Score	Factor	Minimum score	Maximum score
Desire	1, 2	1-5	2-10	0.6	1.2	6.0
Excitation	3, 4, 5, 6	0-5	0-20	0.3	0	6.0
Lubrication	7, 8, 9, 10	0-5	0-20	0.3	0	6.0
Orgasm	11, 12, 13	0-5	0-15	0.4	0	6.0
Satisfaction	14, 15, 16	0-5	0-15	0.4	0.8	6.0
Pain	17, 18, 19	0-5	0-5	0.4	0	6.0
Total score					2.0	36

Statistical analysis. SPSS 12.01 software (SPSS, Chicago, IL, USA) was used in the analysis. The one-sample Kolmogorov-Smirnov test was used to assess sample distribution. The mean values of each variable were compared using analysis of variance. Independent *t*-test was used to compare means between controls and patients groups, and paired *t*-test was used to compare the data within each group. All results are expressed as means ± SD. Statistical significance was fixed at $p < 0.05$.

RESULTS

The results showed that during the six-month study period tibolone relieved significantly the climacteric complaints. The KI in the tibolone group decreased (15.7 ± 9.2 vs. 11.3 ± 6.8, $p < 0.001$) while there was no significant difference in the control group (Fig. 1). A significant improvement of sexual function was found in all domains in the tibolone group: desire – from 2.6±1.0 to 3.1±1.0, $p < 0.001$ (Fig. 2); arousal – from 2.3 ± 1.8 to 3.4 ± 1.1,

Figures 1-8. Changes in KI and FSFI during the treatment with tibolone and in the control group

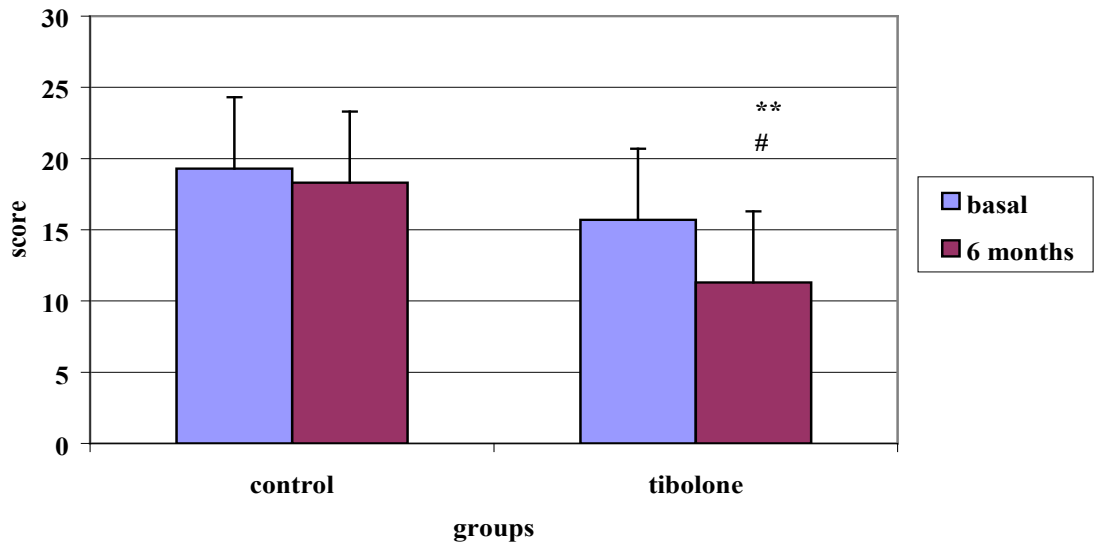

Figure 1. Kupperman Index Score.

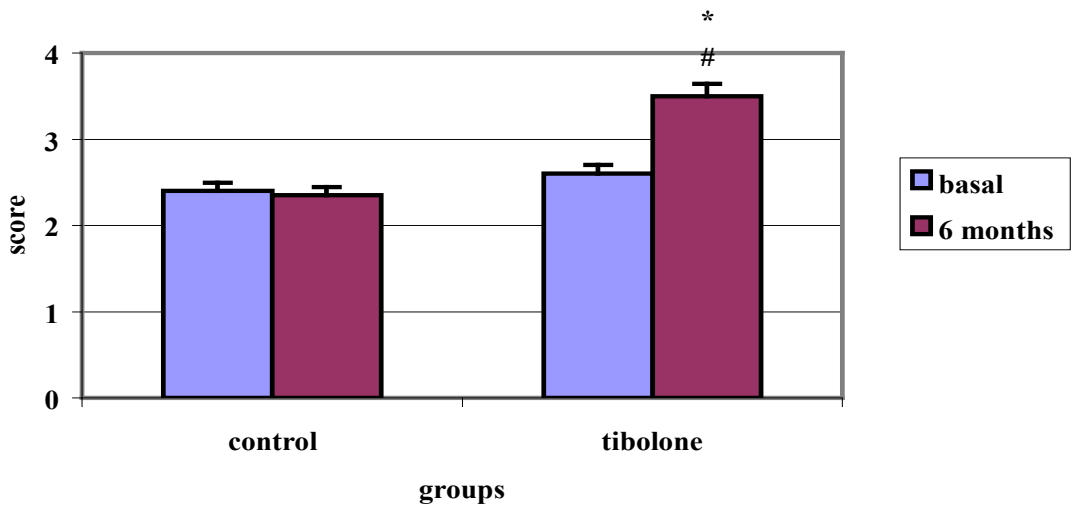

Figure 2. Desire domain.

$p < 0.001$ (Fig. 3); lubrication - 2.6 ± 2.1 and 3.5 ± 1.4, $p < 0.05$ (Fig. 4). The ability to reach orgasm increased from 2.3 to 3.1, $p < 0.001$ (Fig. 5) and pain and discomfort during and after sexual intercourse significantly decreased from 2.9 to 4.3, $p < 0.001$ (Fig. 6). Moreover, the overall satisfaction $p < 0.05$ (Fig. 7) and the total score $p < 0.001$ (Fig. 8) in the treated group changed favorably in a statistically significant manner, while in the control group these parameters underwent no changes.

DISCUSSION

A large Danish population study of 10 458 participants has shown that women have a lower level of sexual desire than men and the level of sexual desire decreases with age in both genders, more steeply in women.[10]

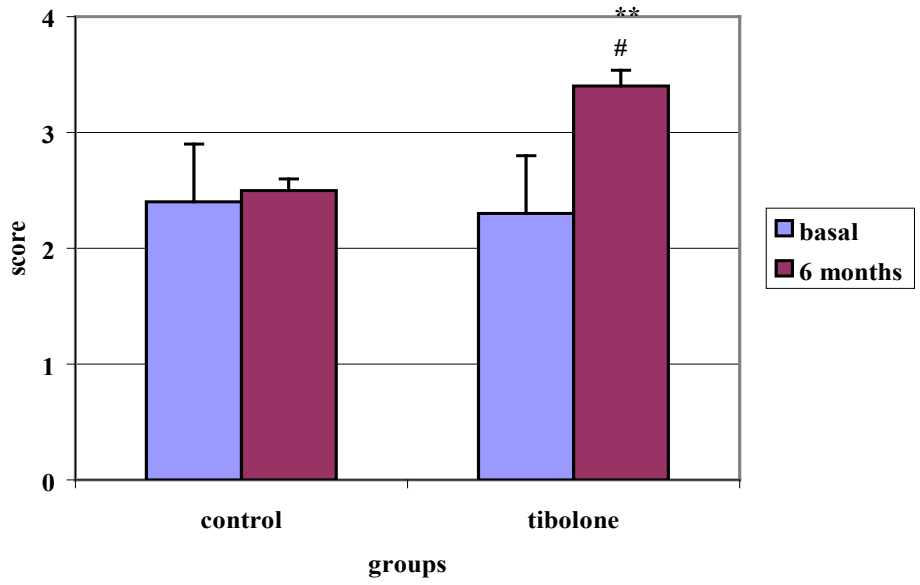

Figure 3. Arousal domain.

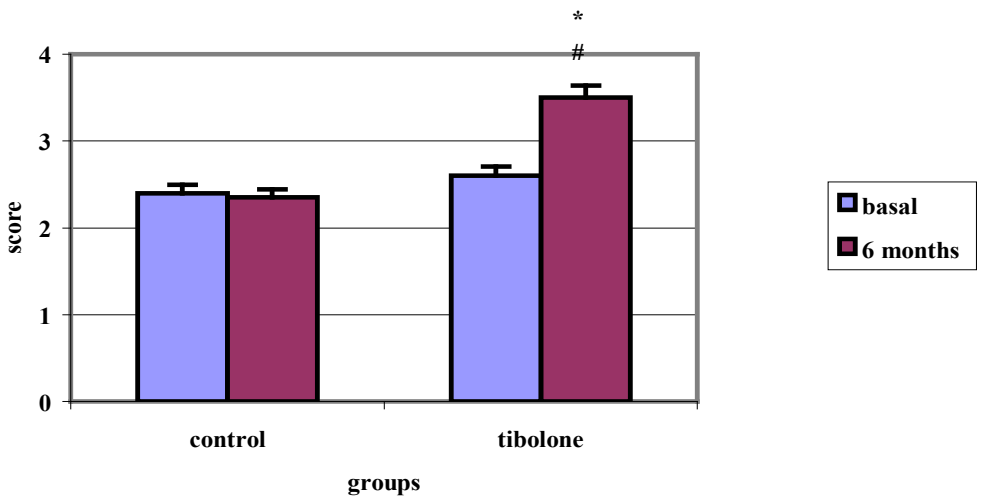

Figure 4. Lubrication domain.

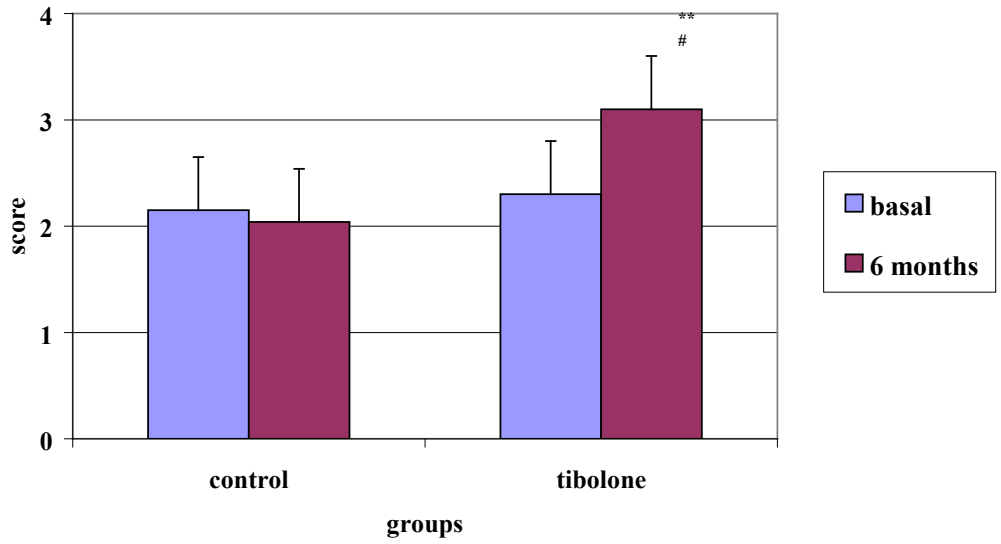

Figure 5. Orgasm domain

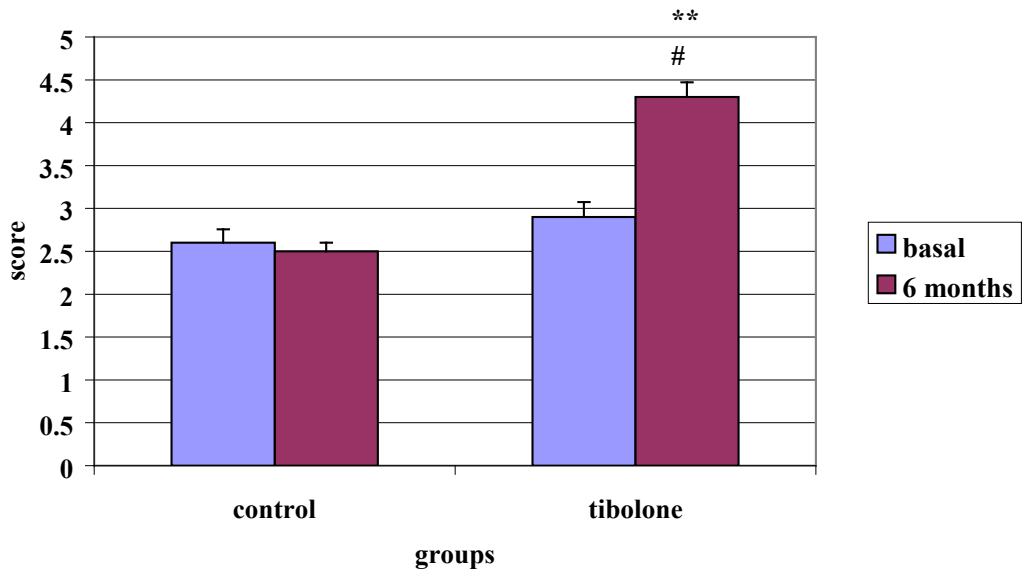

Figure 6. Pain domain.

Women's sexuality is a delicate process depending on numerous factors. Motivations for sexual activity are complex and include mood, sense of well-being, and emotional closeness with the partner. Other factors, which are of no less importance, are the sexual desire and the quality of the experience during sexual performance – arousal, lubrication and pain during and after intercourse, ability to reach orgasm and finally overall satisfaction with sex and sexual relations as a whole.[11] Unfavorable alterations in any of these aspects after menopause lead to sexual dysfunction. Some of these factors are psychological, provoked by stress, family problems or use of medications.[12] Others, however, are endocrine in origin and hormone-dependent. Decreased sexual function in postmenopausal women is due in part to both estrogen and androgen depletion after menopause[13,14] and can be improved by appropriate substitution.

Tibolone is a steroid compound that is selectively metabolized by specific tissues to metabolites with estrogenic, progestagenic and androgenic properties, thus exhibiting tissue-specific hormonal effects.[15] The results of our study showed that tibolone affected favorably both the climacteric symptoms and sexuality. The evaluation of desire and arousal, which are considered the most affected domains after menopause, usually not improved by HRT[3], showed

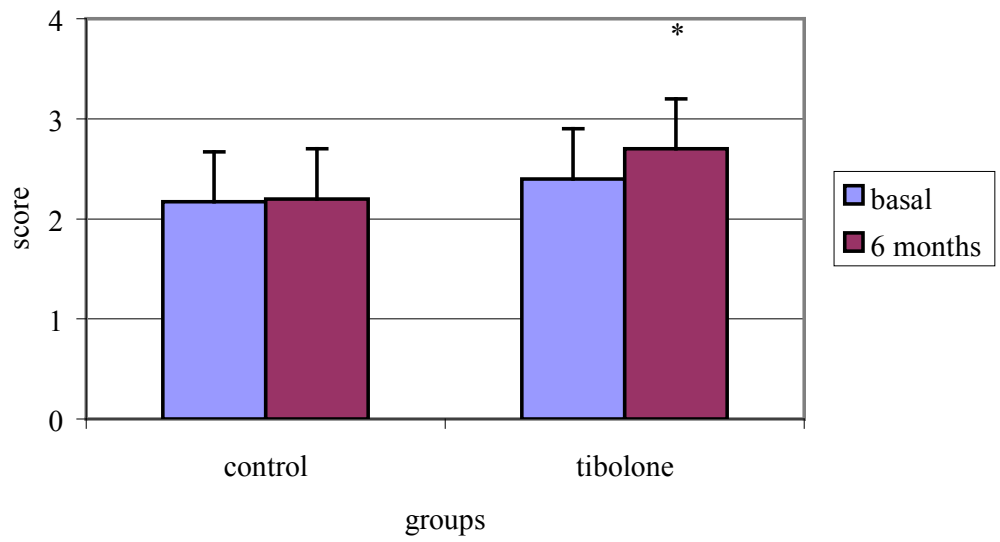

Figure 7. Satisfaction domain.

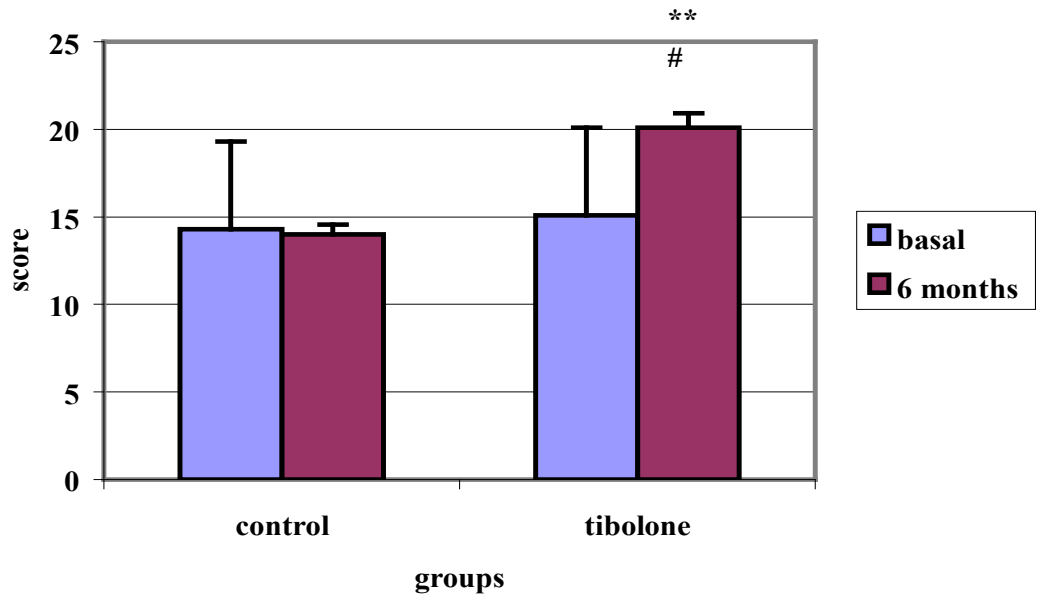

Figure 8. Total score.

- P < 0.05, for the tibolone vs.control group at the 6th month
* - P < 0.05; ** - P < 0.001, for the basal vs. 6 months values of the tibolone group.

that tibolone affected both of them beneficially. The pathogenetic mechanisms are different and still not well understood. It was shown that tibolone can improve mood, anxiety and insomnia, and the overall feelings of well-being via stimulation of production and release of endogenous opioids, specifically beta-endorphin.[16] Another mechanism through which tibolone improves libido is possibly via the weak androgenic activity of tibolone in the central nervous system.[8] As to the androgenic action of tibolone, its metabolite △4 isomer shows a high affinity to the androgen receptor, nearly 40% of that exhibited by the most potent androgen, dihydrotestosterone.[17] Another possible mechanism is the suppression of sex hormone binding globuline (SHBG) levels in plasma, thus increasing the availability of free testosterone.[18] The exact physiological and biochemical role of androgens in female genital sexual arousal function remains controversial and still poorly understood. Some studies suggested that androgens had no effect on sexual arousal *per se* but can influence sexual

desire as fantasies.[19] Other studies, however, report that androgen insufficiency in women is associated with impaired sexual function.[20] Experimental studies show that in ovariectomized rabbits treated with or without androgen replacement therapy androgens enhanced nitric oxide synthase activity and down-regulated arginase activity in proximal vagina, while estrogens down-regulated nitric oxide synthase activity and increased arginase activity in distal vagina.[21] The authors found that vaginal tissues express androgen receptors and the expression of these receptors is regulated differently in the proximal and distal vagina by androgens and estrogens. They concluded that androgens play an important role in the vaginal and clitoral function during genital sexual arousal. Thus, the beneficial effects of tibolone on desire and arousal found in our study can be attributed to the androgenic action of tibolone through both central and local effects on arousal. Another favorable impact of tibolone found in this study can be attributed to its estrogen effects. These comprise the significant alleviation of the climacteric complaints in the tibolone group, evaluated by the KI and improved sexual performance. We found that treatment with tibolone improved lubrication, decreased pain and discomfort during and after intercourse, improved the scores in the orgasm domain and increased satisfaction.

CONCLUSIONS

The results of our study showed that tibolone combining the central and peripheral effects of estrogens and androgens improved sexual function in late postmenopausal women. Treatment with tibolone had a beneficial effect on climacteric symptoms and sexual response and performance. Moreover, it seems that in improving desire and arousal, without the adverse effects on the breast and endometrium tibolone has an advantage in comparison with conventional HRT.

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ЭФФЕКТ TIBOLONE НА СЕКСУАЛЬНУЮ ФУНКЦИЮ У ЖЕНЩИН ПОСЛЕ МЕНОПАУЗЫ (ПОЗДНИЙ ПЕРИОД)

З. Каменов, М. Тодорова, В. Христов

РЕЗЮМЕ

Сексуальная дисфункция после менопаузы серьезно может нарушить качество жизни и фамильные отношения.

***Цель**: Настоящее исследование ставит себе целью определить эффект Tibolone на климактерические симптомы и сексуальные аспекты у женщин с жалобами после менопаузы.*

***Пациенты и методы**: Проведено проспек-тивное исследование двух групп клинически здоровых женщин (поздний период после менопаузы). Контрольная группа включает 18 женщин (средний возраст 57.8 ± 4.1; менопауза в 49.7 ± 2.5; период после менопаузы 8.1 ± 4.0). Группа женщин, принимающая 2.5 mg Tibolone в день в течение 6 мес, состояла из 22 человек (средний возраст 57 ± 4.5; менопауза в 47.7 ± 3.9; период после менопаузы 9.2 ± 4.6). И при обеих группах определен индекс Kupperman (KI) до и после лечения. На этих этапах определена и сексуальная функция с помощью вопросника Female sexual function index (FSFI). Он включает следующие элементы: желание, возбуждение, лубрикацию, оргазм и боль. Регистрированы удовлетворенность и общая оценка.*

***Результаты**: Во время периода наблюдения KI значимо уменьшается в группе, принимающей Tibolone (15.7 ± 9.2 до 11.3 ± 6.8, $p < 0.001$), в то время как в контрольной группе изменение не наблюдалось. В группе, принимающей Tibolone, достоверно установлено улучшение сексуальной функции во всех аспектах (желание – от 2.6 ± 1.0 на 3.1 ± 1.0 - $p < 0.05$; возбуждение – от 2.3 ± 1.8 на 3.4 ± 1.1 – $p < 0.001$; лубрикация – от 2.6 ± 2.1 на 3.5 ± 1.4 – $p < 0.05$). Увеличилась и способность достижения оргазма – $p < 0.001$, а боль и дискомфорт во время и после полового акта значимо уменьшились - $p < 0.001$. Удовлетворенность и общая оценка улучшились достоверно в группе лечившихся и не изменились в контрольной группе.*

***Заключение**: Лечение препаратом Tibolone оказывает благоприятный эффект на симптомы климактерия и сексуальную функцию у женщин после менопаузы (поздний период). Улучшение желания и возбуждения является некоторым преимуществом препарата Tibolone по сравнению с конвенциональным гормонозаместительным лечением.*

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IMMUNOLOGICAL AND MICROBIOLOGICAL INVESTIGATIONS OF PATIENTS WITH BURN INJURIES

Yordanka D. Stoilova, Irina A. Haidushka[1], Mariana A. Murdjeva[1], Ivan Z. Traikov[2], Tatyana A. Popova[1], Ani K. Kevorkyan

Department of Infectious Diseases, Epidemiology, Parasitology, [1]Department of Microbiology and Immunology, [2]Clinic of Burn Injuries and Plastic Surgery, Medical University Plovdiv, St George University Hospital, Plovdiv

ABSTRACT

Susceptibility of patients with burn injuries to infections is associated with disturbances of the humoral and cell-mediated immunity.

The aim of the present study was to investigate the dynamics of phagocytic indices and serum immunoglobulin levels in patients with burn injuries as there is a high risk of colonization and infection of the burn wound.

Patients and methods: Between December 2004 and April 2005, we studied the dynamics of the following indices of the phagocytic system and humoral immunity of 30 burned patients: phagocytic index, phagocytic number, nitro blue tetrazolium chloride test, leukocyte number in the peripheral blood, the percentage of the neutrophilic leukocytes from the differential blood count, serum levels of immunoglobulins A, G and M and the percentage of lymphocytes. In addition, microbiological tests of the wound secretions were performed using conventional microbiological methods.

Results: The changes in the phagocytic indices suggested that it is the decreased phagocytic index and number that are important for the evaluation of wound infection risks. The increased total number of leukocytes and elevation of the neutrophil levels usually observed in burned patients indicate the presence of bacterial infection and the prompt involvement of the phagocytic mechanisms. The lower percentage of lymphocyte in comparison with the healthy controls was indicative also of a possible involvement of the cell-mediated immune response. The characteristic humoral mechanisms of the immune response (serum immunoglobulins) in patients with thermal injuries altered only slightly.

Conclusions: The analyzed data may be of help in evaluating the risk of colonization and infection of the burn wound in patients with burn injuries. We would like to recommend a compulsory battery of microbiological tests and several immunological investigations - phagocytic index, phagocytic number, leukocyte number, differential blood test and serum immunoglobulins with a view to applying an adequate antimicrobial and immune modulating treatment.

Key words: *thermal trauma, phagocytic indices, immunoglobulins A, M and G, microbiological isolates*

INTRODUCTION

Severe thermal traumas can cause various disorders in the immune system functioning and regulation.[1-4] The alterations of the immune response are determined by the loss of proteins, including immunoglobulins. Wound infection in patients with burn injuries has a high incidence – up to 70%.[2,3,5] It plays an important role for the clinical course of a burn injury and appears to be a major problem for the performance of plastic surgery.[3-6] The susceptibility to different infections is associated with disturbances of the humoral and cell-mediated immunity.[7] Phagocytosis is one of the main factors of natural resistance. As cells performing phagocytosis, neutrophils play a key role for the protection of the macro organism. Phagocytic dysfunction is associated with a higher risk of infections in patients with thermal injuries.[8] On the other hand, different reports on the serum

Correspondence and reprint request to: Y. Stoilova, Department of Infectious Diseases, Epidemiology, Parasitology, Medical University Plovdiv
15A Vassil Aprilov St., 4002 Plovdiv, Bulgaria
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immunoglobulin levels characteristic of the humoral immunity in burned patients are quite contradictory.[7] There are little systematic immunological data about this problem in Bulgaria.

The aim of the present study was to investigate the dynamics of the phagocytic indices and the serum immunoglobulin levels in patients with thermal injuries as there is a high risk of colonization and infection of the burn wound.

PATIENTS AND METHODS

We studied several indices of the phagocytic system and humoral immunity in 30 burned patients (5 to 70% body surface area burn) treated in the Clinic of Burn Injuries and Plastic Surgery, St George University Hospital, Plovdiv between December 2004 and April 2005.

Phagocytic activity of blood cells:

- phagocytic index (PI) - the percentage of polymorphonuclear leukocytes (PMN) with phagocytosed bacteria (staphylococci);
- phagocytic number (PN) - the number of bacteria in 1 PMN
- NBT (nitroblue tetrazolium chloride test) - assessment test of the neutrophilic oxidative burst;
- leucocyte number in the peripheral blood;
- percentage of the neutrophilic leucocytes from the differential blood count (DBC).

The serum levels of immunoglobulins A, G and M were assessed using the Manchini radial immunodiffusion test.

The percentage of lymphocytes in the differential blood count was also investigated.

The investigations were performed in dynamics (on the first and seventh days after burn). The results were compared with the established laboratory standards obtained with a control group of 30 asymptomatic patients.

Concurrently with the immunological investigations the wound secretions of the patients were studied using conventional microbiological methods. We included in the study patients whose isolates were immunologically proved on the very first day and on the seventh day. All data were analysed using the analysis of variance (ANOVA) and the alternative analysis. Differences were statistically significant at $P < 0.05$. We applied the standard statistical program SPPS 7.5-96 – SPSS Inc. USA.

RESULTS AND DISCUSSION

The age range of the patients with thermal traumas was 3 to 84 years: 10 patients were 3 to 10 years old, 3 patients - 11 to 18 years, and 17 patients were over 18 years. Twenty-two of the patients were males (73.33 ± 9.42%) and 8 - females (26.67 ± 15.63%) ($t = 2.55$, $P < 0.01$). The total skin surface area of burn varied from 5 to 70%. About half of the patients - 14 (46.66 ± 13.33 %) had more than 30%, followed by 10 patients (33.33 ± 14.91 %) who had 16-30% and 6 patients (20.00 ± 16.33 %) with up to 15% body surface of burn.

The domestic burn injuries were the most numerous – 21 of the patients (70.00 ± 10.00%). Of these, 12 (57.14%) had sustained burns caused by scalding water. The length of hospital stay varied from 11 to 150 days. Long-term hospitalization (60 days at average) carries a considerably higher risk of wound infection in burn patients. We had lethal outcome in one of the patients.

Microbiological investigations

In 29 of all 30 patients with thermal injuries (96.67 ± 4.38%) we isolated various microorganisms from the wound secretions. They were most often etiologically related to hospital acquired pathogens. The total number of the isolates was 71 (Table 1). Thirty-two of them (45.07 ± 8.75%) were Gram-negative and 39 (54.93 ± 7.97%) - Gram-positive. The distribution of the patients by type of isolates was as follows: 9 patients (31.04%) had four species of microorganisms, 6 patients (26.68%) had three, 8 patients (27.58%) had two and 6 patients (20.68%) had one. The results obtained are consistent with the opinion that after a burn is sustained there occurs a complex "competition" among a great number of different species of microorganisms in every wound. It is well known that Gram-positive flora is usually detected 48 hours after a burn injury; it is replaced by Gram-negative flora about 7 days after sustaining burn injuries.[2,3,5,6,9,10] Our study showed that staphylococci dominated in the microbial spectrum, MRSA being of the highest quota. According to a great number of investigators this pathogen has been of increasing importance recently.[9-11] In our study, *P. aeruginosa* (44.80%) and *A. baumanii* (44.82%) appeared to be the leading pathogens; similar etiological structure was found in our previous studies.[5,6] Other authors reported similar data as well.[12,13] The isolated microorganisms were etiologically related to wound infections, which lead to prolonged hospitalization of the patients, severer clinical course, higher lethality, problems in plastic surgery etc.[3-5,11-13] Such characteristics were mainly specific for hospital acquired pathogens; they were

Table 1. Distribution of patients with microbiological isolates

Type of isolates n = 71	Patients n = 29 Number	%	Sp
Pseudomonas aeruginosa	13	44.80	13.79
Acinetobacter baumanii	13	44.82	13.79
Escherichia coli	10	34.48	15.03
Proteus mirabilis	3	10.34	17.57
Methicillin resistant Staphylococcus aureus	12	41.34	14.21
Methicillin sensitive Staphylococcus aureus	9	31.02	15.41
Coagulase negative staphilococcus	11	37.90	14.62

identified using molecular and epidemiological methods. Because of the small number of the patients and the different isolates it was impossible to establish a definite correlation between the isolate specificity and the specific immunologic alterations it would possibly induce.

Results from the investigation of several immunological parameters

Phagocytosis is of greatest significance for the protection of a great number of bacterial infections.[8,14,15] In the studied burn patients, the PI of sample one was significantly lower in all age groups. They were, respectively, 46.00 ± 1.36% for the 3-10-year group ($t = 2.16$; $P < 0.05$); 46.00 ± 3.46% for the 11-17-year group ($t = 2.0$; $P < 0.05$) and 44.36 ± 0.92% for the group over 18 years ($t = 2.41$; $P < 0.05$) compared to the reference value of 62.50 ± 7.5%. Followed dynamically, the indexes showed no changes and remained low (Fig. 1). The average PN values were nearer to the lower reference limits and were similar for the different age groups. They were lower in the 3-10-year group and in the group over 18 years (Fig. 2). Other authors also reported phagocyte activity disturbances, especially in patients with burns in more than 30% body surface area.[4,16,17]

NBT was within reference range (15.00±5.00%). It had a slight increase in all three age groups. The values were as follows: for the 3-10-year group sample one showed 15.50 ± 0.98%, sample two - 17.18 ± 0.44%; for the 11–17-year group sample one showed 18.66 ± 1.38%, sample two - 20.00 ± 1.15%, and for the group over 18 years sample one showed 16.11 ± 0.74%, sample two - 17.00 ± 0.63% (Fig. 3). This test gave information about the PMN functional activity and their oxidation-dependent bactericide systems, associated with the oxidative burst. Sayeed M. found circulating neutrophils in post-burn inflammations whose hyperactivity often lead to an increase in the oxidative burst and hence, to tissue damages.[18]

The average leukocyte number was higher in samples one and two in comparison with the controls (7000 ± 3500) in the three age groups: for the 3-10-year group it was 7911 ± 669.25 in sample one and 11406 ± 2456 in sample two; for the 11-17-year group it was 11866 ± 2126.3 for sample one and 15873 ± 927 for sample two, and for the over-18-year group it was 10792 ± 990.89 in sample one and 12213 ± 1219.7 in sample two. Dynamically, they tended to increase (Fig. 4). Similar results were reported by Conteras JA, who pointed out that the severity of the thermal trauma was associated with leukocyte hyperactivity and overproduction or dysregulation of the immune mediators.[19] According to Boiarska HM, et al. the lower functional activity of the blood leukocytes in burned patients was proportional to the trauma severity.[4]

Segmented neutrophils showed similar values in the three age groups of patients as follows: for

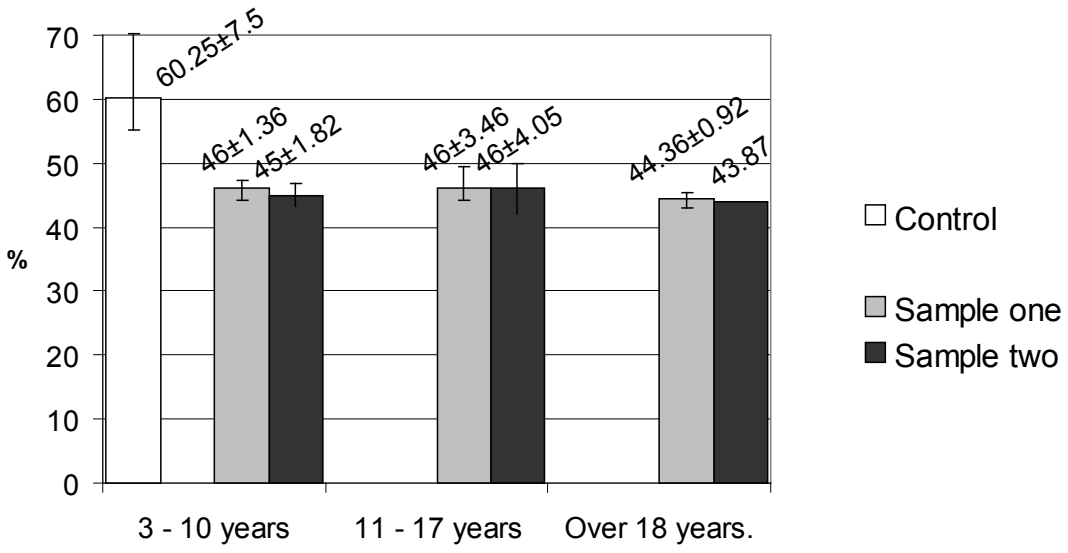

Figure 1. Phagocytic index in burned patients.

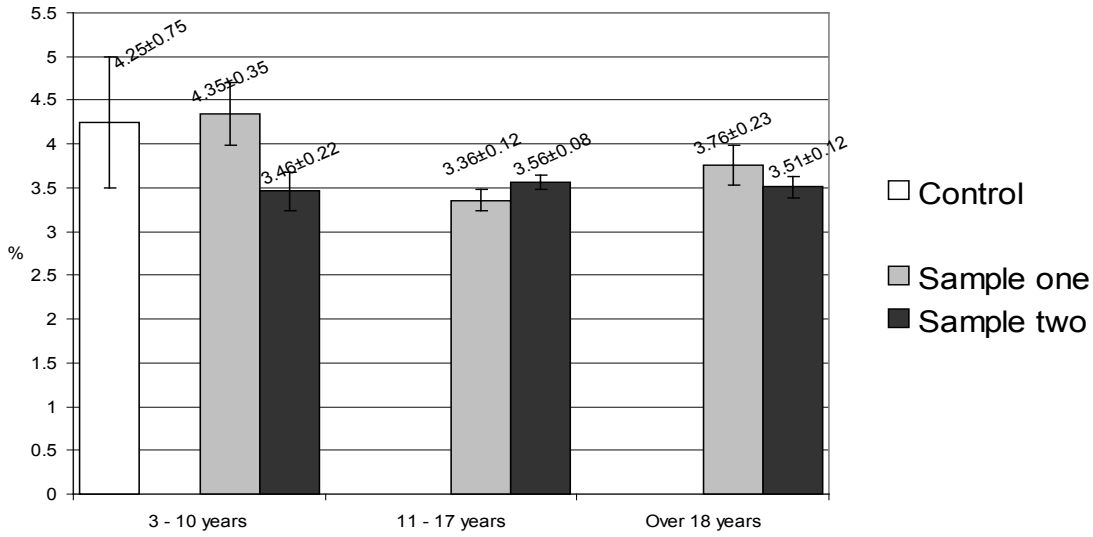

Figure 2. Phagocytic number.

the 3-10-year group they were 50.37 ± 5.96% in sample one and 57.00 ± 5.51% in sample two; for the 11-17-year group - 58.66 ± 6.56% in sample one and 69.33 ± 6.36% in sample two and for the over-18-year group - 65.21 ± 4.19% in sample one and 67. 11 ± 2.99% in sample two. The indexes were slightly higher compared with the controls and their increase was not significant ($P > 0.05$) (Fig. 5). PMN are the most actively phagocytizing cells. Their functional activity comprises several stages – chemotaxis, engulfing, activation of oxygen dependent and oxygen independent mechanisms and intracellular killing. Chronic and recurrent bacterial infections are often associated with disorders of the chemotaxis and the intracellular bactericide activity of the phagocytizing cells.[20] Neutrophils here kept

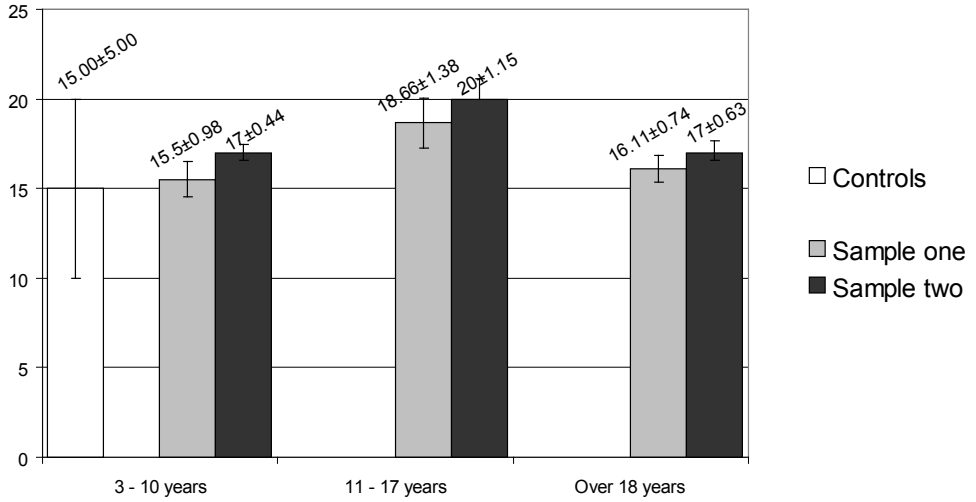

Figure 3. NBT in burned patients.

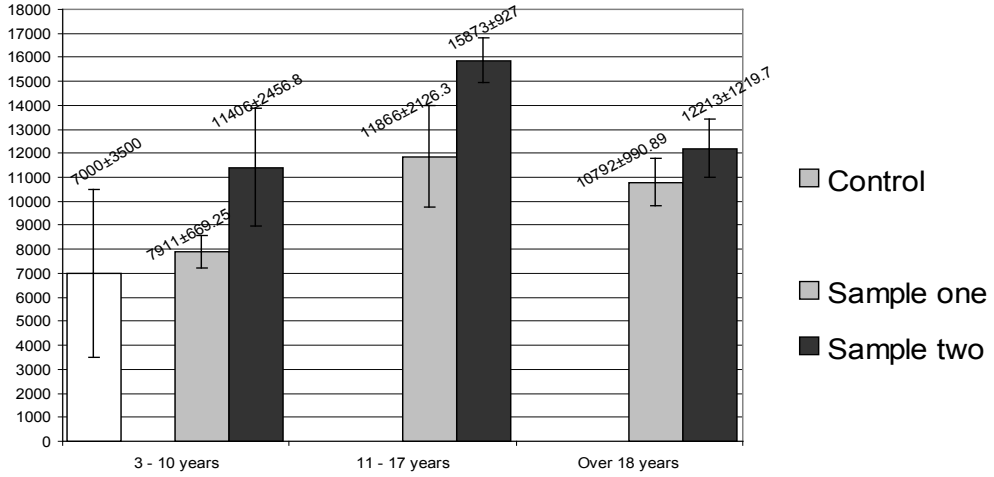

Figure 4. Leucocytic number in burned patients.

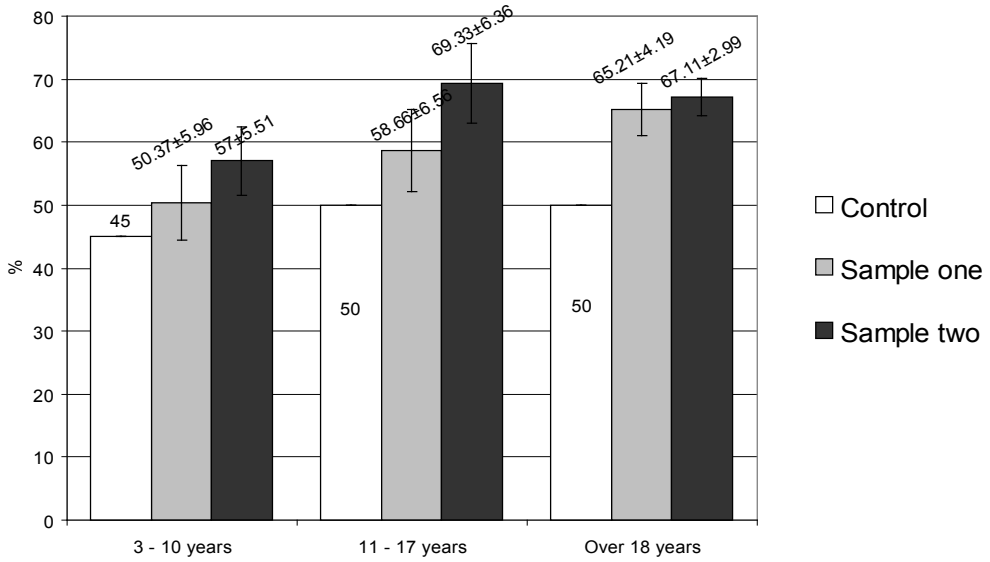

Figure 5. Percentage of segmented leucocytes.

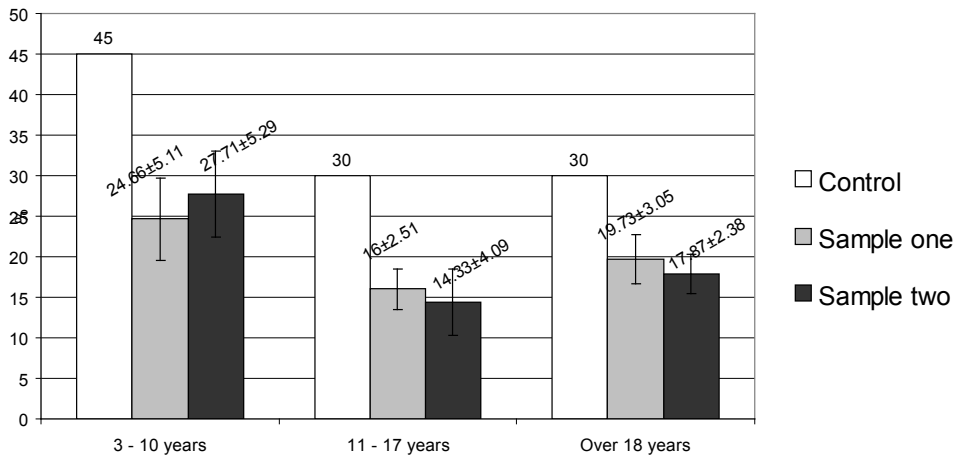

Figure 6. Percentage of lymphocytes (Ly).

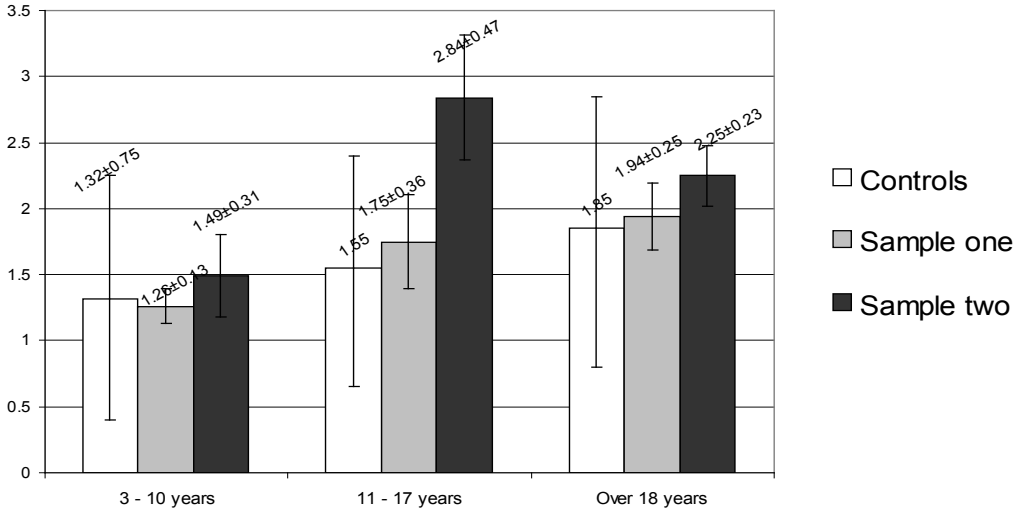

Figure 7. Immunoglobulines A in burned patients.

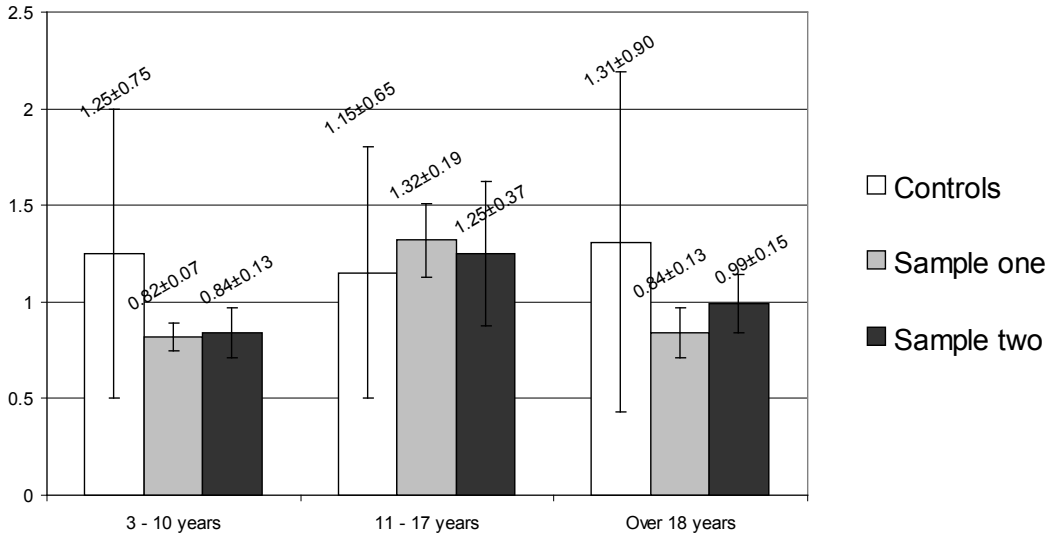

Figure 8. Immunoglobulines M in burned patients.

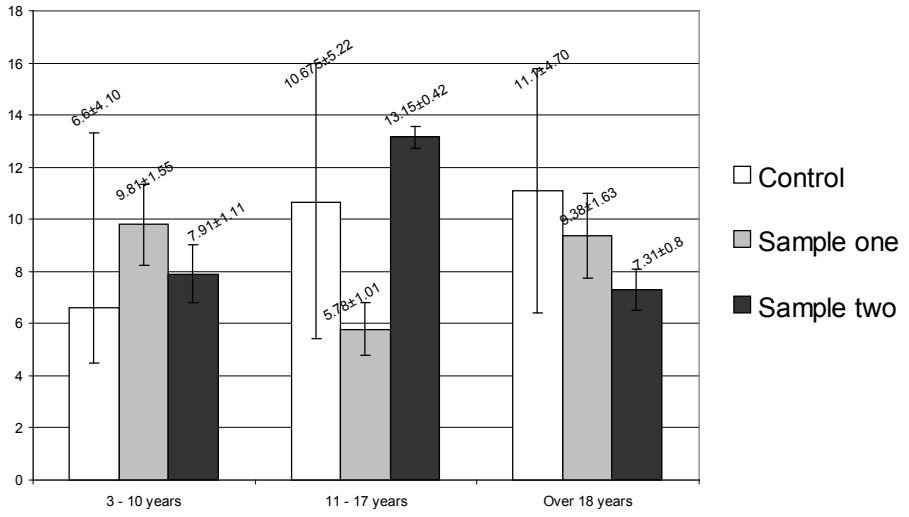

Figure 9. Immunoglobulins G in burned patients.

their important function in the immune response regulation and damaged tissue recovery in burns. It is considered that the pathological activation of the circulating phagocytes in burns is a distinguishing mechanism of phagocyte disorders.[21] Our results showed defects in the phagocyte function up to day 7 after burn. Some authors did not find that kind of disorders in their patients in the early phase of burn injury.[16] Dolgushin II et al. reported interesting data on the role of neutrophils in the immune response regulation and damaged tissue recovery after burn on a rat model. They established suppressed phagocyte function (especially in the neutrophils), suppressed immune response ability to antigens as well as lower recovering ability.[22] Sayeed M. reported hyperactivation and signalling of the neutrophils towards leukocytes as well as elevated production of oxidases accompanying tissue damages in burned patients. It is actually the extent of the neutrophil hyperactivity that signals the alteration – a hostile inflammatory response after burn shock.[23] Acquired defects in the neutrophils, monocytes or macrophages may lead to infection of the thermal trauma and hence, to elevated morbidity and mortality rates.[21]

Studies of the phagocytic system give information not only about deficiency or activation of the anti-infection protective activity. Phagocytic activity is also associated with the cell-mediated and the humoral immune response.[24,25] The percentage of the lymphocytes in the three age groups was lower than the reference values in samples one and samples two. It was, respectively, for the 3-10-year group 24.66 ± 5.11% in sample one and 27.71 ± 5.29% in sample 2 vs 45.00 ± 15% ($P > 0.05$); for the 11-18-year group - 16.00 ± 2.51% in sample one and 14.33 ± 4.09% in sample two vs 30 ± 14% ($P > 0.05$) and for the over-18-year group - 19.73 ± 3.05% in sample one and 17.87 ± 2.38% in sample two vs 30 ± 14% ($P > 0.05$) (Fig. 6). This was probably due to disorders of the cell immune response with the leukocytes as a leading factor. Other authors reported decreased leukocyte total number even in the first week after burn.[26] Detailed examination of the immune response in severe burn injuries demonstrated that it was a disorder of the cell immune response which was strongly suppressed in fact.[27] In 2005, Burd A, et al. also underlined that post burn traumas were accompanied by a decreased cell immune response and one or more organ dysfunction.[28] An interesting study (on mice) presented by Elizabeth J, et al. showed that after burn injuries the macrophages produced a higher quantity of mediators including interleukin-6 and that caused suppressed cell mediated immune response. Recognition of the cell immune response mechanisms after burn may lead to the elaboration of a therapeutic scheme for the treatment of burned patients.[29]

IgA, regardless of the age group, showed elevated values without significant differences in comparison with the reference values in samples one and two. The increase in the values in samples two was not significant (Fig. 7).

IgM were lower in both samples compared with the controls: in the 3-10-year group they were 0.82

± 0.07 in sample one and 0.84 ± 0.13 in sample two vs 1.25 ± 0.75 (P > 0.05) in the controls and in the over-18-year group they were 0.84 ± 0.13 in sample one and 0.99 ± 0.15 in sample two vs 1.31 ± 0.90 (P > 0.05) in the controls. In the 11-17-year group the values in samples one (1.32 ± 0.19) and two (1.25 ± 0.37) were similar to those of the controls (1.15 ± 0.65) (Fig. 8).

IgG did not show considerable deviations in comparison with the control group. They became higher in the 11-17-year group (13.15 ± 0.42) vs the control (10.675 ± 5.22) (P > 0.05), while in the 3-10-year group and the over-18-year group they became statistically insignificantly lower (P > 0.05) (Fig. 9).

Similar insignificant alterations in IgA, IgM and IgG absolute values were found by other authors too.[30] Abnormal immunoglobulin production and antibody functioning disorders were reported in a 20-year study on the humoral immune response in patients with thermal traumas.[31] Another study of burned children established significant elevation of IgM, IgG and IgA levels on day 7 after burn.[32] Based on monitoring of the immunoglobulin levels at admission and two weeks later a hypothesis was put forward on the possibility of prophylactic and therapeutic immunoglobulin treatment that might improve the prognosis of patients with burn injuries.[33]

CONCLUSIONS

A study on the immunological and microbiological findings in patients with burn injuries was conducted for the first time in our country. It is the decrease in PI and PN that proved to be of importance for the evaluation of wound infection risks. The increased total number of the leukocytes as well as the tendency of elevation of the neutrophil levels in burned patients proved to be signs of a bacterial infection and the prompt launch of the phagocytic mechanisms. The lower percentage of the lymphocyte compared with those in the healthy controls suggested a possible participation of the cell-mediated immune response as well. The characteristic humoral mechanisms of the immune response (serum immunoglobulins) in patients with thermal injuries altered very slightly. On the basis of the data obtained in patients with thermal traumas we recommend a mandatory battery of microbiological tests, PI, PN, leukocyte number, differential blood test and serum immunoglobulins with a view to applying an adequate antimicrobial and immune modulating treatment.

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ИММУНОЛОГИЧЕСКИЕ И МИКРО-БИОЛОГИЧЕСКИЕ ИССЛЕДОВАНИЯ ПАЦИЕНТОВ С ТЕРМИЧЕСКИМИ ТРАВМАМИ

Й. Стоилова, И. Хайдушка, М. Мурджева, И. Трайков, Т. Попова, А. Кеворкян

РЕЗЮМЕ

Восприимчивость к инфекции у пациентов с термическими травмами связана с нарушениями гуморального и клеточно-медиированного иммунитетов.

***Цель**: Исследование ставит себе целью в динамике изучить фагоцитарные показатели, как и уровни общих сывороточных иммуноглобулинов у пациентов с термическими травмами в связи с высоким риском колонизации и инфекции раны от обжигания.*

***Пациенты и методы**: За период декабрь 2004 – апрель 2005 г. в динамике у 30 пациентов с обжиганиями исследовали пока-затели фагоцитарной системы и гуморального иммунитета: фагоцитарный индекс, фагоцитарное число,нитроблаутетразолиум хлорид тест, число лейкоцитов в периферической крови, процент нейтрофильных лейкоцитов (дифференциальная картина крови), сывороточные уровни иммуно-глобулинов A,G , M и процент лимфоцитов. Проведены и микробиологические исследования секрет ран с применением конвенциональных методов.*

***Результаты**: От установленных изменений в фагоцитарных показателях относительно риска инфекции раны значение имеет снижение фагоцитарного индекса и фагоцитарного числа. Повышенное общее число лейкоцитов, как и тенденция повышения нейтрофилов у пациентов с термическими травмами говорит о наличии бактериальной инфекции и о немедленном включении механизмов фагоцитоза. Более низкий процент лимфоцитов по сравнению с этим у лиц контрольной группы показывает и возможное включение клеточно-медиированного иммунитета. Специфические гуморальные ме-ханизмы иммунного ответа (сывороточные иммуноглобулины) у пациентов с указанными травмами не изменяются существенно.*

***Заключение**: Анализированные данные могут помочь при оценке риска колонизации и инфекции раны вследствие обжигания у пациентов с термическими травмами. Авторы хотели бы рекомендовать обязательный комплекс микробиологических исследований и некоторые иммунологические исследования (ФИ – фаго-цитарный индекс, ФЧ – фагоцитарное число, число лейкоцитов, ДКК – дифференциальная картина крови и сывороточные иммуноглобулины) с целью включения адекватной антимикробной и иммуномодулирующей терапии.*

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TACKLING RARE DISEASES AT EUROPEAN LEVEL: WHY DO WE NEED A HARMONIZED FRAMEWORK?

Domenica Taruscio, Annalisa Trama, Rumen Stefanov[1]
National Centre Rare Diseases, Istituto Superiore di Sanità, Roma, [1]Department of Social Medicine and Health Management, Medical University of Plovdiv, Bulgaria

ABSTRACT

Since 1999 the European Commission has gradually developed a proactive approach towards rare diseases (RD). Despite the progress made over the last years, a comprehensive and evidence based approach is still missing in many EU Member States (MS), leading to an incomplete and often inadequate framework to address rare diseases. Healthcare systems in EU MS differ to great extent among countries in respect to their organization and funding. In general, they are not ready to face the specific problems and needs of people with rare diseases for possible prevention, timely diagnosis, adequate treatment and rehabilitation. Access to new advanced treatment and approved orphan drugs by EMEA is also a big challenge for many MS. A public health approach is needed to properly tackle rare diseases. It is a while that the idea of a comprehensive approach addressing the different challenges of rare diseases is under discussion. In our opinion, the first step to build a comprehensive approach is to properly plan the activities to undertake accordingly to needs, gaps and resources available in a Country. It is therefore important to develop a strategic plan.

Adopting a strategic planning approach to rare diseases implies taking advantage of ongoing actions and building on it to adjust, re-orient or expand the response. So far only France has developed a national strategic plan for rare diseases, Bulgaria is in the process of approving its national plan for RD and Spain is in the process of developing it. In this context, considering the importance of developing national plans for RD, it would be very useful to develop recommendations for RD national plan development in order to provide an instrument to support Countries in designing their national plans. The three MS initiatives presented in this paper confirmed the availability of great experiences and expertises among many EU MS and supported the idea that all these different experiences available at the EU level should form the basis for developing recommendations on how to develop strategic plans for RD. The recommendations will provide an instrument to support Countries in designing national plans contributing to the development of a harmonised and evidence based approach for addressing RD in EU MS. The elaboration of a European Commission Communication on rare diseases will ensure that common policy guidelines are shared everywhere in Europe. The availability of recommendations for developing national plans on rare diseases will link MS efforts with a common strategy at European level.

Key words: *rare diseases, orphan drugs, health policy, European legislation*

Rare diseases (RD) are diseases of low prevalence: not more than 5 per 10000 persons according to the EU definition. It is estimated that between 5000 and 8000 rare diseases are presently recognised, affecting more than 30 million people in the EU. RD are characterised by broad diversity of symptoms that vary from disease to disease and also within the same disease. The impact on life expectancy varies from one disease to the other; 80% of RD are recognised to have genetic origins, however, for most RD the pathogenic mechanisms are still unknown. Despite this great diversity, RD have common characteristics and challenges: chronic, often degenerative and life-threatening; disabling, in most cases devoid of effective treatments; difficult to manage; insufficient information and scientific

Correspondence and reprint request to: Domenica Taruscio, National Centre Rare Diseases, Istituto Superiore di Sanità, Roma
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knowledge leading to delay in diagnosis and inappropriate healthcare; high cost of treatment; inequities in availability of treatment and care.

Since 1999 the European Commission has gradually developed a proactive approach towards RD. This includes the Orphan Drugs Regulation[1], as well as being priority topics in the 7th Framework Programme of Research & Development and in the Community Action Programme on Public Health.[2,3] In addition, some Member States (MS) have developed specific policies on RD and orphan drugs at national level. However, the situation in EU MS is very heterogeneous: many countries have public-funded structures for RD; however, MS have different policy attitudes; epidemiological data at EU level are scarce; there is poor collaboration among MS in the research field and more effort should be given to exploit the role of patients' organisations[4] (Table 1).

Despite the progress made over the last years in the field of RD, a comprehensive and evidence based approach is still missing in many EU MS leading to an incomplete and often inadequate framework to address RD. It's a while that the idea of a comprehensive approach addressing the different challenges of RD is under discussion; however, only few MS are trying to transfer this concept into co-ordinated programmes.

Italy has solid national policies, France has a national strategic plan for RD, Bulgaria is in the process of approving its national plan for RD, Spain and Portugal are in the process of developing it. The Netherlands Orphan Drugs Steering Committee is an innovative example for addressing orphan drugs issues. UK and Sweden are actively involved in EU discussion on orphan drugs and networks of reference centres for RD.

However, the healthcare systems in EU MS differ to a great extent among countries in respect to their organization and funding. In general, they are not ready to face the specific problems and needs of people with RD for possible prevention, timely diagnosis, adequate treatment and rehabilitation. Access to new advanced treatment and approved orphan drugs by EMEA is also a big challenge for many countries.

Previous and current EU funded projects addressed major RD challenges promoting exchange of information, developing surveillance system, building public policy on RD. In addition, the Rare Diseases Task Force at the DG-SANCO provides a forum for exchange of views and experiences.

There is consensus among stakeholders that a comprehensive public health approach is needed to provide, building on the current and past experiences, an adequate response to RD challenges and patients needs.[5]

In our opinion, one of the first steps to build a comprehensive approach supporting, at operational level, a public health policy is to properly plan the activities accordingly to needs, gaps and resources available in each Country. It is therefore important to develop a strategic plan for RD. EU MS are different, culturally, economically and differences do exist with regard to the health care system organisation and functioning. An appropri-

Table 1. Status of rare diseases in some EU Member States

Actions	BE	DE	DK	EE	ES	FR	IT	NL	SW	UK
National plans/national centres			X		X	X	X			
National networks / national registries		X			X	X	X			
Public funded structures on RD	X	X	X	X	X	X	X	X	X	X
Steering Committee on RD at Ministry level						X	X			
Steering committee on Orphan Drugs			X			X	X	X		
Databases on RD				X	X	X	X	X		
Databases on orphan drugs		X		X	X	X	(X)	X		
Research: specific schemes	X	X				X	X	X		
Research: RD as priority topic				X	X	X	X		X	
Public support to patients' organisations			X	X	X	X	X	X	X	

Source: European Conference on Rare Diseases, Luxembourg, 2005 [4]

ate strategy for tackling RD should be developed on the basis of the gaps, needs and opportunities specific of each individual Country: the development of a strategic plan for RD should be **a Country driven process**.

Adopting a strategic planning approach to RD implies taking advantage of ongoing actions and building on it to adjust, re-orient or expand the response. Strategic planners want to draw from the experience that has been built up so as not to reinvent the wheel at every turn. Taking examples from some national experiences (including a comprehensive set of best known practices) will save time and increase the chances of achieving success. With regard to RD there are several MS from which experiences and best practices can be taken. Three examples follow:

Italy

The Italian Government promulgated Regulations (Ministerial Decree 279/2001, «Institution of the national network of rare diseases» [6]) on RD inspired by the 1998-2000 National Health Plan and in line with the decision of the European Union. The National Health Plan emphasises the importance of supporting scientific research on RD, promoting prevention activities as well as making available appropriate and timely diagnosis and treatment for RD patients. The Ministerial Decree 279/2001 provides to exempt RD patients from health service costs establishing a national network of regional and inter-regional referral centres, including the institution of the National Register of RD. Certified Centres are expressly identified by the Regions among those possessing:

- documented experience in diagnostic and/or specific therapeutic activities for single or groups of rare diseases
- adequate support structures and complementary services (emergency services and services for biochemical and genetic-molecular diagnosis)

After several years from the publication of the Decree on RD, regional centres have been identified in nearly the entire national territory. The list of the regional centres identified in the national territory is available on the National Centre Rare Diseases (ISS) website (http://www.iss.it/cnmr). A dedicated database has been developed to provide information on the regional centers (by geographic location and/or by RD) to patients and clinicians.

In July 2002, as part of the State-Regions conference a permanent inter-regional technical group including Regional Representatives, the Ministry of Health (MoH) and the National Health Institute (Istituto Superiore di Sanità - ISS) was established with the objective of optimising the function of the regional networks and to safeguard the principle of equality in healthcare for all citizens.

It is worth noting that following the Ministerial Decree, RD and the implementation of the national network have been included as priorities in all Italian National Health Plans (2003-2005; 2006-2008).

The National Register of RD was established at the National Centre Rare Diseases (ISS) in order to contribute to the national and regional health planning and to ensure the surveillance of RD. In particular the National Register aims at defining the prevalence/incidence of RD, identifying possible risk factors and fostering collaboration among healthcare providers.

The flow of the epidemiological data is presented in Fig. 1.

Epidemiological data are collected from all regional centres officially identified by the Region. From the different centres data are sent to the regional coordination centre that will then send data to the national register of the ISS. When a regional coordination centre is not yet available, the Region will be responsible for sending the data to the ISS. This flow has been recently confirmed by a specific agreement signed at the State-Regions Conference (May, 2007)[7].

The National Centre Rare Diseases, established at the ISS, in addition to ensuring the functioning of the National Register, leads several activities ranging from laboratory research on specific RD (including rare tumors), to prevention, and information to health care professionals, patients, families and to the general population [8].

The main activities of the Italian National Centre Rare Diseases (http://www.iss.it/cnmr) are briefly described as follows:

- Research activities on selected RD
- Quality assurance and standardization of genetic tests
- Primary prevention of congenital defects and folic acid
- Surveillance of rare diseases at national level: Rare Diseases National Register
- National register of orphan drugs
- Development of guidelines for diagnosis and treatment of specific RD

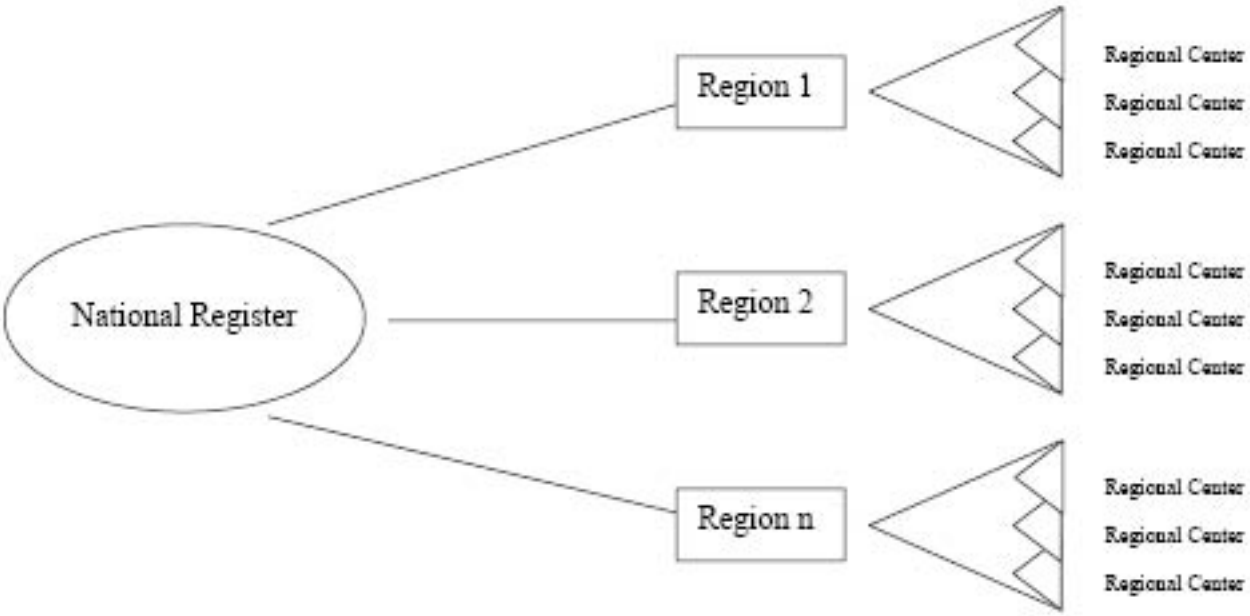

Figure 1. Italian National Register of RD: epidemiological data flow.

- Studies on accessibility and quality of health and social services for the patients with rare diseases
- European project (NEPHIRD http://www.iss.it/cnmr; E-Rare http://www.e-rare.eu; TEDDY http://www.teddynoe.org)
- Development of narrative medicine
- Provision of information to patients, families and to the general population
- Training and continuous education of health care operators

With regards to the research, it is worth mentioning the important initiatives of the Italian Drug Agency (AIFA) that launched two calls for national projects (in 2005 and in 2006) including orphan drugs among the priority areas. The Law 326/2003 requires that the Italian pharmaceutical companies donate 5% of their promotional expenditure to a specific fund established within the AIFA. These funds have been used to launch the calls for proposal representing an innovative scheme to support research, care and treatment of diseases too often neglected.

In addition, the MoH and the ISS launched two national calls for proposals (in 2004-2006 and in 2006-2008) on RD in the frame of a bilateral Italy (ISS-National Centre Rare Diseases) – USA (NIH-Office for Rare Diseases) agreement on joint research and development of public health actions. Moreover, the MoH and the ISS contributed with financial support to the first trans-national call for European projects on RD launched within the framework of the E-Rare project.

It is worth mentioning that the Italian Minister of Health has formally established a "Consulta" of RD patients organisations in order to provide a forum for strengthening the dialogue between patients organisations and MoH. The Consulta met for the first time on June 5th 2007 with an open and healthy discussion in presence of the Minister. The Consulta, upon request of the Minister, is currently working with the scientific technical support of the National Centre Rare Diseases on the definition of priority needs and gaps that need to be addressed by the Ministry of Health.

France

Over the last ten years, France has taken many initiatives in the area of RD. The collaboration with patients' associations was strategic and of primary importance, in particular the engagement and collaboration with the *Association Française contre les Myopathies* (AFM). Major initiatives follow:

1. creation of an orphan drugs mission (1995);
2. financing of a telephone help line service for patients (1995) and of an online information server,
3. Orphanet (1997) (http://www.orpha.net)
4. financing of clinical research on rare diseases in the programme of clinical research in hospitals (since 2001);
5. providing access to the list of clinical trials by the *Agence française de sécurité sanitaire des*

produits de santé (AFSSAPS5, 2002);
6. creation of a *Groupement d'Intérêt Scientifique - Institut des Maladies Rares* (2002).

In this framework, France was the first EU MS to develop a National plan for RD. The National Rare Diseases Plan 2004 – 2008 as part of the 9 August 2004 law relating to public health policy, is seen as the instrument to develop, reinforce and bring coherence to the different initiatives undertaken and ongoing in France. Its priority is "to ensure equity in the access to diagnosis, to treatment and to provision of care" for people suffering from a rare disease through ten strategic priorities:

1. Increase knowledge of the epidemiology of rare diseases
2. Recognise the specificity of rare diseases
3. Develop information for patients, health professionals and the general public concerning rare diseases
4. Train professionals to better identify them
5. Organise screening and access to diagnostic tests
6. Improve access to treatment and the quality of healthcare provision for patients
7. Continue efforts in favor of orphan drugs
8. Respond to the specific needs of accompaniment of people suffering from rare diseases and develop support for patients' associations
9. Promote research and innovation on RD , notably for treatments
10. Develop national and European partnerships in the domain of RD

Accordingly to the National Plan, "labelled centres of reference" will be established by the end of the plan. These centres, made up of multidisciplinary teams, have the following missions:

- to facilitate diagnosis and define a strategy of therapeutic and psychological care and of social accompaniment;
- to define and circulate care protocols, in association with the National union of national health insurance funds (*Haute Autorité de Santé* and the *Union Nationale des Caisses d'Assurance Maladie* (UNCAM);
- to coordinate research and participate in epidemiological surveillance, in association with the *Institut de Veille Sanitaire* (InVS);
- to participate in training and information initiatives for health professionals, patients and their families, in association with the *Institut National de Prévention et d'Education pour la Santé* (INPES);
- to manage and coordinate the networks of health and socio-medical care providers
- to be the main interlocutors for the ministries and patients associations.

Up to June 2007, 132 Centres have been established (Newsletter of the Rare Diseases Task Force, June 14, 2007).

Other interesting and strategic actions are those related to priority one: increase knowledge of the epidemiology of rare diseases. Different and parallel activities are foreseen in order to gather information on rare diseases:

- develop a nomenclature and a classification adapted to rare diseases, in collaboration with the international bodies: the World Health Organisation and the European bodies, notably the Rare Disease Task Force which brings together European experts on rare diseases;
- support centre of reference for the setting up of a collection of consistent and appropriate data, databases, and appropriate statistical methods, with the collaboration of the hospital-based public health services;
- mobilise all existing databases
- continue the development of registers of rare diseases
- set up an epidemiological study on mortality due to rare diseases using death certificates.

Following up the implementation of the National Plan, a committee has been created under the auspices of the Ministry of Health and Social Protection and in association with the different partners including patients associations.

An evaluation of the proper distribution of the funds allocated to different establishments in the Plan will be undertaken regularly. In addition, the committee will also be responsible for evaluating the impact of the different measures and for making propositions. The importance of such evaluation and continue monitoring is straightforward.

Bulgaria

Bulgaria is an example of a new MS with a substantial work and activities in the area of RD policy and organization done in a very short time. At the end of 2004, the Information Centre for Rare Diseases and Orphan Drugs (ICRDOD) was started as a project of a non-government non-profit organization (www.raredis.org).[9] This Centre stressed on the importance of working in 6 main directions – information, education, awareness, support,

networking and lobbying. As a result of its activities, a working group has been established at the Ministry of health in 2006, that drafted and submitted a project for National Plan for Rare Diseases (NPRD). This national plan is currently (by July 2007) under review and approval by the Bulgarian Ministry of Health. The aim of the Plan is to create a uniform national health policy, ensuring that the citizens of Republic of Bulgaria are provided with prevention, timely diagnosis, adequate treatment and rehabilitation of rare diseases.

The National Program for RareDiseases foresees the following 7 priority tasks:

1. Establishing an adequate institutional framework and mechanisms for ensuring timely prevention, diagnosis, optimal treatment and rehabilitation for the patients with rare diseases in Bulgaria.
2. Providing epidemiological data about rare diseases in Bulgaria by creation of a National register.
3. Training of medical students and doctors for increasing their knowledge and professional qualification in the field of early diagnosis and prevention of rare diseases.
4. Stimulating the scientific work and research on rare diseases and establishing a referent research centre for rare diseases in Bulgaria.
5. Organizing a national awareness campaign to introduce rare diseases and their prevention to the society.
6. Support for establishing and working of patient associations for rare diseases.
7. Close collaboration with other EU member states and Rare Diseases Task Force (RDTF) at DG SANCO – European commission.

For the realization of NPRD, the current healthcare system must be amended and adjusted. The following institutional framework is going to be established (Fig. 2):

- National Consultative Council for Rare Diseases (NCCRD) at the Ministry of Health
- Expert groups at the medical scientific societies
- National network from nine referent counselling offices for rare diseases in the university hospitals
- National register of patients with rare diseases

The expected flow of patients as a result of the realization of the proposed institutional framework is presented in Fig. 3. As a result of this national

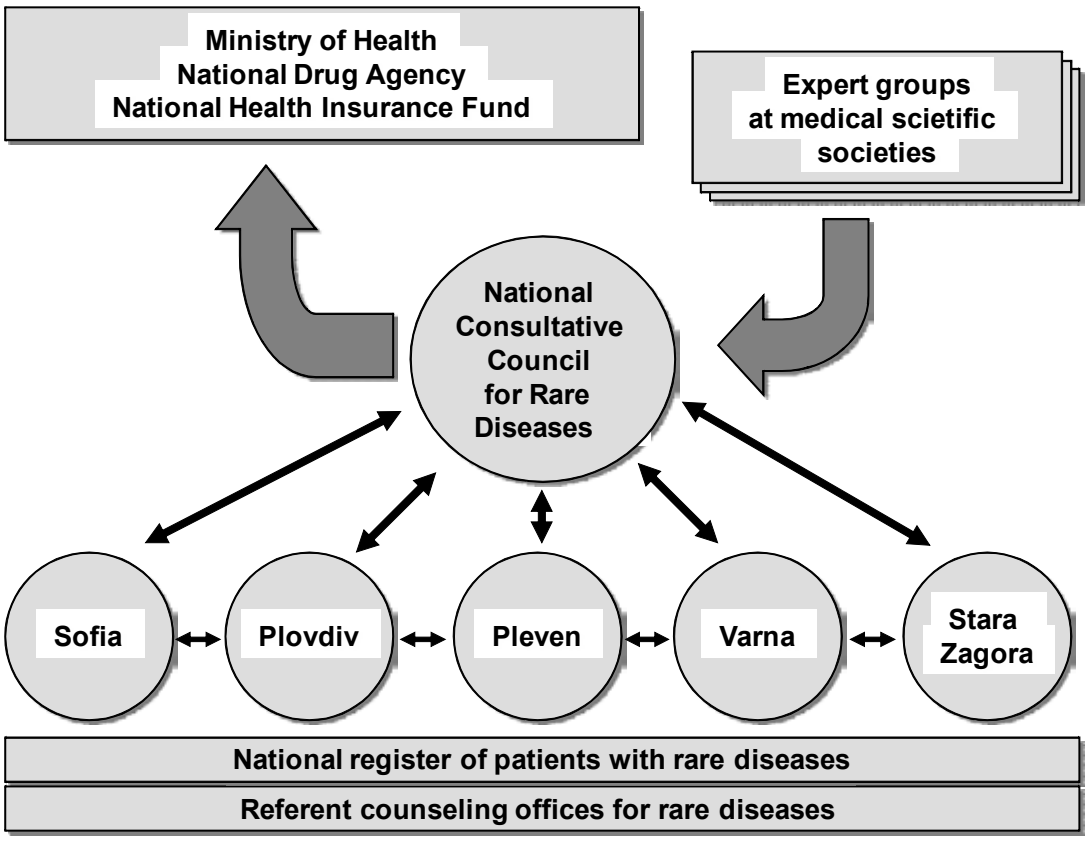

Figure 2. Institutional framework for implementation of the National Program for Rare Diseases in Bulgaria.

plan, a uniform national health policy, ensuring that the citizens of Bulgaria are provided with prevention, timely diagnosis, adequate treatment and rehabilitation of rare diseases is expected, providing boost in several directions - national policy for reimbursement of prevention, diagnosis, treatment and rehabilitation of patients with rare diseases; national network for rare diseases; national register of patients with rare diseases, providing correct epidemiological data about prevalence of rare diseases in Bulgaria; identified and labelled expert centres in Bulgaria for prevention, diagnosis, treatment and rehabilitation of a single rare disease or group of diseases; establishing of effective expert groups at the medical scientific societies; increasing the knowledge and professional qualification of medical students and doctors in the field of early prevention and diagnosis of rare diseases; significant increase of scientific work and research in the field of rare diseases and establishing a referent research centre for rare diseases in Bulgaria; awareness of rare diseases and the ways for their prevention to society; attracting patient associations and NGOs as active partners in the healthcare system; establishing collaboration with other EU member states and raising the international prestige of Bulgaria as a country actively engaged with the problems of people with rare diseases.

These examples confirm the availability of great experiences and expertises among EU MS. Thus, it is important and is becoming urgent, in our opinion, to identify and describe best practices in order to share information, models and data on effective strategies to address RD. A critical analysis should be performed of the functioning of the current activities in key areas of intervention for RD (institutional framework; provision of care; surveillance system; support to patients organisations); this will serve as a guidance to support EU MS in the development of their national plans for RD, targeted on national needs as well as based on EU harmonized general criteria. The development of national plans, based on evidence and best practices, will contribute to putting into practices, in individual MS, the EU priorities/policies: regulation on orphan drugs and paediatric drugs regulation, patient mobility, active participation of patients' groups.

In this context, it would be very important to develop recommendations for RD national plan development in order to provide an instrument to support Countries in designing national plans. The recommendations should propose best practices in

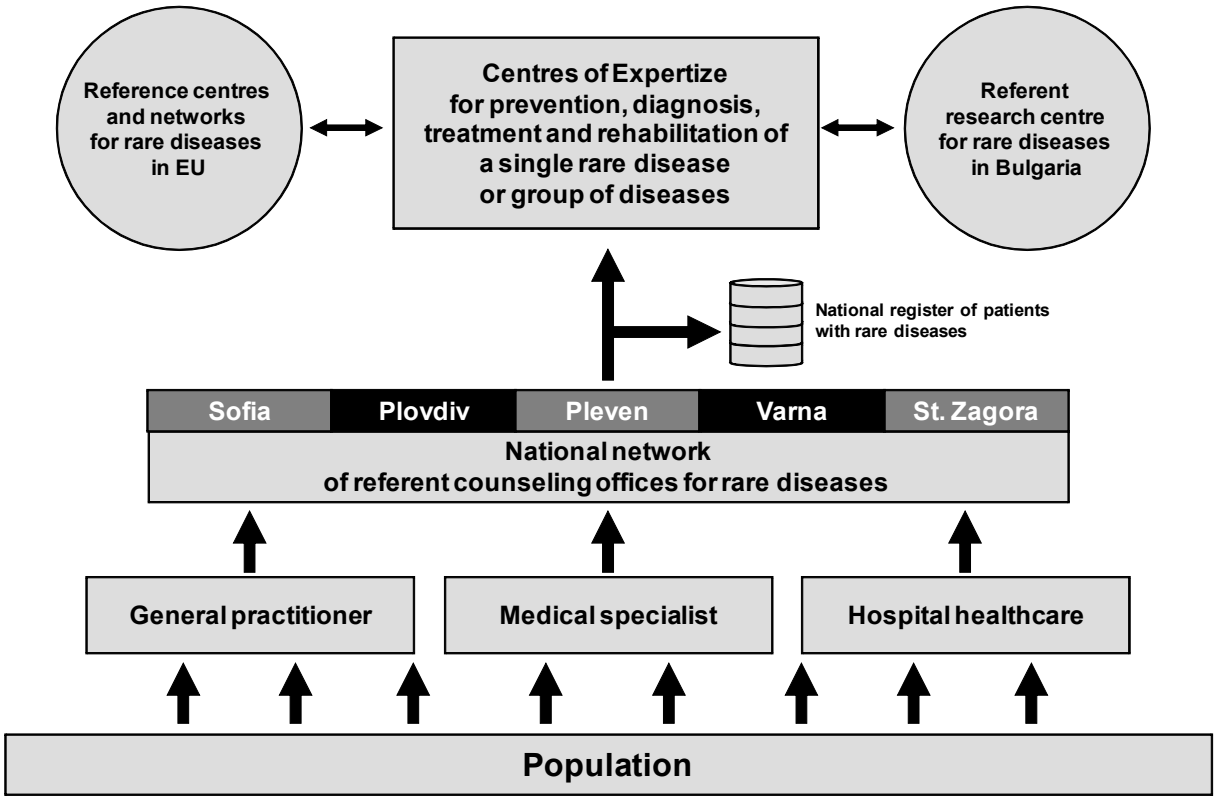

Figure 3. Expected flow of patients with rare diseases.

order to offer possible effective models to consider when developing a programme for RD. This will contribute to the development of an harmonised and evidence based approach for addressing RD in EU MS, through consensus on an overall minimum common strategy for RD. The recommendations will represent an EU-based tool for MS to develop effective national responses designed to meet the specific needs of the country and to improve the health of the citizens. The recommendations aiming at introducing the main concepts of strategic planning should remain flexible enough in order to be adaptable for planning in different contexts and serve as a practical assistance for planners in different MS.

The European Commission, DG SANCO, was requested by the Council of Ministers to prepare a Communication on RD during the first semester of 2007. This Communication is expected to be submitted to the European Parliament and the Council of Ministers during the second semester for possible adoption at the end of 2007 (RDTF Newsletter, 21 December 2006). The elaboration of a Commission Communication on RD will ensure that common policy guidelines are shared everywhere in Europe. In this framework, we believe that it will be essential to create a culture of strategic planning for RD, the availability of recommendations for RD national plan development promoting national policies and best practices for RD within EU MS will link national efforts with a common strategy at European level. This "double-level" approach will ensure that progress is globally coherent and follows common orientations throughout Europe.[5] This will ultimately contribute to ensuring access to diagnosis, treatment and care for patients with RD.

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СПРАВЛЕНИЕ С РЕДКИМИ БОЛЕЗ-НЯМИ НА ЕВРОПЕЙСКОМ УРОВНЕ: ПОЧЕМУ СУЩЕСТВУЕТ НЕОБХОДИ-МОСТЬ В ГАРМОНИЗИРОВАННОЙ РАМКЕ?

Д. Тарушо, А. Трама, Р. Стефанов

РЕЗЮМЕ

Начиная с 1999 года Европейская комиссия (ЕК) постепенно развивает проактивный подход к редким болезням (РБ). Несмотря на прогресс в последние годы, все еще во многих странах-членах Европейского союза (СЧЕС) отсутствует целостный и основывающийся на доказательстве подход, что приводит к незаконченной и часто неадекватной институционной рамке по отношению к редким болезням. Системы здравоохранения в СЧЕС значительно отличаются друг от друга по отношению своей организации и финансирования. Одним словом, они не готовы встретить специфические проблемы и нужды людей с редкими болезнями в целях превенции, своевременного диагноза, адекватного лечения и реабилитации. Доступ к новым современным методам лечения и к одобренным Европейским Лекарственным агентством (EMEA) лекарствам-"сиротам" тоже представляет большим вызовым для многих СЧЕС. Необходим общественноздравоохраняющий подход, направленный к различным аспектам, связанным с редкими болезнями.

По мнению авторов, первый шаг в этом направлении - это правильная планировка необходимых деятельностей в зависимости от конкретных нужд, пропусков и наличных ресурсов в каждой конкретной стране. С этой целью важно разработать стратегический план.

Принятие стратегического планирования насчет редких болезней требует использования в качестве базы уже накопленного опыта; эта база впоследствии следует быть адаптированной, преориентированной и обогащенной.

До настоящего момента только Франция уже разработала и приняла национальную программу по проблемам редких болезней. Болгария находится в процессе одобрения своей национальной программы по этой проб-леме, а Испания – в процессе подготовки. В этом контексте, принимая во внимание исключительная необходимость в разработке национальных программ по проблемам редких болезней, особенно важным будет создание Руководства (Справочника) с рекомендациями по созданию национальных планов по борьбе с РБ в СЧЕС. Эти три европейские страны, о которых уже шла речь в настоящей статье, утверждают наличие различного опыта и экспертизы среди СЧЕС и подкрепляют тезу, что все эти различные опыты в Европейском союзе следует послужить основой при выро-ботке Руководства с рекомендациями создать такие национальные программы для РБ. Эти рекомендации окажутся вспомогательным инструментом для европейских стран при проектировании их национальных программ, что приведет к единому гармонизированному и основывающемуся на доказательствах подходу к редким заболеваниям в странах Европейского союза.

Публикация комюнике о редких болезнях со стороны ЕК будет гарантировать, что общие направления в политике здравоохранения будут распространены среди всех стран ЕС. Наличие рекомендаций по созданию национальных планов свяжет усилия отдельных стран ЕС в единой общей стратегии на европейском уровне.

Unabridged Summaries of Successfully Defended Dissertations in Medical University, Plovdiv in 2005 - 2006

CARCINOMA OF THE PENIS – EPIDEMIOLOGICAL, DIAGNOSTIC AND SURGICAL ASPECTS

Ivan Y. Dechev
Department of Urology, Medical University Plovdiv

Key words: *carcinoma of the penis, epidemiology, primary tumor surgery, lymph node dissection, survival*

AIM

The aim of the present study was to make a complex evaluation of the dynamics in the incidence of carcinoma of the penis (CP) in Bulgaria and lay down criteria for diagnosis and adequate surgical treatment.

MATERIAL AND METHODS

A retrospective descriptive epidemiological study of the incidence of CP on populational and regional level (Districts of Plovdiv, Pazardzhik and Smolyan - DPPS) and ambispective longitudinal clinical study (retrospective from 1972 through 1990 and prospective from 1991 through 2003) were performed. Data from the National Cancer Registry for newly diagnosed cases (488 cases between 1990 and 2000) and deaths (83 cases between 1995 and 2000) from CP in Bulgaria, as well as data for patients recently entered in the follow-up register (51) of the Oncological Dispensary in Plovdiv (1993-2002) were used. World standard incidence rates (numbers of patients per 100 000 males) were used for comparison. The changes in the crude and standardized incidence rate in Bulgaria and crude incidence and mortality in the region for a 10-year period were followed. Regression analysis was used to reveal the continuous tendency of the CP incidence in the population. Sixty-seven patients with histologically verified CP (primary tumor or local recurrence) diagnosed, treated and followed up in the Clinic of Urology of the Medical University Plovdiv (1972-2003) were also enrolled in the study. "Life table" method for evaluation of 5 and 10-year survival rates and Kaplan-Meier method for evaluation of the mean survival time in the regional and clinical groups of patients were used. The data were processed and analyzed using statistical software products SPSS 11.0 and Microsoft Excel 2000.

RESULTS

Although CP is a rare male malignancy in Bulgaria, the number of newly diagnosed cases has increased by 57.14% between 1990 and 2000. In 2000 CP constituted 0.34% of all malignancies in males. Both crude and standardized incidence rates increased with 0.0467 and 0.0264, respectively, per 100 000 males. The mean crude incidence for the examined period in Bulgaria was 1.07 and mean standardized morbidity – 0.67 per 100 000 individuals. The ratio between the highest crude incidence – 1.65/100 000 (1999) and the lowest crude incidence – 0.72/100 000 (1993) was 2.29, i.e., the crude incidence had increased more than twice. The incidence increased with age and most of the afflicted males were in the age range of 70 - 85. The crude incidence rate for the period 1995-2000 remained relatively constant. An interesting epidemiological phenomenon - a tendency to a parallel increase of the crude incidence and mortality rates (0.1355 and 0.0392 per 100 000 males, respectively), was observed in DPPS for the period 1993-2002. The regional sample presented with rather wide age range – from 40 to 90 years (66.35±12.08). The greatest percentage and greatest number of patients was found in the age groups 60-64 and 70-74. The situation is very alarming with respect to the early diagnosis and adequate treatment of the disease. High proportion of patients diagnosed in a stage of invasive local tumor was evident (82.35%). The stage distribution of the cases was also extremely unfavorable – more than 33% were diagnosed late (advanced stages III and IV) and only 17.65% were diagnosed in stage I. In the

Correspondence and reprint request to: Iv. Dechev, Department of Urology, Medical University, Plovdiv 15A Vassil Aprilov blvd., 4002 Plovdiv, Bulgaria

greatest proportion of cases (70.59%) single surgical treatment was applied, combine therapy was applied in 23.53% and non-treated patients in curable stages II and III were 5.88%. The mean survival time of diagnosed patients in DPPS was 7 years (95% CI [5-8]), 5-year survival rate was just about 56.86% and 10-year survival rate 50.87%.

The mean age of the clinical group patients was 62.84±7.86 (range 28-88) and the follow-up period varied from 0 to 26 years (mean 7.06±0.97, Me=4 yrs.). The mean delay in diagnosis was 14.00±2.28 months (range 1-120) and in 74.64% of the examined patients one-year delay was observed. The delay was mostly because of the patient (59.62%) and this dominance was statistically significant (u=1.998, P<0.05). The fear of invalidating operation was one of the most common reasons (32.69%) for a delayed medical request. The primary cause for delayed diagnosis by the medical doctor was a preceding dermatological treatment without histological verification of the penile lesion (34.62%).

Diagnostic biopsy of the primary tumor and modified inguinal lymph node dissection (Catalona, 1988) in patients with clinically negative inguinal lymph nodes are established as crucial and sufficient techniques for primary clinical diagnosis due to their high informative value, safety and absence of major complications. The results obtained show that localization of the primary tumor and type of local surgery appear particularly important factors for development of local recurrence and the relative risk after penis conserving operation is 3.84 times greater compared with penis amputation (RR=3.84, P=0.049). Our study determined extent of dissection and surgical approach as statistically significant factors for morbidity after lymph node dissection. The overall survival in the clinical group of CP patients and the prognostic value of some clinical and pathological characteristics of the primary tumor for the general survival were evaluated. The mean survival rate of the examined 59 patients (8 were excluded because of failed follow-up) was 12.56 years (95% CI [9.46-15.67]) and Me=14 years (95% CI [3-25]), 5-year survival rate was 62.19% and 10-year survival rate 54.39%. An algorithm for diagnostics and surgical treatment of primary tumor and regional lymph nodes in CP was made.

CONCLUSIONS

The unfavorable tendencies to morbidity rise and mortality stabilization indicate that CP is going to become a substantial home nosology in the near future. Although this study is a pilot one and one of the first on the epidemiological aspects of CP in Bulgaria and in the region, in particular, it suggests a necessity of screening programs. The data presented about the distribution, diagnosis and treatment characterize the disease as one of the most serious problems in the oncological practice in the region. The parallel increase in the morbidity and mortality in DPPS, unfavorable condition in terms of timely diagnosis and great number of cases diagnosed in late stage result in low effectiveness of the therapeutics attempts in this localization and implausibly low 5-year survival of the penile malignancy. The regularities in the local recurrences onset according to the type of surgery of the primary tumor are embedded in a specific follow-up program, especially during the first year when most of them appear (61.54%). Significant difference in the mean time of the local recurrences onset following penis conserving surgery and amputation is a criterion for the extent of the surgery of the primary tumor and can be discussed as a marker for success or failure of treatment. To reduce the postoperative morbidity a modified inguinal lymphadenectomy is recommended as a prophylactic procedure for staging the nodal status in clinically indeterminate inguinal lymph nodes. Presently, in the lack of standardized protocol for diagnostics and surgical treatment of the disease, the algorithm we introduced in our home practice appears a favorable decision for the patients allowing early stage diagnosis and adequate surgical treatment and increasing the survival rate.

CLINICAL VALUE OF THE CHANGES OF PLACENTAL ANGIOGENIC PROTEINS AND COMPONENTS OF THE IMMUNE SYSTEM IN PREECLAMPSIA

Elena D. Dimitrakova
Clinic of Obstetrics and Gynecology, University Hospital St. George, Plovdiv

INTRODUCTION

Placental ischemia as a cause for endothelial cell dysfunction is a hypothesis that provides more evidence of a wide spectrum of important molecules whose balance is crucial during pregnancy. Studies of some markers of endothelial activation have recently shed more light on the problem. The serum levels of Fms-like tyrosine kinase-1 receptor have been found to be elevated. This receptor binds the vascular endothelial growth factor (VEGF) and the placental growth factor (PLGF) and causes endothelial dysfunction by changing the angiogenic balance.

Investigation of the T-lymphocyte populations and cytokine profile as components of cell-mediated immune response sheds new light on the problem of the impaired immune adaptation and its relation to the vascular endothelial dysfunction.

AIM

To study the serum levels of placental angiogenic proteins and components of the immune system in condition of preeclampsia and assess their clinical value to determine the severity of the disorder and treatment approach.

MATERIAL AND METHODS

Seventy pregnant women aged 18-36 years were examined, 40 with preeclampsia and 30 with normal pregnancy. The control group included 30 age-matched healthy non-pregnant women.

The serum levels of Fms-like tyrosine kinase-1 and the placental growth factor were determined using the ELISA test (R&D Systems, USA). The analysis was done in the Leth Israel Deaconees Medical Centre, Boston, USA.

Immunophenotyping of the peripheral blood lymphocytes was done with laser flowcytometry. The tests were performed in the Central Clinical Laboratory of the Medical University, Plovdiv.

The serum levels of cytokines were determined with the ELISA test (Bender Medsystems Austria) in the Central Clinical Laboratory of the Medical University, Plovdiv.

The data were analyzed statistically using SPSS 11.0 for Windows.

RESULTS

I. Studies of the serum levels of sFlt-1 and PLGF in women with normal pregnancy and pregnant women with preeclampsia.

Our study results confirmed that in preeclamptic pregnant women the serum level of sFlt-1 was elevated while the level of the circulating free PLGF was low. Between-group comparison using the criterion of normal distribution showed a statistically significant difference in the serum levels of PLGF between preeclamptic women and women with normal pregnancy (CI=99.99%, t=10.08). This was due to more than fourfold lower levels of PLGF in women with preeclampsia. Statistically significant negative correlation was found between sFlt-1 and PLGF. The tendency to quick return to normal levels of the variables after delivery confirmed the placental origin of this disorder.

II. Study of the serum levels of cytokines in pregnant women with preeclampsia.

The serum levels of the proinflammatory cytokines IL-2, IL-6, and TNF-α were significantly increased in the preeclamptic pregnant women:

1. The serum level of IL-2 was increased in all pregnant women with severe preeclampsia. Statistically significant correlation was found between the serum IL-2 level and arterial blood pressure components.
2. Statistically significant correlation was found between the serum levels of IL-6, TNF-α and diastolic arterial blood pressure in the preeclamptic pregnant women.
3. There was no statistically significant difference in the serum level of the anti-inflammatory cytokine IL-10 between the preeclamptic and normal pregnancy women.

The IL-2, IL-6, IL-10, TNF-α constellation characterizes the immunological activity in pre-

Correspondence and reprint request to:E. Dimitrakova, Clinic of Obstetrics and Gynecology, Medical University Plovdiv, 66, Peshtersko shosse Str.
15A Vassil Aprilov blvd., 4002 Plovdiv, Bulgaria

eclampsia and can be used as a diagnostic and prognostic index.

III. Lymphocyte populations and subpopulations in women with normal pregnancy and preeclampsia.

The basic lymphocyte subsets (cell count/ml) were considerably changed in the examined preeclamptic women. The counts of B-lymphocytes (CD19+), total T-lymphocytes (CD2+) and immunocompetent cells (CD3+) were significantly higher in the preeclamptics than in the normally pregnant women.

T-helpers (CD3+CD4+) showed statistically significant increase in the preeclamptic women compared with those with normal pregnancy, while suppressor cytotoxic T-lymphocytes (CD3+CD8+) did not change.

On the background of considerable quantitative changes in the T-helper lymphocytes statistically significant increase was found for the suppressor-inducer lymphocytes (CD4+CD45+RA+). This was related to the stimulation of the suppressor T-lymphocytes by the massive placental antigenemia.

The proportion of the basic lymphocyte subsets (%) showed insignificant changes for B-lymphocytes, total T-lymphocytes (CD2+) and immunocompetent T-cells (CD3+). Statistically significant increase was evident in the proportion of T-helpers (CD3+CD4+) coinciding with decrease in the proportion of the suppressor-cytotoxic subpopulation (CD3+CD8+).

IV. Lymphocytes with primary killer function and activated immunocompetent lymphocytes.

The lymphocytes with MHC limited killer function (CD8+CD56+) were increased in the preeclamptics but the differences with normal pregnancy women did not reach statistical significance. The same finding was observed for the lymphocytes with MHC unlimited killer function (CD8+CD56+). The activated immunocompetent T-lymphocytes (CD3+HLA-DR+) were increased in the preeclamptic women compared with those with normal pregnancy. Statistical significance was reached both for the absolute count and relative proportion. The increased expression CD3+CD25+ validates activation and clonal proliferation of T-lymphocytes. The ratio of CD4+/CD8+ was significantly higher in the preeclamptic group than in the women with normal pregnancy. This increase is connected with deviations in the helper and suppressor-cytotoxic subpopulation T-cells and indicates that the immune potential as a whole in preeclampsia is activated with dominance of the helper subset.

HLA-antigen (class II MHC) is a marker of the activated T-cells. The significant increase in CD3+HLA-DR+ indicates availability of lymphocytes prepared for antigen maturation, i.e., vigilance of the organism to react to antigen stimulation.

V. Hypothesis of the preeclampsia pathogenesis in pregnant women

Using literature data of the preeclampsia pathogenesis and our own results we put forward a hypothesis of the pathogenetic circuit of this toxicosis. Due to anomalies in the trophoblasts and/or myometrium the appropriate trophoblast penetration into the uterus is restrained. The perispiral trophoblast deficit holds back the placental angiogenesis. Placental hypoperfusion that does not meet the functional requirements develops. This placental hypoperfusion activates mechanisms directed at optimization and rearrangement of the blood flow function-structure mass in the placenta. One of these mechanisms is increased production of Fms-like tyrosine kinase (sFlt-1). The increased concentration of this kinase in the serum lowers the level of PLGF in pregnant women by the mechanism of binding and elimination. The impaired balance disrupts the endothelial function. The generalized endothelial dysfunction in preeclamptic women is manifested with vasoconstriction, increased vascular permeability, decreased platelet resistance and increased antigen expression. These pathogenic factors determine the functional manifestations of preeclampsia – arterial hypertension, edema, proteinuria, and HELLP syndrome.

The endothelial dysfunction activates secondarily, by antigen expression, the cytokine production (IL-2, IL-6, TNF-α) and the immune reactions in pregnant woman. These in turn can further hamper trophoblast invasion and aggravate the observed preeclamptic manifestations – a peculiar vicious circle is formed. Some women adapt to these changes and do not develop preeclampsia. In others the vicious circle gains strength and preeclampsia develops.

CONCLUSIONS

On the basis of the observed changes in the serum concentrations of placental angiogenic proteins (sFlt1, PLGF), lymphocyte populations and subpopulations and serum cytokines (IL-2, IL-6, TNF-α, IL-10) we think that these could be used in a constellation with well-established clinical and laboratory methods for studying the biological and clinical characteristics of preeclampsia.

DIABETES MELLITUS AND MICROELEMENTS – EXPERIMENTAL AND CLINICAL INVESTIGATIONS

Stefka V. Vladeva
Clinic of Endocrinology and Metabolic Diseases, St George University Hospital, Plovdiv

INTRODUCTION

There are conflicting, insufficient data about the role of the essential microelements selenium, chromium, zinc, vanadium and manganese in the versatile manifestation of type 2 diabetes mellitus. We therefore developed and conducted a number of experimental and clinical studies on this issue.

AIM

We aimed at studying experimentally and clinically the role of the trace elements selenium, chromium, zinc, vanadium and manganese in the manifestation, course and treatment of type 2 diabetes.

Tasks

A. *Experimental* – a study of the extrareceptor insulin-like effect of selenium, chromium, zinc, vanadium and manganese on the smooth muscles of Wistar rats.

B. *Clinical* – a study of the serum levels of selenium, chromium, zinc, vanadium and manganese in type 2 diabetes patients and their correlation with the metabolic control, duration of disease and age of patients; evaluation of the glycemic control prior to and after adding selenium, chromium and zinc; analysis of the serum levels of immunoreactive insulin and HOMA-index in patients with type 2 diabetes prior to and after selenium, chromium and zinc; elucidation of the total antioxidant status in healthy subjects and type 2 diabetes patients before and after adding selenium, chromium and zinc; follow-up of the lipid status before and after selenium, chromium and zinc intake; determination of the reference values of chromium and vanadium.

MATERIAL AND METHODS

The experimental part included measuring the contractile activity of smooth muscle from rat stomach after administration of selenium, chromium, zinc, manganese, vanadium and substances, the microelements which affect the membrane voltage-gated calcium channels.

The clinical study included 130 patients with type 2 diabetes and 35 healthy control subjects. It was conducted in two stages – before and after a 60-day administration of selenium, chromium and zinc. Simultaneously, several laboratory variables were examined: serum levels of selenium, chromium, zinc, manganese and vanadium, blood sugar, glycated hemoglobin A1c, immunoreactive insulin, HOMA-index, total antioxidative status, total cholesterol, HDL-cholesterol, triglycerides, and the index of lipid atherogenic risk.

RESULTS

The experimental study proved cogently that selenium, chromium, zinc, manganese and vanadium increase the intracellular calcium concentration. Inside the cell calcium ions are an important factor for realization of postreceptor insulin effects. These microelements can prove a valuable and highly specific achievement for the pharmacotherapeutic regulation of the insulin effect. We suppose that the insulin-like effects of selenium, chromium, zinc, manganese and vanadium observed by us can be useful for early prophylaxis of type 2 diabetes in subjects with signs of metabolic syndrome.

The clinical findings indicated marked deficiency of the examined microelements in patients with type 2 diabetes. After a 60-day administration of selenium, chromium, zinc, manganese and vanadium improvement in the parameters of carbohydrate and lipid metabolism, as well as in the total antioxidative status, was observed.

CONCLUSIONS

Within certain concentration limits selenium, chromium, zinc, manganese and vanadium increase cellular influx of calcium and lead to increase in the level of intracellular calcium ions. Increased cytoplasmic calcium is a powerful secondary messenger in the insulin effect, which allows the studied microelements to take an important place in the treatment of type 2 diabetes.

Serum levels of selenium, chromium, zinc, manganese and vanadium are significantly lower

Correspondence and reprint request to: St. Vladeva, Clinic of Endocrinology and Metabolic Diseases, University Hospital St. George, Medical University Plovdiv; E-mail: stefka_vladeva@abv.bg
15A Vassil Aprilov blvd., 4002 Plovdiv, Bulgaria

in patients with type 2 diabetes than in the control group of healthy subjects, which is a serious argument for their therapeutic correction. Decrease correlates with duration of diabetes and poor metabolic control. There is no causal relation between this deficiency and the manifestation of the disease, but is a consequence of diabetes development. Addition of selenium, chromium and zinc has a positive effect on the parameters of carbohydrate and lipid metabolism as well as those of the total antioxidative status. Reference limits of serum chromium and vanadium were determined during this study.

MEDICO-SOCIAL EVALUATION OF THE REPRODUCTIVE BEHAVIOR OF THE ROMA PEOPLE

Maria A. Semerdzhieva
Department of Social Medicine and health Management, Medical University, Plovdiv

INTRODUCTION

It is predominantly ethnographers, folklore researchers, and linguists that show some scientific interest in the Roma people in Bulgaria. There is also the commonly discussed problem of the isolation and discrimination of this minority in all spheres of social life.

The health problems and needs of this ethnic minority have been little studied, and most of the research tackles just one specific aspect. There are no studies that specifically target the demographic development of the Roma. Most often, the information presented about them in this respect is scanty and is used as a base in comparison studies. The Roma in Bulgaria are the second most numerous minority group after Turkish. Analysis of the population size dynamics by ethnic groups shows that the Roma population has increased more than four-fold for 100 years, while the increase in the Bulgarian and Turkish populations is much lower (2.3 and 1.4 times, respectively).

AIM

To analyze the demographic status of the Roma people and identify the specific factors influencing their reproductive behavior.

MATERIAL AND METHODS

The study was a retrospective anamnestic survey. The reproductive behaviour of the Roma population was examined in a transverse study. Five hundred women were interviewed. Twenty two women were excluded from the study for deficient or missing data in the questionnaires, so the real number of the examined subjects was 478. We used basically two methods in the study: the historical method and the direct individual interview.

RESULTS

We studied the basic factors that have an impact on the formation of the family model in the Roma minority. The young age at marriage and birth of the first child, the low degree of literacy and the low economic status in this minority leads to prolongation of the reproductive period and greater number of childbirths. The primary motives here to give birth to a child are the need for perpetuation of the family line and preservation of the Roma community in the conditions of continuous assimilation. The reproductive behavior in this ethnic population is largely determined by the husband while, traditionally, the wife has little choice.

The Romas are currently oriented to a three-child family model, which stands in stark contrast to the currently decreased birthrate in this country. The mean number of live born children in the examined Roma families is considerably lower than that of their parents, which is considered to be a great change in the reproductive behaviour of this ethnic minority. The mean number of pregnancies in the Roma population is high but most of the pregnancies are terminated voluntarily by abortions.

The length of first birth interval in the Roma people depends practically only on the age of the interviewee which implies that there is actually no conscious first childbirth planning in this minority. The intergenetic intervals in the studied ethnic group are longer than two years which has a favourable impact on the mother's health and the normal development of the fetus.

The Romas tend to have quite an expanded idea of how big a "big family" is, or how many the retrospective and ideal number of children are, and as a result they usually raise a larger number of children in their families. The preferred number of children in the Roma families is not associated with the quality of life they have; it is the husband in the family that determines how many there will be. The rather scanty information the Romas have about the basic methods of contraception and their irregular use because of the high prices of the drugs and low sexual culture make abortion on request the primary family planning method in the Roma people.

The education, social affiliation and reproductive behavior of the mothers of the interviewed women serve as a model for their daughters and determine their reproductive behavior.

Correspondence and reprint request to: M. Semerdzhieva, Department of Social Medicine and Health Management
15A Vassil Aprilov blvd., 4002 Plovdiv, Bulgaria

CONCLUSIONS

- A thorough review of the demographic status of the Rome population in a number of countries is made. A profound social and medical study of the reproductive behavior of one of the most numerous minority groups in Bulgaria is made for the first time in this country.
- On the basis of a wide range of factorial and resultant indices we establish social and psychological laws that are important for the health-care and demographic policy in Bulgaria and determine the reproductive behavior of the Roma population.
- The influence of the reproductive behavior of the mothers of the interviewed subjects on the reproductive behavior of their daughters was analyzed thoroughly and meticulously, which demonstrates in a very original way of the role of continuity in the Roma population.

SURVIVAL PREDICTION AND MODELING IN PATIENTS WITH CHRONIC LYMPHOCYTIC LEUKEMIA

Nonka G. Mateva
Department of Social Medicine and Health Management, Medical University, Plovdiv

Key words: *survival rate, prognostic models, chronic lymphocytic leukemia*

INTRODUCTION

Patient survival is a primary index of the efficacy of the prophylaxis, early diagnosis and clinical management. Prognosis of the disease duration, course and outcome in a patient is an essential part of the medical practice.

The aim of the present study was to make an epidemiological characteristic and analyze the overall survival in patients with chronic lymphocyte leukemia, to unveil the factor dependencies and propose prognostic mathematical and statistical models.

PATIENTS AND METHODS

Epidemiological characterization of chronic lymphocytic leukemia (CLL) in Bulgaria and in the districts of Plovdiv, Pazardzhik and Smolyan was done using data of the National Cancer Registry and the Oncological Dispensary in Plovdiv. The case histories and outpatient records of 150 patients with CLL treated and followed up in the Clinic of Hematology of the St George University Hospital from January 1978 through April 2004. The data were analyzed using the following statistical methods: descriptive statistics and analysis of proportions, Life table method and Kaplan-Meier method for patients' overall survival estimation, Log rank test for comparison of the survival functions, and Cox proportional hazard model for evaluation of the effect of the examined factors on the patients' survival.

RESULTS AND DISCUSSION

Crude and standardized incidence rate increased in both genders between 1991 and 2000. The mean annual rate of increase in the standardized rates in the males was approximately 0.1/100 000 and in the females 0.05/100 000. In comparison with the standardized incidence rate, the crude incidence rate had higher values with mean annual increase of 0.18/100 000 for the males and 0.11/100 000 for the females during the examined 10-year period and the highest levels were in 1999. The mean annual rate of increase was more clearly expressed in the crude than in the standardized rates. Essential role in these differences had the increasing aging of the population.

Of 150 followed up patients 96 (64%) died during the examined period. The mean survival time of these 150 patients evaluated by the method of Kaplan-Meier was 76 months (CI 95% [65,87]) and the median 61 months (CI 95% [49,73]), i.e., approximately 5 years. Using the life table method the 5-year survival rate was found to be 43.93% and the 10-year survival rate 18.23%. Results of different studies made recently in Europe have shown better survival rate of the patients with CLL (mean duration 51 months) than two decades earlier. This tendency can be explained with the improved early diagnosis and elaborated treatment methods.

A statistically significant difference was found in the overall survival between patients with mediastinal lymphadenopathy and extranodal involvement and patients without these symptoms. The morphological type, clinical stage (Rai) and ECOG PS are statistically significant prognostic factors. A statistically significant difference in the functions of overall survival was found between the patients with different absolute lymphocyte count ($\leq 100.10^9$/l vs. $>100.10^9$/l). The overall survival of the patients with abdominal lymphadenopathy was lower than that of the patients without this symptom, but the difference did not reach statistical significance. No statistically significant difference was found in the survival rate with respect to gender, age and presence of autoimmune hemolytic anemia. The mean survival time of the patients indicated for treatment (77 months, CI 95% [64,91]) was greater than that

Correspondence and reprint request to: Nonka G. Mateva, Department of Social Medicine and Health Management, Medical University, Plovdiv; E-mail: nonka_mateva@yahoo.com
15A Vassil Aprilov blvd., 4002 Plovdiv, Bulgaria

of the patients indicated for follow-up (73 months, CI 95% [53,92]), but the difference did not reach statistical significance (Log rank=0.09, P=0.7580). This finding can be explained with the heterogeneity of the follow-up group, in which patients rejecting treatment were involved. In the group of treated patients statistically significant difference in positive aspect was found in the survival rate between patients with effect from the initial chemotherapy and chemotherapy resistant patients (Log rank=11.84, P=0.0006).

Variables significant in univariate analysis were tested to construct a prognostic mathematical model using Cox proportional hazards model. The stepwise regression analysis selected ECOG PS, extranodal involvement and morphology type as the most important prognostic variables. The analysis was done in initial and validating sample, which confirmed the reproducibility of the model. Using a regression model groups of patients with different prognostic indices – low, moderate, moderately high and high, were defined. Multiple Cox regression analysis was performed additionally in the group of treated patients (n=103), including variable expressing the therapeutical response, for prognosis of the overall observed survival. In this group factors with the best prognostic value for overall survival were ECOG PS, clinical stage (Rai) and response to therapy. Using the regression coefficients and evaluation of the baseline survival function $S_{0(t)}$, for cumulative survival function we determined the survival probability for a single patient in different periods of time. For a more convenient use of the prognostic models and support of the evaluation of the prognosis in each patients program module Onco Hem Pro was constructed which represented Graphical User Interface (GUI) application, running under Windows platform.

CONCLUSIONS

The present study analyzes, interprets, and discusses some epidemiological and clinical aspects of CLL. It presents the dynamics in the disease incidence and unveils certain essential problems in the registration of the primary data in the examined region.

The marked heterogeneity of CLL determines the interest in the factors influencing the clinical course and outcome of the disease and possibility for evaluation of the survival expectancy of the patients during the follow-up period. That is why some important for the patient survival clinical symptoms, observed at the time of disease diagnosing, were analyzed with single factor and multifactor analysis. The prognostic models presented for overall survival and therapeutic effect define patient groups with different prognostic risk, requiring selective therapeutic approach. These models are easily applicable and interpretable in the everyday clinical practice and require only routine hematological tests and clinical examination.

On the other hand, the prognostic models and variables included give important information, necessary for the correct planning and carrying out of clinical examinations in CLL patients. The computer program proposed supports the everyday clinical practice by improvement of the registration and alleviates the evaluation of the personal prognosis in any patient.

ROOT SURFACE INSTRUMENTATION IN CHRONIC PERIODONTITIS

Elena I. Firkova
Department of Periodontology and Oral Diseases, Medical University, Plovdiv

INTRODUCTION

Chronic inflammatory process in the periodontal tissues causes pathological changes of root surfaces, which become biologically incompatible. Scaling and root planning is a basic element of the periodontal therapy, which appears to be a serious challenge for the dentists. Removal of the dental biofilm and calculus is traditionally done with hand instruments, sonic and ultrasonic devices. Their efficacy, potential and shortcomings are currently investigated.

The AIM of the present dissertation is to evaluate the efficacy of the various methods and instruments for root surface debridement in patients with chronic periodontitis.

MATERIAL AND METHODS

1. Root surfaces of teeth with advanced chronic periodontitis and root surfaces after treatment with manual curettes and with Piezon Master 400 and Vector™-ultrasonic system in terms of the instrumental efficacy and damages are examined with scanning electron microscopy.
2. Morphological changes of periodontally affected root surfaces are examined after conditioning with tetracycline and EDTA.
3. Bacterial endotoxin concentration on the root surfaces is determined before and after scaling and root planning.
4. Comparison of the gingival response to hand and ultrasonic treatment modalities is done in patients with early to moderate chronic periodontitis.

RESULTS

1. The efficacy of scaling and root planning depends on the instrumental method applied. The most effective dental calculus removal is achieved with Vector™-system, followed by Piezon Master 400 system 402 and Gracey curettes.
2. Root surface damages highly correlate with the type of the applied instruments. The most sparing treatment is achieved with Vector™-system, followed by Gracey curettes and Piezon Master 400 system 402.
3. Optimal conditioning of the root surfaces is achieved by a 5-minute application of tetracycline hydrochloride (100 mg/ml) and PrefGel (25% EDTA).
4. Endotoxin concentration on the diseased root surfaces is 0.825 EU/ml, after manual debridement the concentration is reduced to 0.240 EU/ml, but statistically significant reduction is achieved with the ultrasonic devices – 0.184 EU/ml.
5. The greatest reduction in pocket probing depths in shallow pockets (up to 4 mm) is observed one month after completion of the treatment with Vector™-system. No statistically significant differences are found between this approach, Gracey curettes and Piezon Master 400 system 402 in months 3 and 6. In treatment of moderately deep periodontal pockets (4-6 mm) marked improvement appears later – in months 3 and 6. The highest efficacy is again achieved with Vector™-method and the poorest with Piezon Master 400 system 402.
6. During the examined period of chronic periodontitis treatment favorable gingival response, evaluated by the gingival index, is achieved with the three instrumental methods. In months 3 and 6 the lowest index values are observed in the patients treated with Vector™-system, although the difference is short of statistical significance.

CONTRIBUTIONS

1. Electron microscopic description of diseased root surfaces is done.
2. Three methods of instrumental treatment of root surfaces with chronic periodontitis are used and the results obtained are verified with scanning electron microscopy.
3. Comparative evaluation of the efficacy of dental biofilm disruption and calculus removal with Gracey curettes, Piezon Master 400 and Vector™-system is made. Comparative evaluation of the instrument aggressiveness and risk of damage during root instrumentation is also made.
4. The concentration and duration of application of two conditioning agents for optimal conditioning of the root surfaces is established.
5. Presence of bacterial endotoxin on the root surface of chronic periodontitis teeth is verified. The endotoxin concentration before and after treatment with manual instruments and ultrasonic appliances is correctly detected.

Correspondence and reprint request to: El. Firkova, Department of Periodontology and Oral Diseases, Medical University, Plovdiv
24 Veliko Tarnovo St., 4002 Plovdiv, Bulgaria

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EXTRAPERITONEAL APPROACH OF CHEATLE-HENRY IN OPERATIVE TREATMENT OF CRYPTORCHIDISM

Alexander S. Yonkov
Clinic of Pediatric Surgery, Medical University, Plovdiv

Key words: *cryptorchidism, extraperitoneal approach*

The AIM of the dissertation is to determine and evaluate the therapeutic possibilities of the extraperitoneal approach of Cheatle-Henry in the operative treatment of cryptorchidism.

OBJECTIVES

1. To examine and master the extraperitoneal approach of Cheatle-Henry in children with high arrested cryptorchid testes.
2. To specify the indications for application of the extraperitoneal approach of Cheatle-Henry.
3. To analyze and compare the early postoperative results between the orchidopexia with extraperitoneal approach of Cheatle-Henry and the standard inguinal approach.

MATERIAL AND METHODS

The study was conducted as a case-control study in patients with cryptorchidism. The principal group included 46 children with 50 high arrested testicles, mainly non-palpable, treated by extraperitoneal approach after Cheatle-Henry between 1997 and 2004. The control group was composed of 50 children with the same number of non-palpable testicles that had been treated by standard inguinal approach between 1989 and 1997.

The methods of investigation included clinical methods like history of disease, clinical examination, laboratory examinations, palpation under narcosis, hormone therapy with chorionic gonadotropin (Pregnyl) - 500 U intramuscularly twice weekly to a total dose of 5000 U. Preoperative ultrasonography, histological examination and a 6-month follow-up of the patients were also applied. The data were analyzed using analysis of variance, alternative, non-parametric analysis and Student-Fisher t-test.

RESULTS

1. Examination and mastering the extraperitoneal approach of Cheatle-Henry in children with high arrested cryptorchid testes.

The operative technique is based on the mathematical rule that the shortest distance between two points is a straight line. The testis descends in the scrotum through the shortest distance but not through the inguinal canal. Forty-six children with 50 cryptorchid testes (47 non-palpated and 3 palpated) were treated by this approach. Most of the cases were of abdominal retention of the testis 32 (64%). All 36 testes found during the operation were fixed in the scrotum using the operative techniques of Ombredanne, Petrivalsky-Shoemaker or Gross. There was no case of orchiectomy or subcutaneous fixation of the testis. Excision biopsy was done in 5 cases of aplasia (10%) verifying histological absence of functionally competent testicular tissue.

All 50 children from the control group were treated for unilateral non-palpable testes. Of all cryptorchid testes those with abdominal position were most frequently found – 26 (52%). In abdominal testicle retention subcutaneous fixation of the testis was most frequently applied – 14 cases (53.85%). The subcutaneous fixation was not used only in inguinal retention of the testis. The Gross method was only used for scrotal fixation of the testis – 20 cases. Neither Ombredanne nor Petrivalsky-Shoemaker method was used.

2. Indications for extraperitoneal approach of Cheatle-Henry.

Standard inguinal approach for treatment of non-palpable testes was used from 1989 through 1997. The mean age of the operated children was 5.98 ± 0.326 years, varying between 2 and 15 years. After 1997 when the extraperitoneal approach of Cheatle-Hentry was introduced in high testicle retention the mean age of the operated children was 4.76 ± 0.26, varying between 2 and 12 years. Statistically significant difference was found in the mean age of

Correspondence and reprint request to: Al. Yonkov, Clinic of Pediatric Surgery, Medical University, Plovdiv 15A Vassil Aprilov St., 4002 Plovdiv, Bulgaria

the two groups of children (t = 2.929, P = 0.0043). On the opposite side of the pathology high testicular retention at different level (i.e. bilateral cryptorchidism), concomitant hernia or hydrocele were found. These conditions were additional indication for the extraperitoneal approach of Cheatle-Hentry with bilateral surgery performed at one and the same surgery time. Using this technique bilateral orchidopexy at one operation time was performed in 4 children. In another child surgical correction of concomitant hydrocele was performed.

3. Early postoperative results in orchidopexies performed using extraperitoneal approach of Cheatle-Henry and standard inguinal approach

Subcutaneous fixation of the testis was the principal technique in the standard inguinal approach – 14 cases (41.18%). In the extraperitoneal approach of Cheatle-Henry scrotal implantation of the testis was the primary procedure – 28 cases (77.78%), and subcutaneous fixation of the testis was not done. Of the more significant postoperative complications wound infections developed in 3 children operated by the extraperitoneal approach of Cheatle-Henry and in 2 children operated by the standard inguinal approach. No statistically significant difference was found in the wound infection incidence between the two groups, although the extraperitoneal approach is more traumatic (t = 0.46, P > 0.05). There were no lethal outcomes in both groups.

It is worth noting that there were many successful orchidopexies using the extraperitoneal approach of Cheatle-Henry – 28 cases (77.78%), and only one case of failure (2.78%) because of testicular atrophy. In the standard inguinal approach the operation was successful only in 8 cases (23.53%) and in 3 cases (8.82%) the treatment failed again because of testicular atrophy. In this approach satisfactory result was the most consistent outcome achieved in 23 cases (67.65%).

CONCLUSIONS

- The extraperitoneal approach of Cheatle-Henry through a Pfannenstiel incision cut in operative treatment of high-retained cryptorchid testes creates a wide field to free off the cord elements without distorting the inguinal canal.
- Maximal release of the spermatic vessels and vas deferens and direct placement of the testis in the scrotum through a “new inguinal canal” ensure the best possible lengthening of the cord without impairment of the testis blood supply and passing it through the shortest way.
- The most convenient age for operation of cryptorchidism is about 2 years, when the extraperitoneal approach of Cheatle-Henry can be successfully applied.
- The physical examination of the patient according to our clinical classification of cryptorchidism, supported by ultrasound diagnostics and palpation under anesthesia are sufficient enough to make a choice of operative approach.
- The extraperitoneal approach of Cheatle-Henry is especially suitable in bilateral cryptorchidism, which can be corrected at the same operative time.
- The extraperitoneal approach of Cheatle-Henry allows to deal with the surgically most difficult cases of undescended testes and place them in the scrotum without any difficlties. The highly encouraging early results using this approach give us ground to recommend it as a method of choice in the treatment of high-arrested cryptorchid testes.

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Medical University
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DISSERTATION. List the name of author, title of dissertation, specified in square brackets [dissertation], the locale where the dissertation was written, the name of institution and year of writing.

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Back cover:

One of the most beautiful reliefs of the healing cults in the Greco-Roman pantheon is the unique "Frieze of the healing family" exhibited in the Archeology Museum in Plovdiv.
It was excavated in the foundations of an old ruined Turkish mosque in 1921. The correct identification of all figures was performed by Professor Zapryanov in 1964 - Department of Social Medicine. The frieze, according to him, used to adorn a Roman valetudinaria - a military hospital - off the walls of the east entrance of the ancient city which was called Trimontium by the Romans in the late III century. It weighs about 3000 kg and is 2.80 m long and 1.08 m high. The figures on it are framed in a wide rim; it bears the personified images of the Moon (on the left) and the Sun (on the right).*
Presented on the frieze are (from left to right): Jaso and Panacea - Asclepios' daughters, Telesphor - the fortunate genius of the healing process, Asclepios - the god of healing art, Hygeia - his daughter, Epione - Asclepios' wife, Machaon and Podaleirios - his sons worshipped as military physicians.
All figures, except Panacea, are entirely in full face which is very rare in a general composition picture. The frieze's sculptor depicted in great detail the figures' anatomic features, clothes and peculiar attributes. All deities in the composition are on a par with the only association seen between Panacea and Asclepios (Panacea touches a bundle of herbs next to Telesphor's cowl with her left hand, while pouring the cure all (panacea) in Asclepios' bowl).

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* Folia Medica 1964; 6(3): 152 - 156